TSA FIELD ENABLEMENT · WORKSHOP MODULE

Cortex Cloud
A Day In The Life

Read the environment. Understand the drivers. Define the operational tasks.

WORKING SCENARIO · “Customer Name Here”

Agenda - Day 1 - General Overview

Overview of Cortex Cloud

Overview of Modules (In Entitlements)

- Posture Management
- Runtime
- Appsec
- Data Security
- CIEM

Overview of XSIAM (If Cortex Cloud Licensed)

- Automation and Playbooks
- XQL
- Dashboards and Widgets
- Data Ingestion

Operations

Overview of Teams

- SecOps Team
- Cloud Sec Team
- DevOps Team

Overview of Components

- ASM
- Exposure Management
- ITDR
- RBAC and SBAC

Agenda - Day 2 - Cortex Cloud Operations

Understanding Cortex Cloud

Findings and Their Origins (Detection Engines)

- Posture Management
- Runtime
- Appsec
- Data Security
- CIEM

Automation and Response Techniques

- Posture Management
- Runtime
- Appsec
- Data Security
- CIEM

Issue and Case Lifecycle Management

- Assigning Severities
- ITSM (SNOW & JIRA)
  - Integration and Automation
- Issue and Case Closure
- Reports

Operations Through DevSecOps

- Integrations
  - DevOps Pipeline Tools
- Configuration as Code (Mgt and Maint)
  - Policy Management
  - Issues Management
  - Reporting and Alerting
- Reports

Agenda - Day 3 - XSIAM Operations (Based on Entitlements)

Data Ingestion

Understanding Data Ingestion

- Data Sets
- Data Modelling
- DFE
- XSIAM Detection Engines
  - Analytics
  - Correlations
  - IOC
  - BIOC
  - Cortex Cloud (Modules)

Investigation

- Issues and Severities
- Cases and Severities

Response

- Issues and Cases
- Issue Domains
- RBAC and SBAC
- Dashboards and Widgets
- Custom Layouts and Buttons

Gotcha's between XSIAM and Cortex Cloud

- Issue Grouping
- 50k Issues / Day per module

Agenda - Day 4 - Optimization and Architecture Considerations

Optimization

Leveraging Agentix and AI	What Good Looks Like

Next Steps and Recommendations

Review Parking Lot Items	Drafting Operations Roadmap Based on Recommendations

TSA FIELD ENABLEMENT · WORKSHOP MODULE

DAY 1 - General Overview

Read the environment. Understand the drivers. Define the operational tasks.

General Overview - Cortex Cloud (C1 or C3 License and AppSec Module)

Overview of Modules (In Entitlements)

We will walk through your entitlements, and the platform to get a better understanding of how it’s being operationalized currently.

General Overview - Overview of Modules (Based On Entitlements)

Entitlement Items	Used	Not-Used	Why	Team Ownership
xxxx	xxxx	xxxx	xxxx	xxxx
xxxx	xxxx	xxxx	xxxx	xxxx
xxxx	xxxx	xxxx	xxxx	xxxx
xxxx	xxxx	xxxx	xxxx	xxxx

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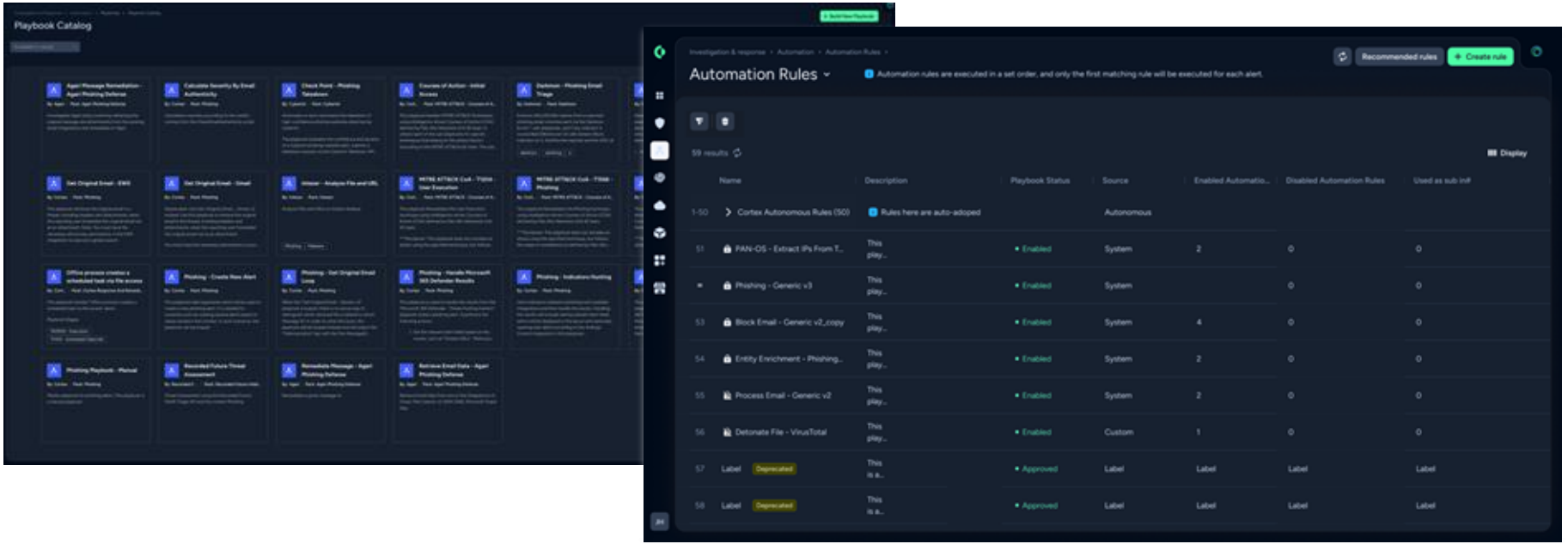
Automation & Playbooks

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General Overview - Overview of XSIAM Capabilities

Automation and Playbooks

Cortex XSIAM automates threat response by using **prebuilt**, **autonomous** and **customizable playbooks** to enrich security context, orchestrate workflows across third-party tools and execute remediation actions at machine speed.



Out of the Box Content

	Alert Handling PBs *Reduce alert volume prior to analyst involvement*	Enrichment PBs *Show analyst best data in timely fashion*	Containment PBs *Simplify task by allowing "1-click" decisions*
Recommended Playbooks	**Playbook Trigger Recommendations** 1. Local Analysis 2. Wildfire Malware 3. Identity Analytics - Alert Handling 4. Ransomware Response 5. NGFW Scan 6. IOC ALert 7. Cortex ASM - Alert 8. Impossible Traveler 9. And more!	**Core Playbooks** 1. Enrichment for Verdict 2. Dedup - Generic v4 3. Get prevalence for IOCs 4. Endpoint Investigation Plan 5. Handle False Positive Alerts **Enrichment Playbooks** 1. IP Enrichment - Generic v2 2. URL Enrichment - Generic V2 3. File Reputation 4. Account Enrichment - Generic v2.1 5. Wildfire Detonate and Analyze File 6. Get entity alerts by MITRE tactics 7. CVE Enrichment - Generic v2 **Action Playbooks** 1. Handle False Positive Alerts	**Core Playbooks** 1. Containment Plan 2. Block Indicators - Generic v3 3. Eradication Plan 4. Recovery Plan 5. Ticket Management - Generic **Containment Playbooks** 1. Isolate Device 2. Disable Account 3. Quarantine File 4. Block Indicators **Blocking Playbooks** 1. Block Domain - Generic v2 2. Block File - Generic v2 3. Block IP - Generic v2 **Eradication Playbooks** 1. Eradication Plan - Delete File 2. Eradication Plan - Reset Password 3. Eradication Plan - Terminate Process
Value	• Only require tuning of inputs based on desired outcomes. • Quick time to value.	• Core Playbooks call one or more sub-playbooks. • Immediate value when used in custom playbooks. • Template for customized versions.	• Tuned to work with common actions in XSIAM. • Simplify response and recovery actions. • Template for customized versions.

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General Overview - Overview of Components

Automation and Playbooks

Automation Requirements	Dependent Scripts & Data Sources
xxx	xxxx
xxx	xxxx
xxx	xxxx
xxx	xxxx

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XQL

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General Overview - Overview of XSIAM Capabilities

XQL

XQL (Cortex Query Language) is Cortex XSIAM's flexible, SQL-like search language that enables security teams to parse, correlate and analyze massive volume of multi-vendor telemetry in real-time for accelerated threat hunting, custom detection creation and deep forensic investigations.

EXAMPLE

Top 50 High Risk Assets → Use-case: The remediation shortlist: the 50 assets with the most independent risk types and findings, with account, region and issues.

```
config timeframe = 30d
| dataset = findings
| filter xdm.finding.is_active = true
| alter age_days = timestamp_diff(current_time(), xdm.finding.first_observed, "DAY")
// per asset: findings, number of different categories, the category list, and the oldest finding age
| comp count() as active_findings, count_distinct(xdm.finding.category) as risk_categories,
      values(xdm.finding.category) as categories, max(age_days) as oldest_finding_days
  by xdm.finding.asset_id, xdm.finding.asset_name, xdm.finding.asset_type

// attach provider / account / region (one inventory row per asset)
| join type = left (
    dataset = asset_inventory
    | fields xdm.asset.id as a_asset_id, xdm.asset.provider as provider, xdm.asset.realm as account, xdm.asset.cloud.region as region
    | dedup a_asset_id
  ) as a a.a_asset_id = xdm.finding.asset_id

// attach number of issues on the asset
| join type = left (
    dataset = issue_to_asset
    | fields xdm.asset.id as ita_asset_id, xdm.issue.id as ita_issue_id
    | comp count_distinct(ita_issue_id) as issues_on_asset by ita_asset_id
  ) as ita ita.ita_asset_id = xdm.finding.asset_id
| alter issues_on_asset = coalesce(issues_on_asset, 0)

// most different risk types first, then most findings
| sort desc risk_categories, desc active_findings
| limit 50
| fields xdm.finding.asset_name as asset_name, xdm.finding.asset_type as asset_type, provider, account, region,
        risk_categories, categories, active_findings, issues_on_asset, oldest_finding_days, xdm.finding.asset_id as asset_id
```

EXAMPLE

Risk Concentration (top 1% / 5% / 10% of assets) → How concentrated risk is: share of active findings held by the top 1%, 5% and 10% of assets, and how many of them carry 3+ risk types.

```
config timeframe = 30d
| dataset = findings
| filter xdm.finding.is_active = true
// step 1: per asset, number of active findings and number of different risk categories
| comp count() as findings, count_distinct(xdm.finding.category) as risk_categories by xdm.finding.asset_id
// step 2: totals over all assets (windowcomp without sort = whole result, every row kept)
| windowcomp sum(findings) as total_findings
| windowcomp count() as total_assets
// step 3: rank assets from most to fewest findings
| windowcomp row_number() sort desc findings as asset_rank
// step 4: running total of findings from the top asset down (between null and 0 = first row to current row)
| windowcomp sum(findings) sort desc findings between null and 0 as cum_findings
// step 5: asset_pct = rank as % of all assets; cum_pct = % of all findings held by assets up to this rank
| alter asset_pct = multiply(divide(to_float(asset_rank), total_assets), 100),
      cum_pct = multiply(divide(to_float(cum_findings), total_findings), 100),
      multi_risk = if(risk_categories >= 3, 1, 0)
// step 6: bucket by rank position
| alter bucket = if(asset_pct <= 1, "1: top 1%", if(asset_pct <= 5, "2: top 1-5%", if(asset_pct <= 10, "3: top 5-10%", "4: remaining 90%")))
// step 7: per bucket; max(cum_pct) = cumulative share of findings up to the end of the bucket
| comp count() as assets, sum(findings) as findings_in_bucket, max(cum_pct) as cumulative_findings_pct,
    avg(risk_categories) as avg_risk_categories, sum(multi_risk) as assets_3plus_risk_types by bucket
// tidy: whole % and 1-decimal average (x10, round, /10)
| alter cumulative_findings_pct = round(cumulative_findings_pct),
      avg_risk_categories = divide(round(multiply(avg_risk_categories, 10)), 10)
| sort asc bucket
```

EXAMPLE

AppSec Shift-Left Profile Discovery → Use-case: Which code-side findings exist (IaC, secrets, SCA / open-source packages, pipeline, code) by asset class/type and category, how many were fixed, and how fast.

```
config timeframe = 90d
| dataset = findings
// keep code-side findings: the asset or the finding mentions code, repositories, IaC, secrets, packages or pipelines
// (~= is a regex match, (?i) = case-insensitive). Replace with the real finding-source field once D-01b shows it.
| filter xdm.finding.asset_class ~= "(?i)code|repo|pipeline|cicd"
    or xdm.finding.asset_type ~= "(?i)code|repo|pipeline|cicd|branch"
    or xdm.finding.category ~= "(?i)iac|secret|sca|package|dependenc|code|pipeline|cicd|license"
    or xdm.finding.name ~= "(?i)secret|hard-?coded|terraform|cloudformation|kubernetes manifest|package|dependenc"
// active flag, and days-to-fix only for findings that are no longer active
| alter is_active_n = if(xdm.finding.is_active = true, 1, 0),
      days_to_fix = if(xdm.finding.is_active = false, timestamp_diff(xdm.finding.last_observed, xdm.finding.first_observed, "DAY"), null),
      new_90 = if(timestamp_diff(current_time(), xdm.finding.first_observed, "DAY") <= 90, 1, 0)
| comp count() as findings, sum(is_active_n) as active, count(days_to_fix) as fixed,
    median(days_to_fix) as median_days_to_fix, sum(new_90) as new_last_90d,
    count_distinct(xdm.finding.asset_id) as assets, values(xdm.finding.name) as example_findings
  by xdm.finding.asset_class, xdm.finding.asset_type, xdm.finding.category
// fix rate = fixed / all findings x 100
| alter fix_rate_pct = if(findings > 0, round(multiply(divide(to_float(fixed), findings), 100)), 0)
| fields xdm.finding.asset_class as asset_class, xdm.finding.asset_type as asset_type, xdm.finding.category as category,
      findings, active, fixed, fix_rate_pct, median_days_to_fix, new_last_90d, assets, example_findings
| sort desc findings
| limit 200
```

General Overview - Overview of Components

XQL

Query Requirements	
xxx	xxxx
xxx	xxxx
xxx	xxxx
xxx	xxxx

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Dashboards and Widgets

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General Overview - Overview of XSIAM Capabilities

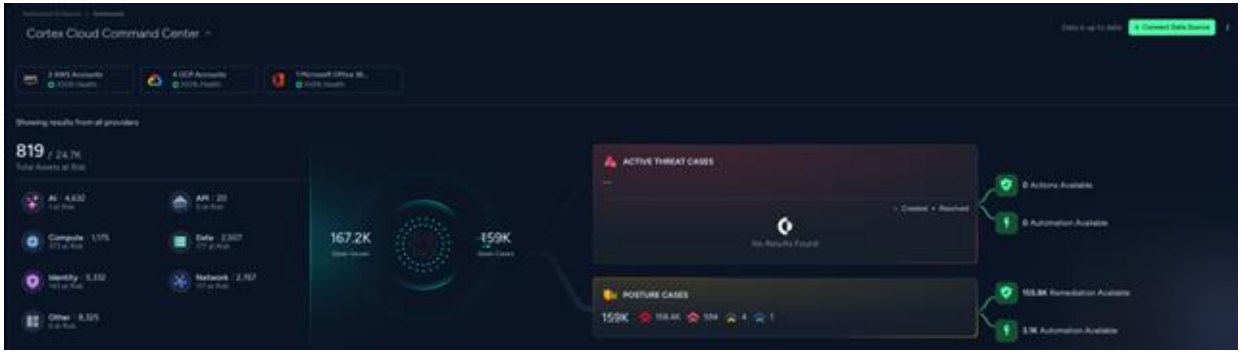
Dashboards and Widgets

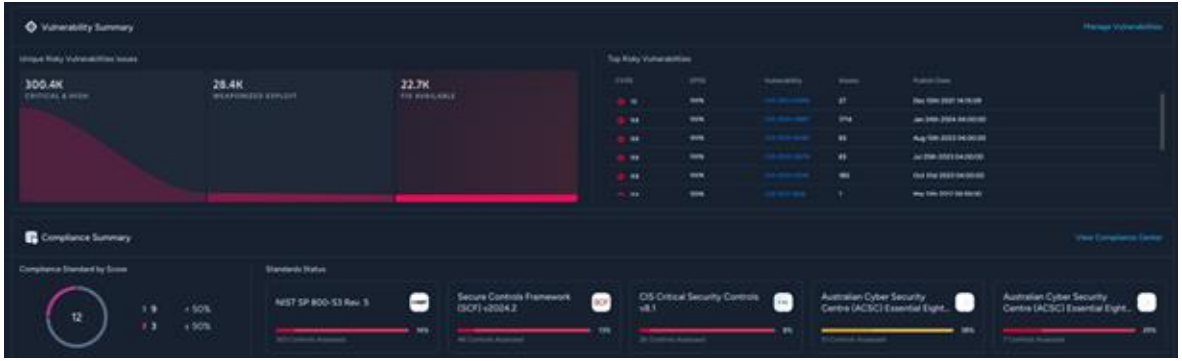
Dashboard and **Widgets** provide dynamic, real-time visual representations of security telemetry powered by XQL queries, AI and customizable templates, enabling security teams to monitor system activity, track operational metrics and analyze threat trends at a glance.

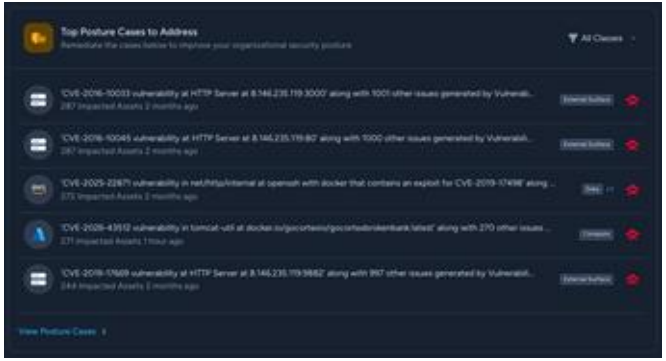
General Overview - Overview of XSIAM Capabilities

Dashboards and Widgets

Cloud Security Operations Dashboard







General Overview - Overview of XSIAM Capabilities

Dashboards and Widgets

Cloud Threat Identity (CIEM) Dashboard

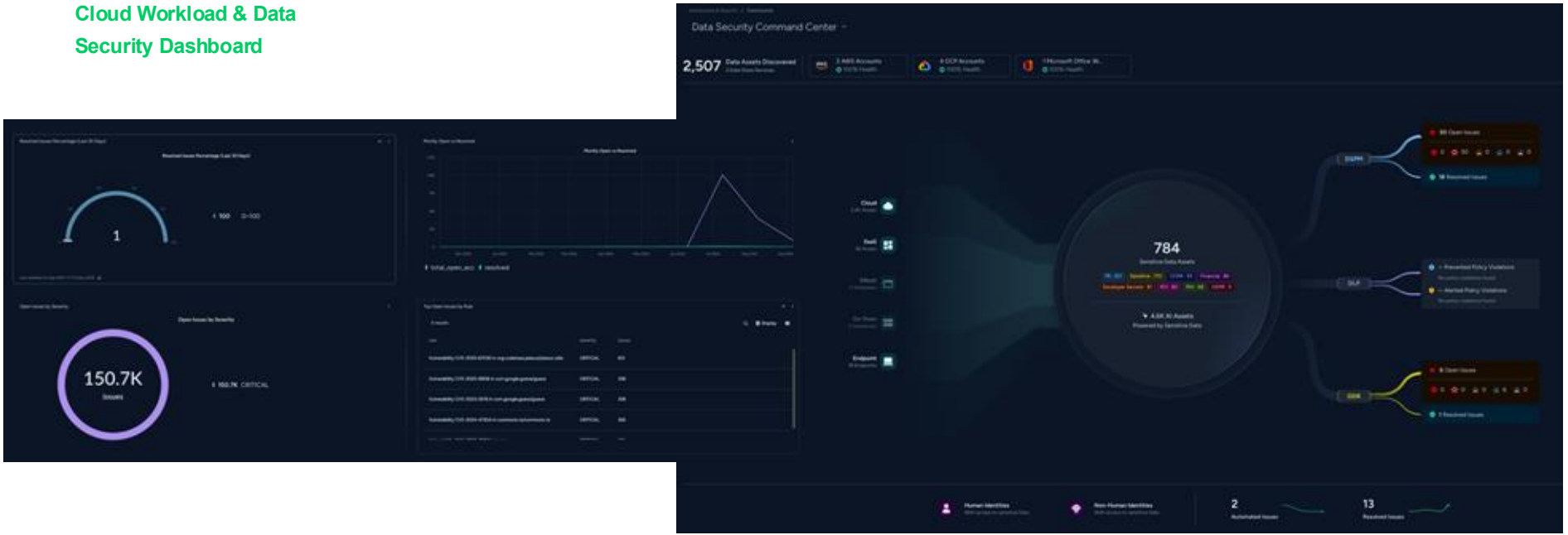




General Overview - Overview of XSIAM Capabilities

Dashboards and Widgets

Cloud Workload & Data Security Dashboard



General Overview - Overview of Components

Dashboards & Widgets

Dashboard Requirements	Dependent Widgets
xxx	xxxx
xxx	xxxx
xxx	xxxx
xxx	xxxx

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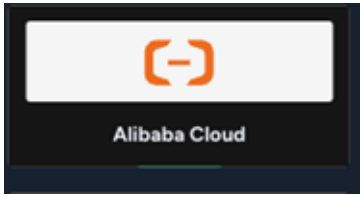
Data Ingestion

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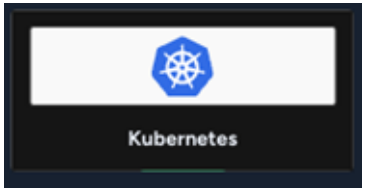
General Overview - Overview of XSIAM Capabilities

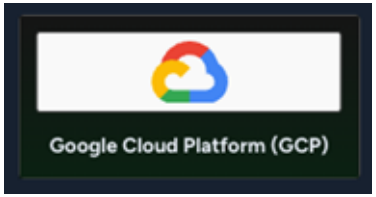
Data Ingestion

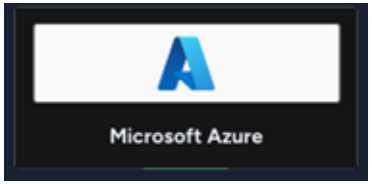
Cortex XSIAM simplifies cloud onboarding and data ingestion by using automated wizards and infrastructure-as-code (IaC) templates to securely connect cloud environments, continuously collecting multi-cloud logs and telemetry into Cortex Data Lake, and normalizing them into the Cortex Data Model (XDM) for real-time threat detection and analytics.

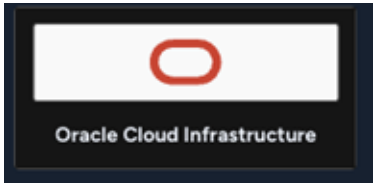




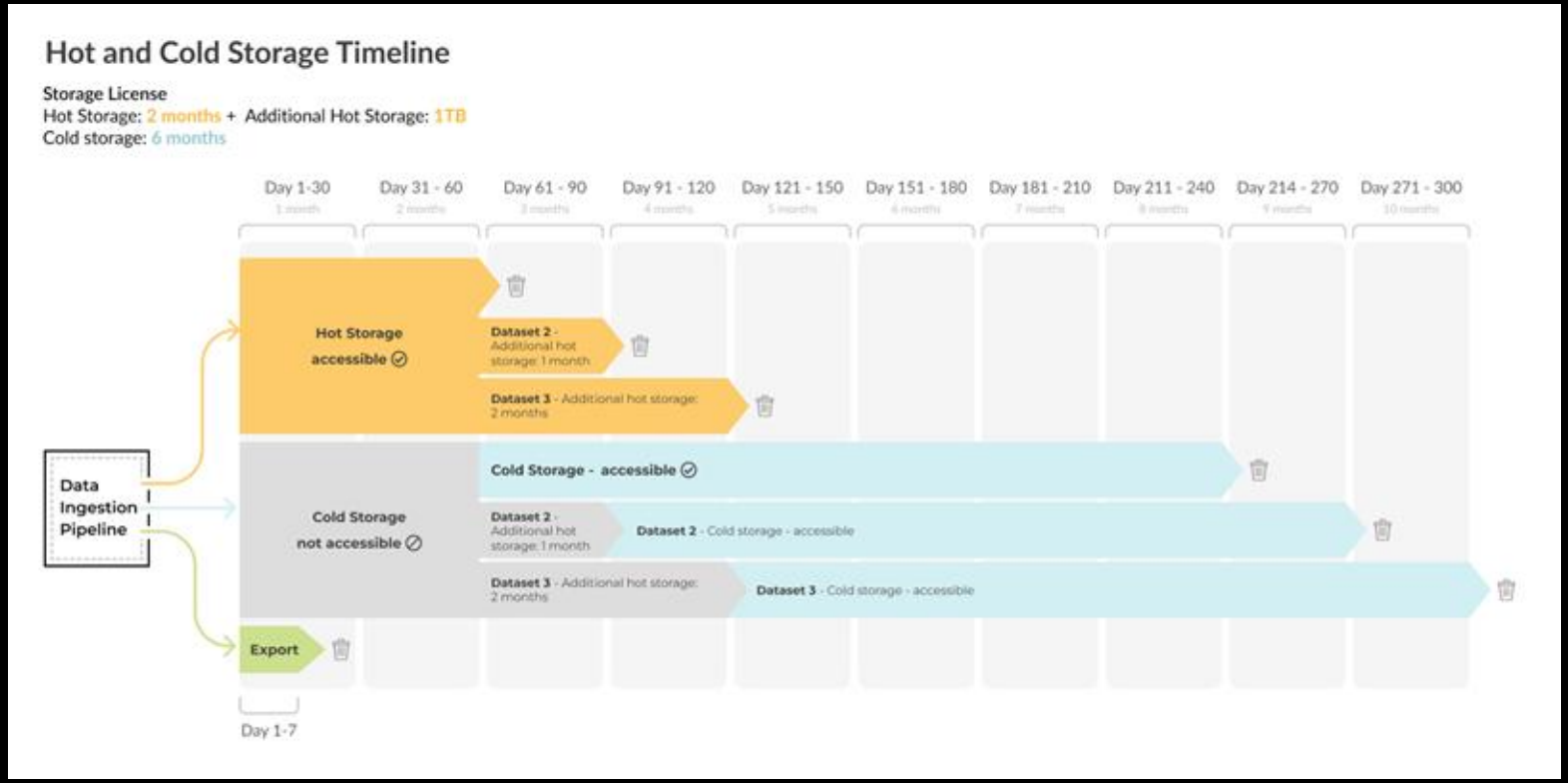








Data Storage Pipeline



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General Overview - Overview of Components

Data Ingestion

Cloud Provider	Required
GCP	xxxx
AWS	xxxx
Azure	xxxx
OCI	xxxx

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ASM

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General Overview - Overview of Components

ASM

Attack Surface Management (ASM) in Cortex XSIAM continuously discovers, attributes, and monitors unmanaged, internet-facing assets and security risks, automatically converting external exposures into actionable issues for automated triage and remediation.

Attack Surface Discovery

SCAN



Indexing Infrastructure (IPv4 and IPv6)

IDENTIFY



IPs (IPv4/IPv6)



Certificates



Domains



Services

ATTRIBUTE



AI attributes assets

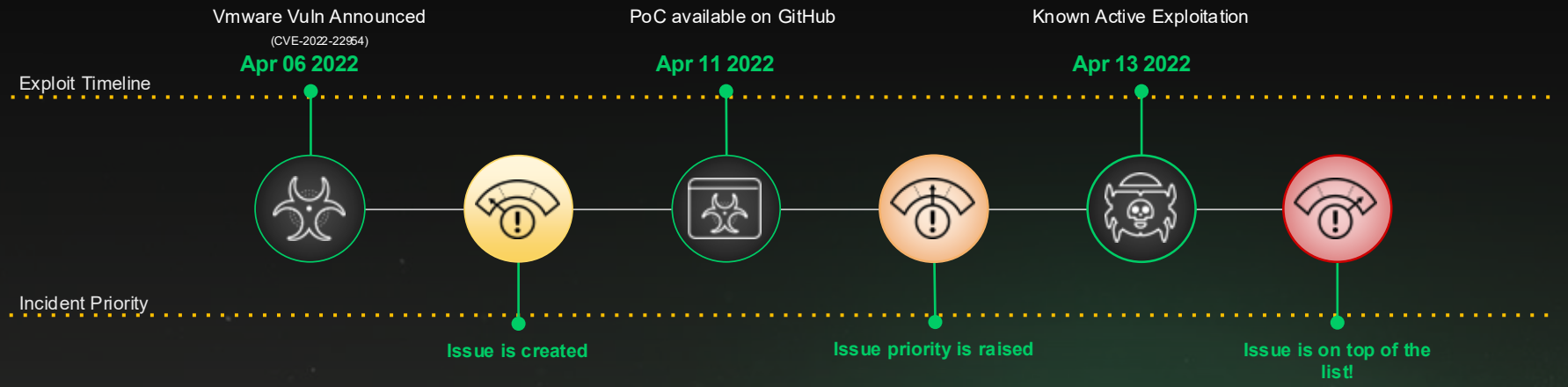
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Human review

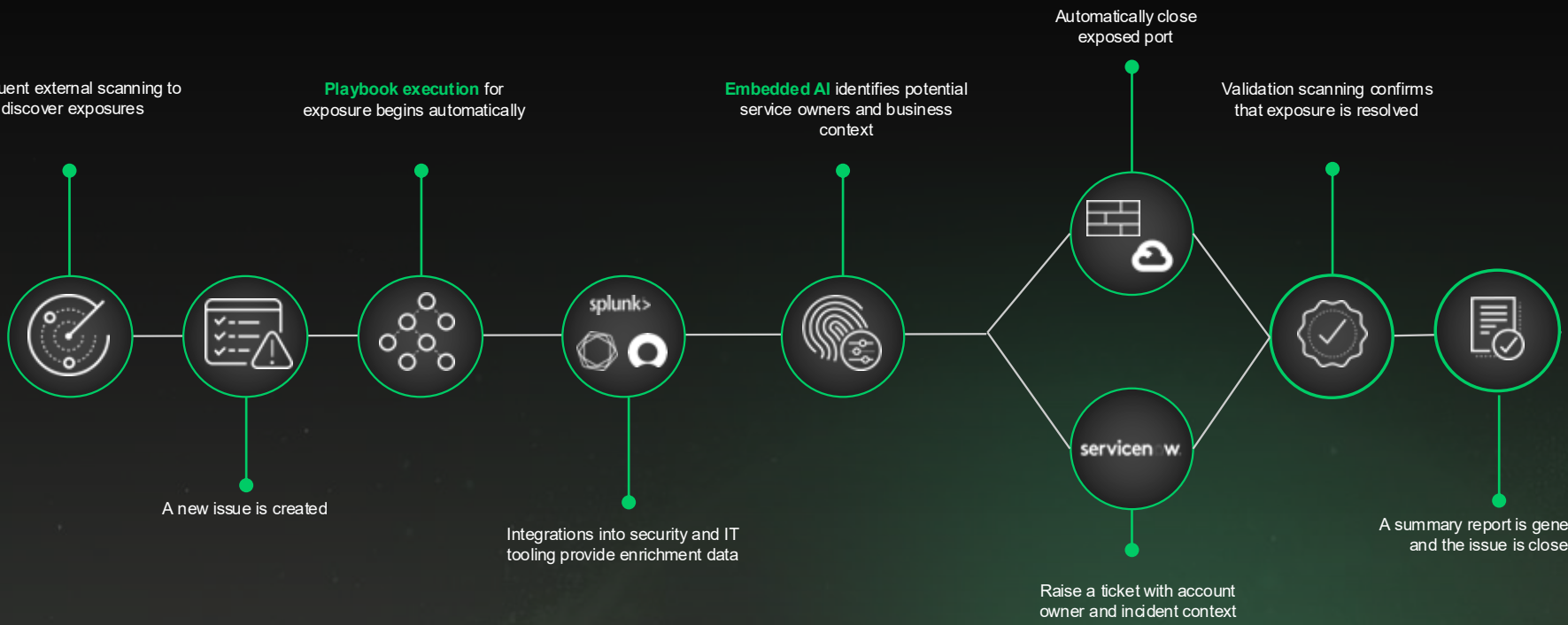
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Attack Surface Assessment



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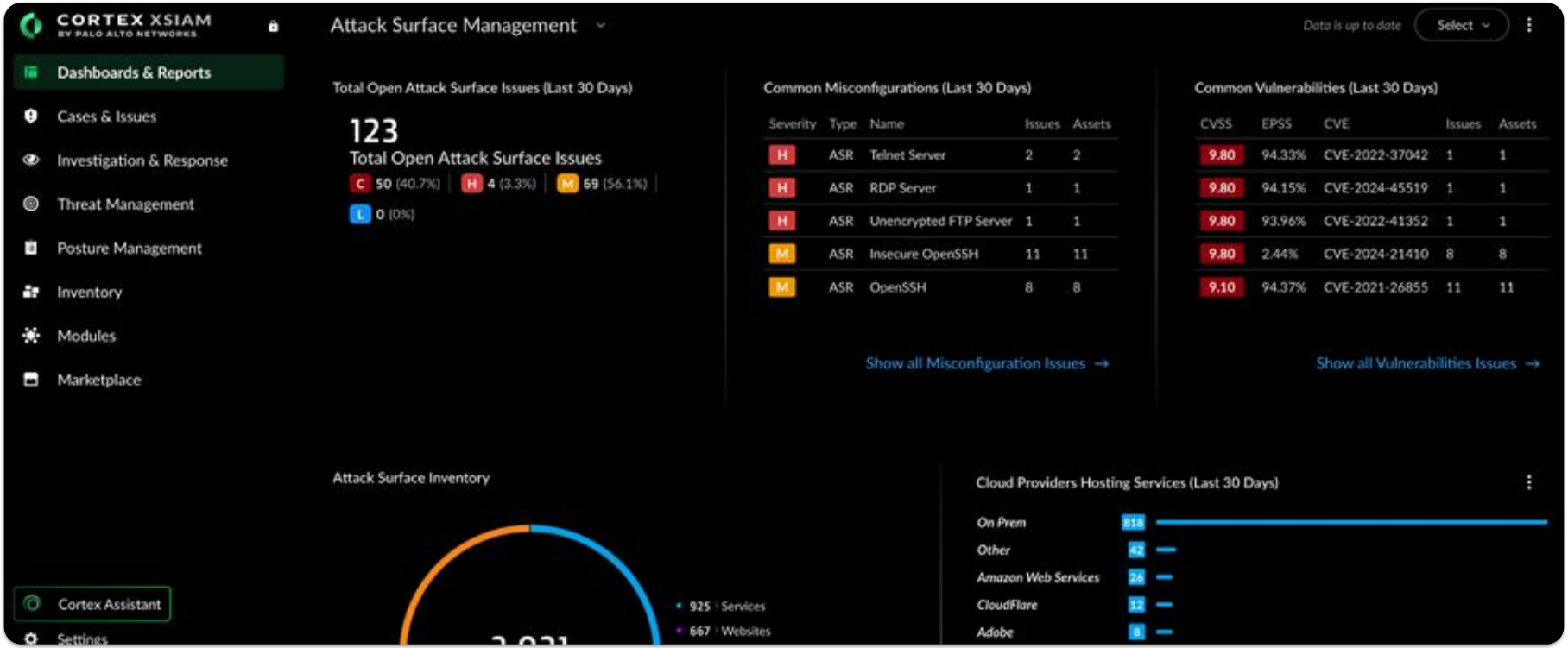
Attack Surface Remediation



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Attack Surface Management Dashboard

Use ASM Dashboard to see overview of total ASM issues, common misconfigurations and vulnerabilities, attack surface inventory by asset type, cloud providers hosting services, and geolocation.



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Exposure Management

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General Overview - Overview of Components

Exposure Management

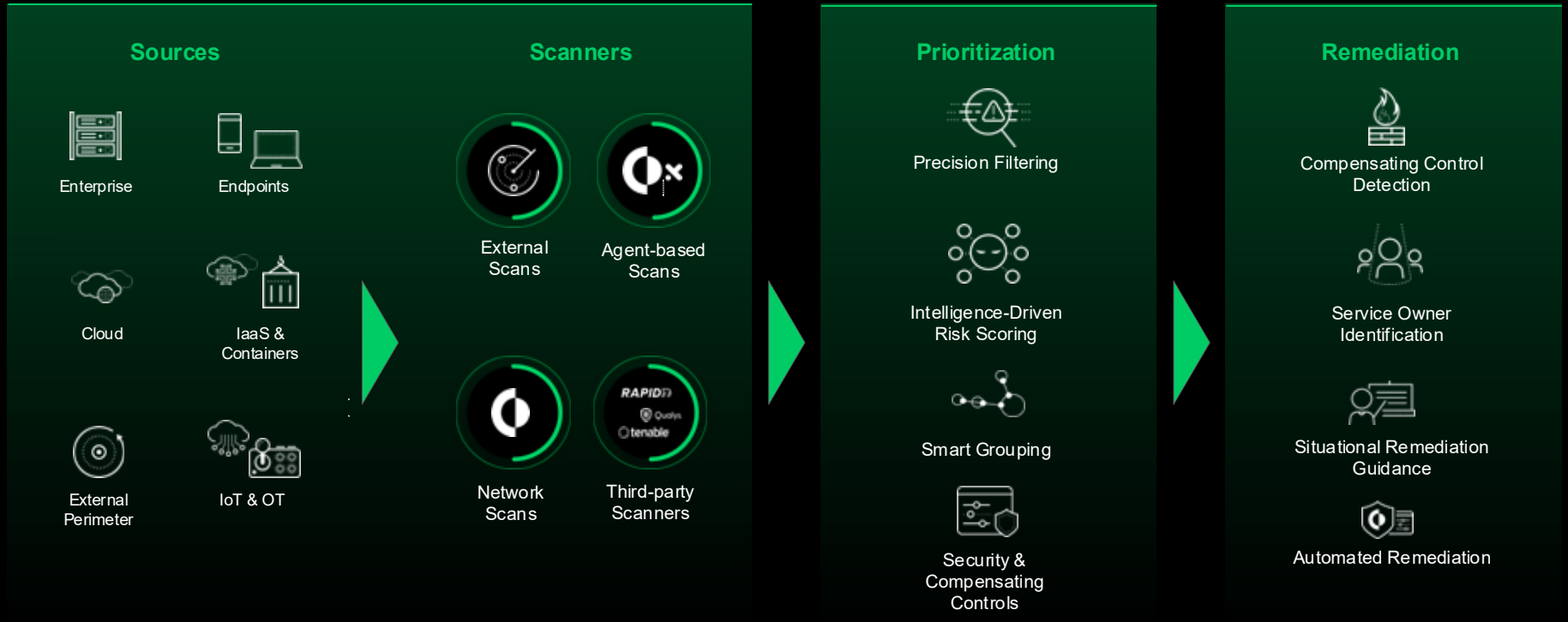
Exposure Management in Cortex XSIAM unifies internal vulnerability data, cloud posture findings, and external attack surface discoveries into a continuous, risk-prioritized framework to proactively identify and remediate security gaps across the entire enterprise footprint.

Modern Organizations Require Exposure Management at Scale

Consolidate vulnerability data in one place

Prioritize vulnerabilities that matter

Remediate at scale with minimal human involvement

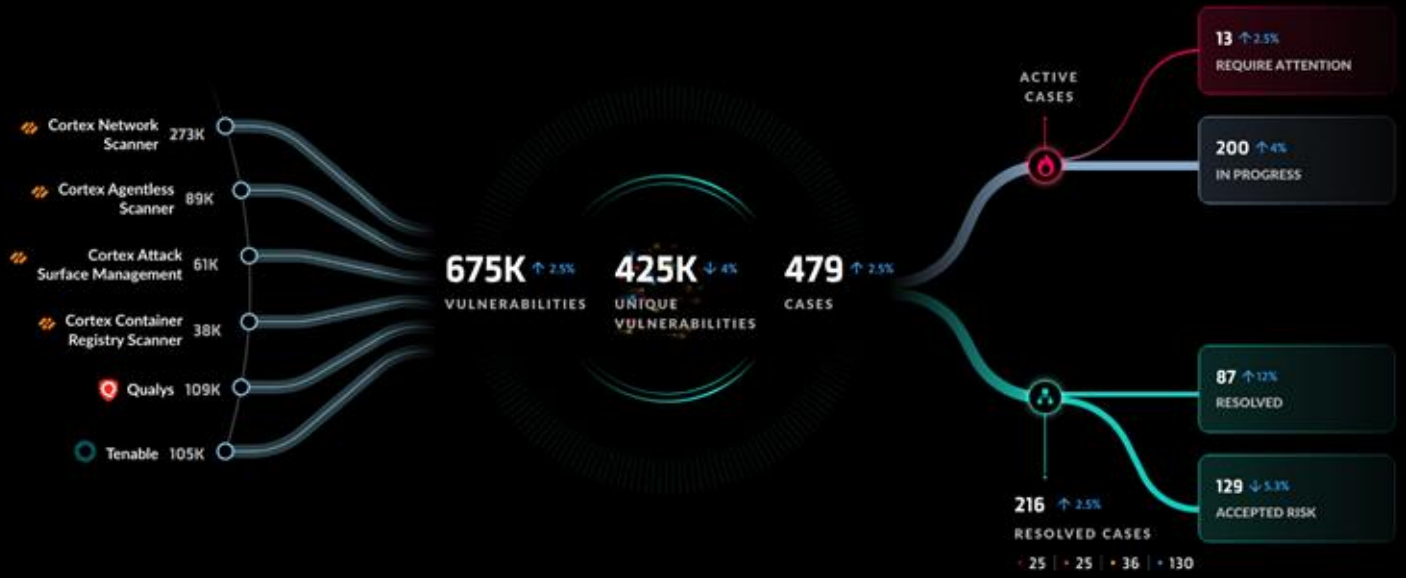


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Exposure Management Command Center

Good Afternoon, marshall

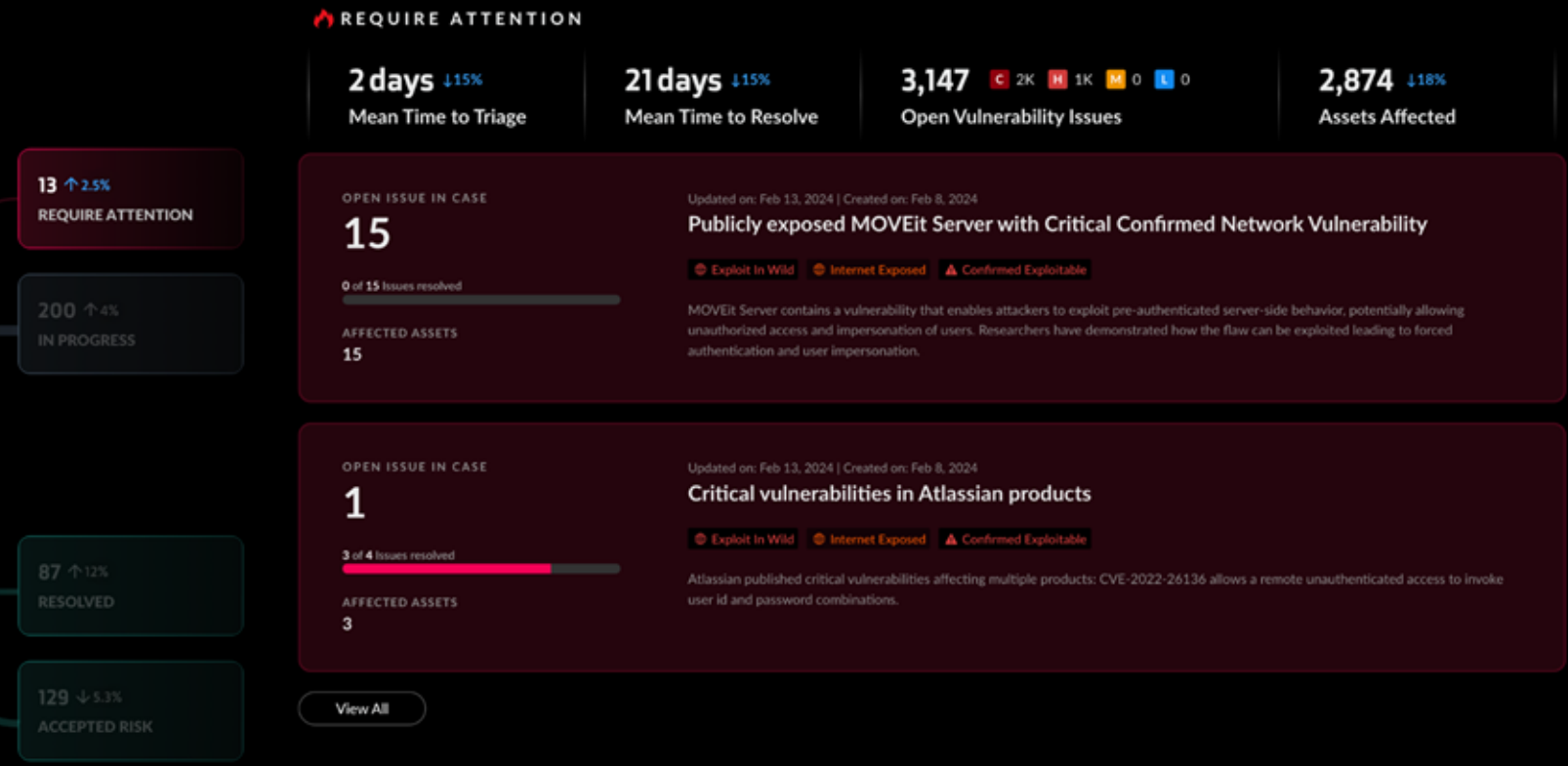


Vulnerable Assets
480 ↑ 104.3%

Active Cases
263
C 101 H 132 M 29 L 1

Mean Time to Resolve
4 DAYS
C 4 DAYS H 4 DAYS M 6 DAYS L 6 DAYS

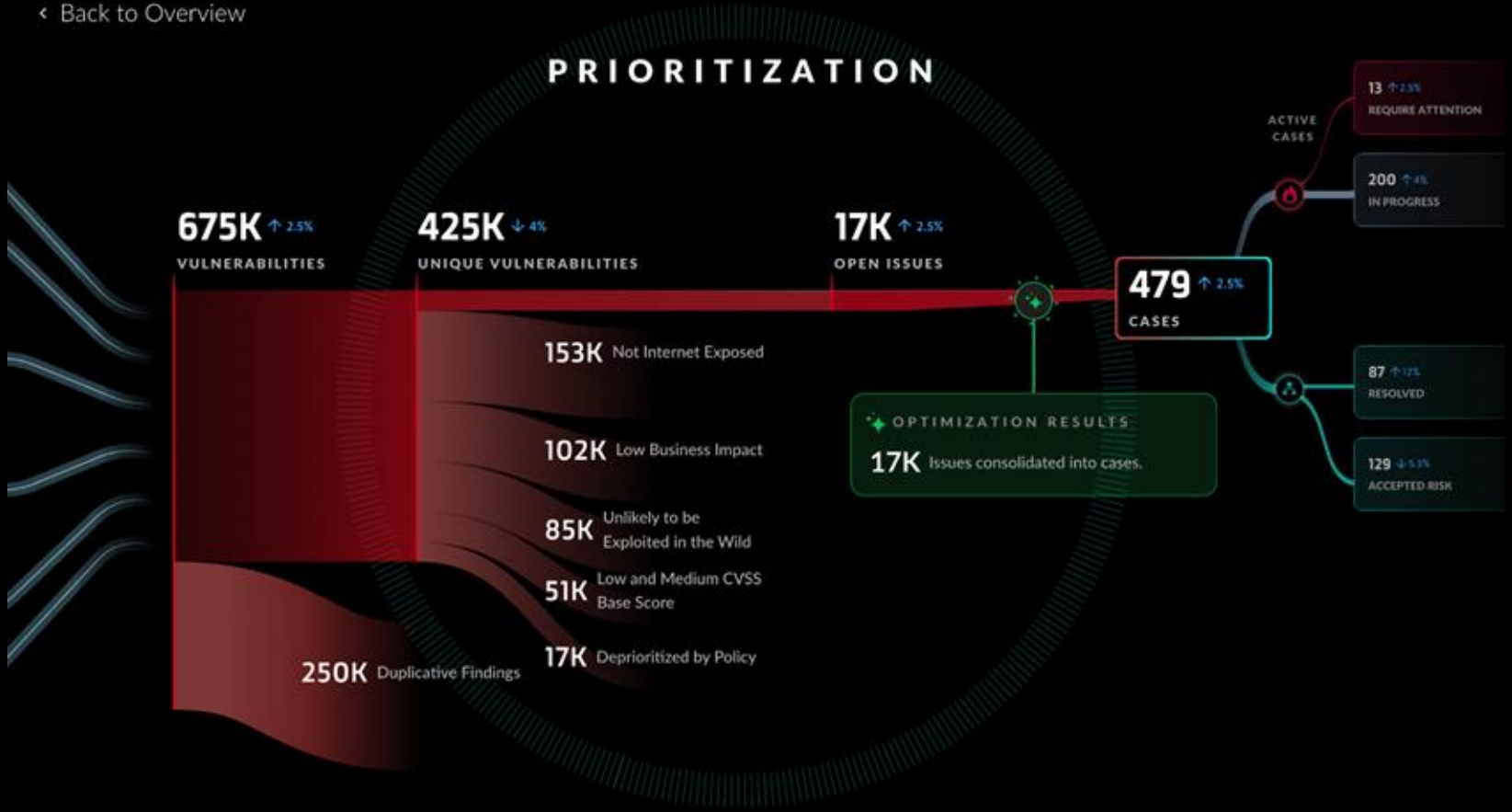
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Exposure Management Command Center

‹ Back to Overview



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Discovery

Quick Host Discovery (Recommended)

Nov 6th 2024 | By: Cortex

Fast and efficient scan to identify live hosts within the network. Best suited for a quick overview or preliminary assessment of the network inventory.

Speed: Fastest *Scope: Minimal*

Advanced Discovery

Nov 6th 2024 | By: Cortex

More comprehensive host and system detection across the network. Ideal for in-depth network environment assessment.

Speed: Fast *Scope: Limited*

Vulnerability Assessment

Balanced Scan (Recommended)

Dec 25th 2024 | By: Cortex

A balance between speed and scope; deeper assessment without resource-intensive tests. Suitable for regular scans in complex environments.

Speed: Moderate - Slow *Scope: Extended*

Express Scan

Nov 6th 2024 | By: Cortex

Fast and efficient inventory assessment with basic vulnerability checks. Ideal for routine authenticated scans or when time is of priority.

Speed: Moderate *Scope: Standard*

Comprehensive Scan

Nov 6th 2024 | By: Cortex

Complete and thorough in-depth vulnerability scan that leaves no stone unturned. Best suited for critical systems and environments that require maximum security coverage.

Speed: Slow *Scope: Complete*

Focused Vulnerability Scan

By: Cortex

Select vulnerabilities to scan for, including emerging threats and zero-day vulnerabilities

Speed: Varies *Scope: Limited*

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ITDR

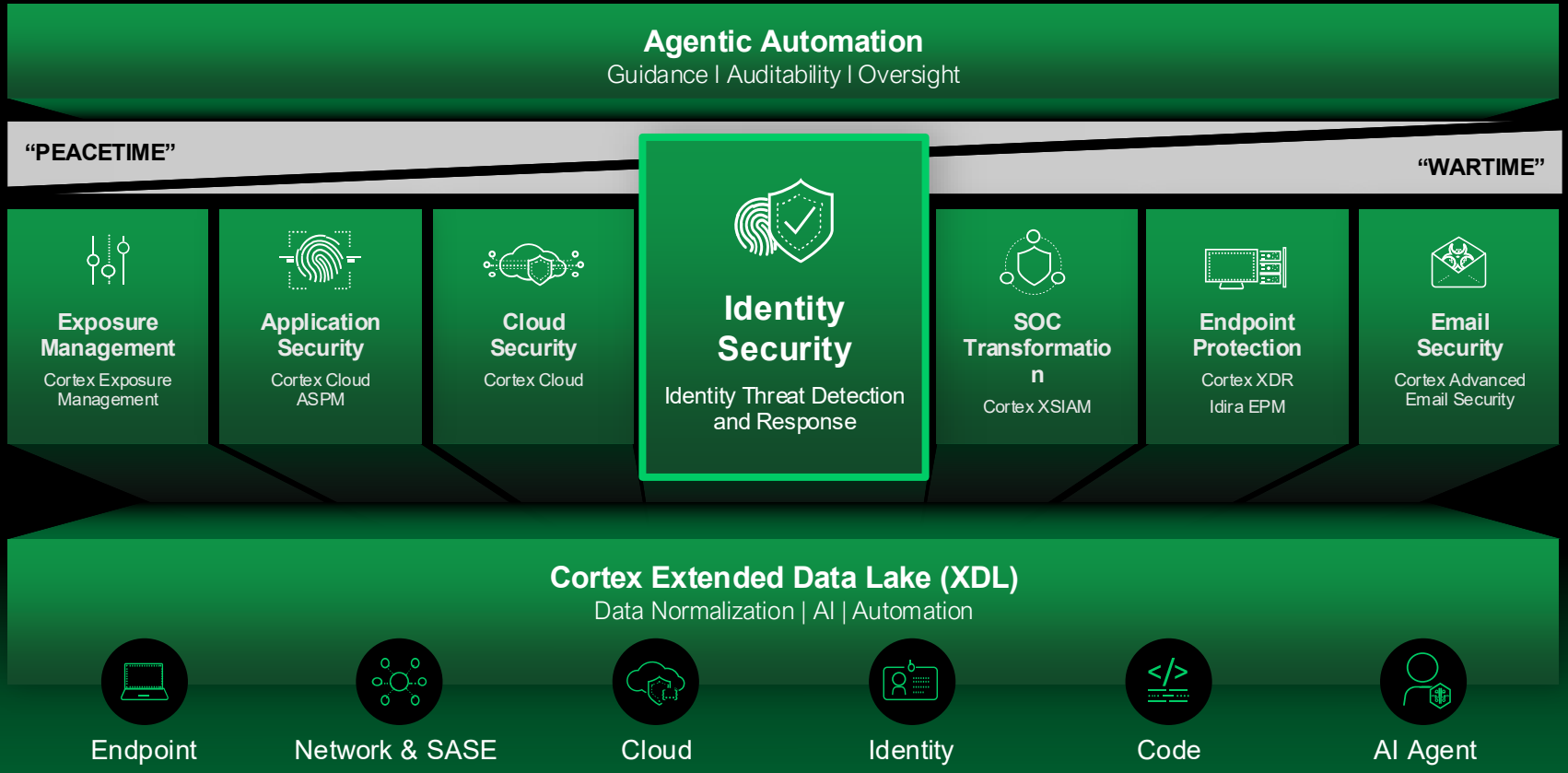
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General Overview - Overview of Components

ITDR

Identity Threat Detection and Response (ITDR) in Cortex XSIAM continuously monitors identity infrastructure, user activity, and access permissions across hybrid environments to detect, triage, and automatically respond to identity-based risks, credential theft, and privilege escalation.

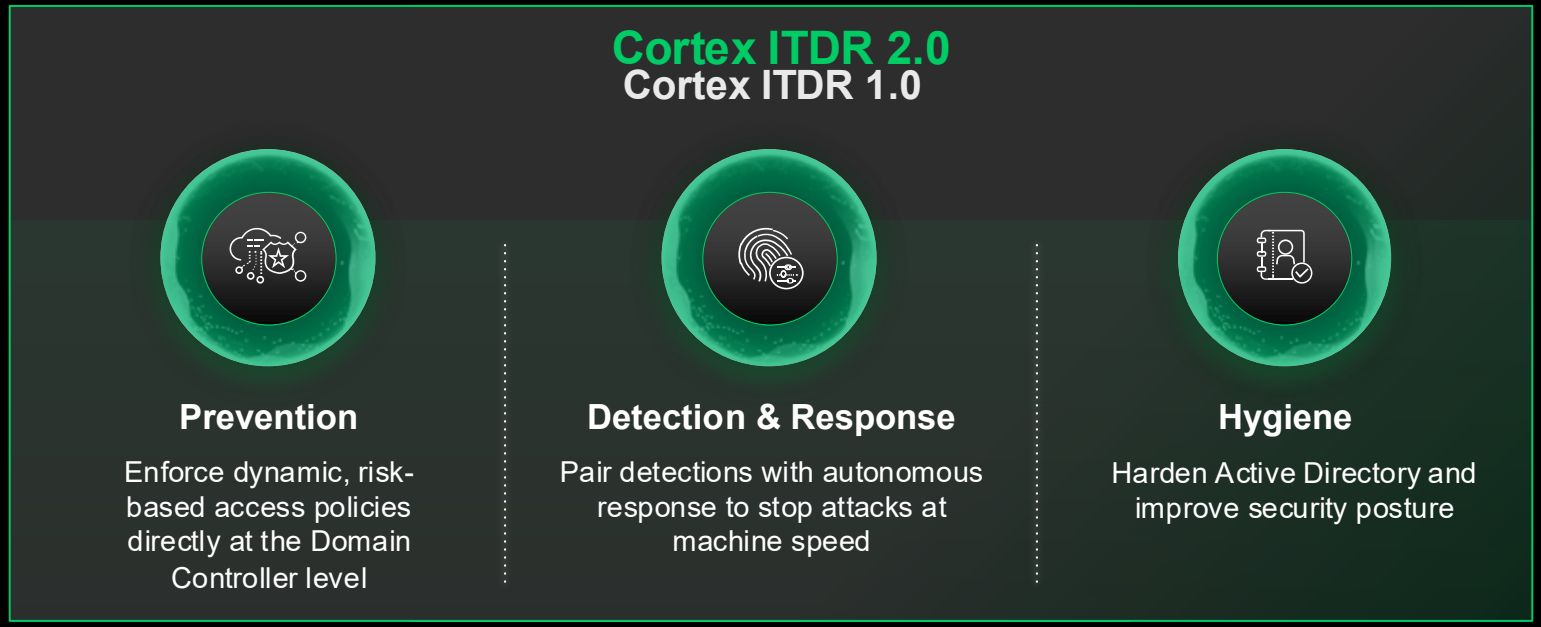
Cortex ITDR: The identity layer of a unified security platform



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Introducing Cortex ITDR 2.0

Redefining identity threat detection and response for the agentic world



Natively integrated with best-in-class architecture for IdentitySecOps

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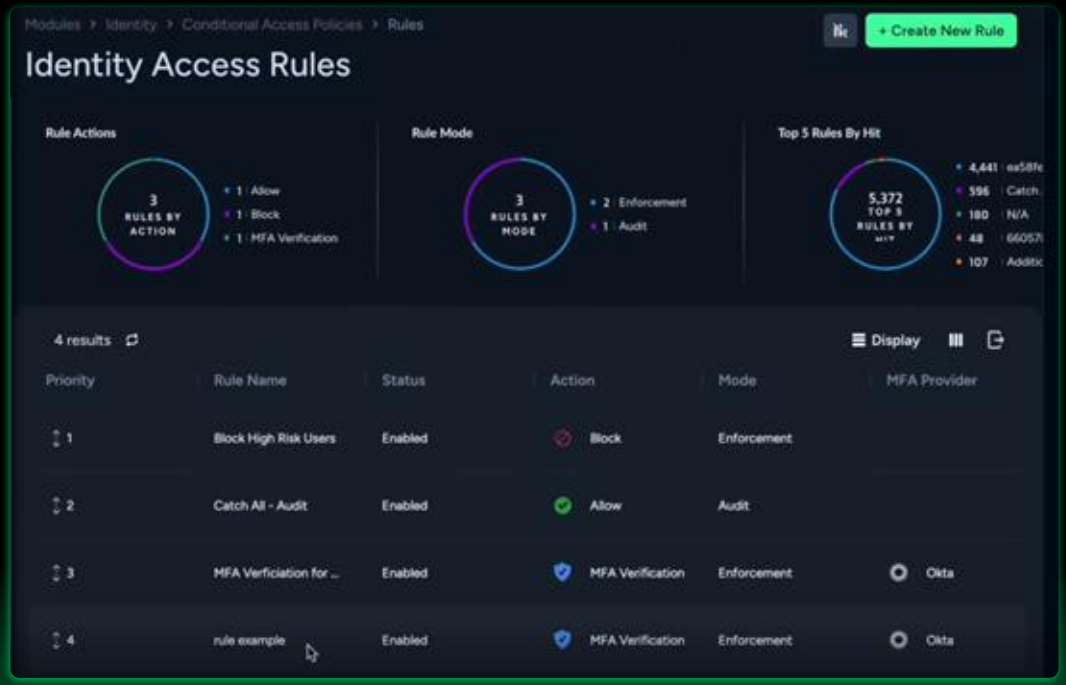
Create guardrails with adaptive access policies

Prevention

Enforce secure access
Block access or trigger MFA on risky authentication attempts

Dynamic risk-based controls
Customize policies and adjust access based on risk levels

Prioritize and manage guardrails
Dynamically prioritize and focus on what's important

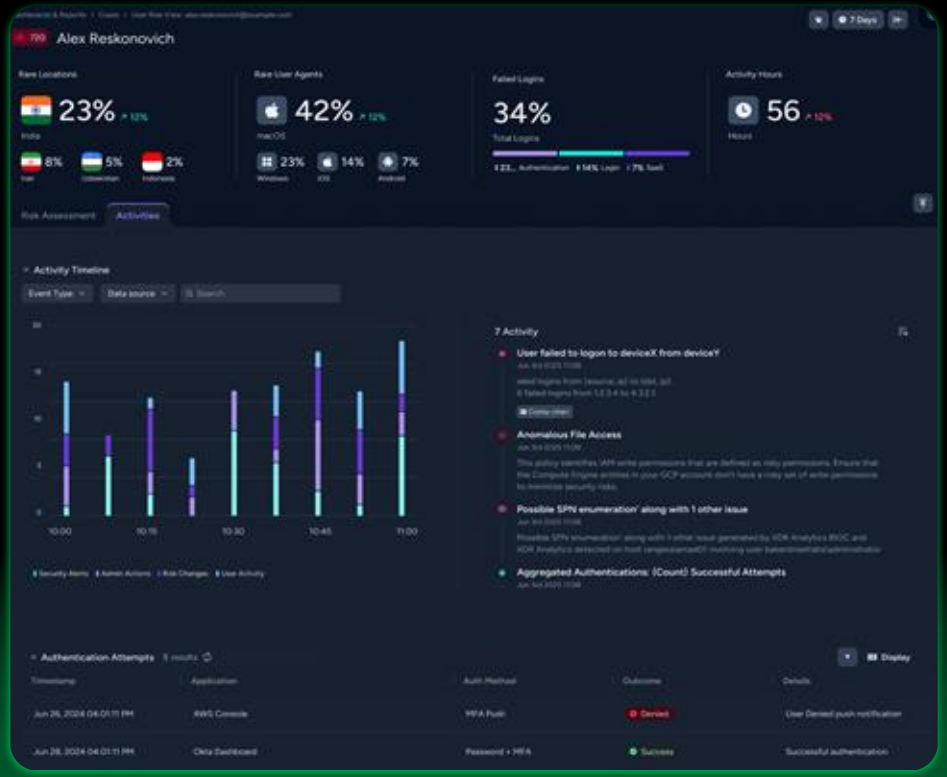


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Dynamically defend against threats with behavioral intelligence

Detection & Response



Investigate with Behavioral Intelligence
Stitches suspect logins and behavioral anomalies

Unified dashboard with Zero Blind Spots
Single view of user activity, timeline, and alerts. Eliminate blind spots during an identity crisis

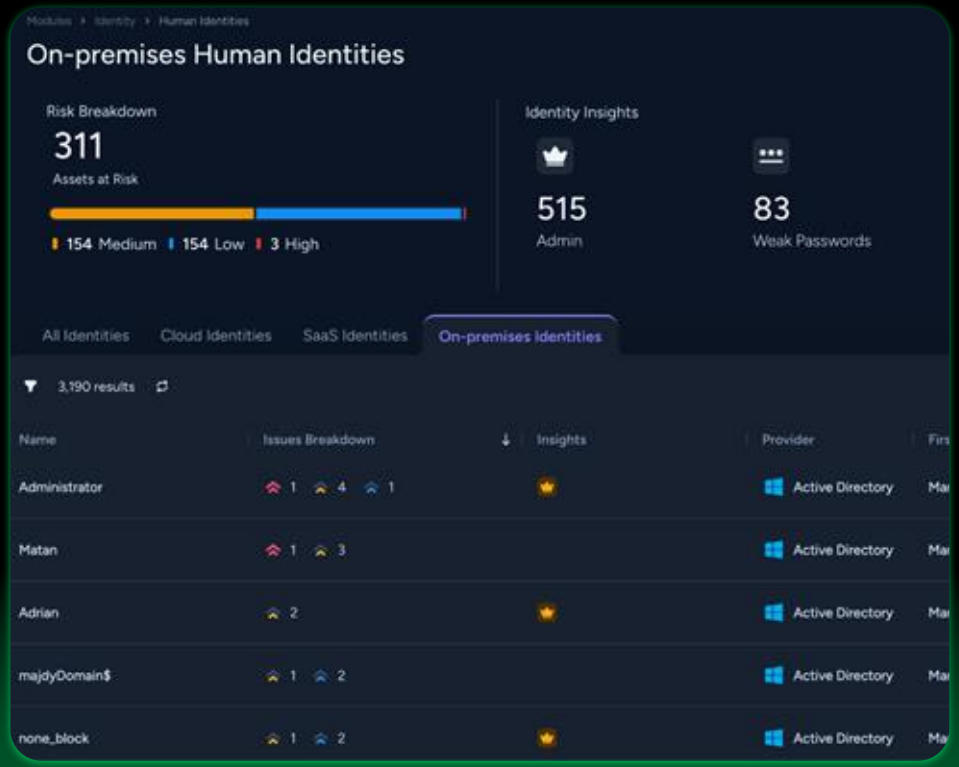
Maximize Analyst Impact
See latest log-in attempts and risk activity

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Harden Active Directory & improve identity posture

Hygiene



Prevent LDAP Reconnaissance
Block LDAP reconnaissance and data enumeration at the Domain Controller

Protect weak credentials
Discover and remediate compromised credentials before exploitation

Enhance Active Directory Posture
Shrink the attack surface by proactively improving your security hygiene

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RBAC and SBAC

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General Overview - Overview of Components

RBAC and SBAC

In Cortex XSIAM, **Role-Based Access Control (RBAC)** determines what actions a user can perform based on assigned permissions, while **Scope-Based Access Control (SBAC)** restricts which data and assets they can view or manage based on organizational boundaries.

Role-Based Access Control (RBAC)

Role-Based Access Control assigns permissions to users based on their role (job function or responsibility). Instead of granting permissions directly, you attach them to roles such as Admin, Developer, Security Analyst, and then assign those roles to users.

Why is it important?

- Simplifies user management at scale.
- Enforces least privilege by aligning access with responsibilities.
- Provides auditability for compliance (PCI, HIPAA, NIST, etc.).

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Scope-Based Access Control (SBAC)

Scope-Based Access Control applies restrictions on where a user can exercise their role's permissions. Scopes can represent cloud accounts, regions, projects, namespaces, or clusters.

Why is it important?

- Adds granularity beyond RBAC.
- Prevents admins from unintentionally or maliciously acting outside of their assigned environment.
- Supports segregation of duties across business units or customers.
- Helps maintain multi-cloud or multi-tenant security by ensuring roles only apply to designated scopes.

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How They’re Similar / Different

Similarities:

- Both are access control mechanisms used in Prisma Cloud / Cortex Cloud.
- Both support least privilege and compliance goals.
- Both can be combined for a layered security model.

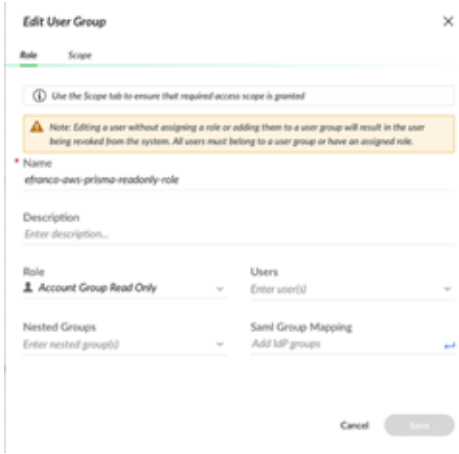
Differences:

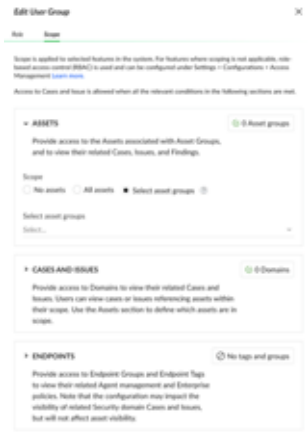
- **RBAC** = defines what actions a user can perform (create, read, update, delete).
- **SBAC** = defines where those actions can be performed (specific account, region, or resource).
- RBAC is about ***function***, SBAC is about ***scope***.
- Together, they create a matrix of access: Role (what) + Scope (where).

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RBAC
Prisma Cloud VS Cortex

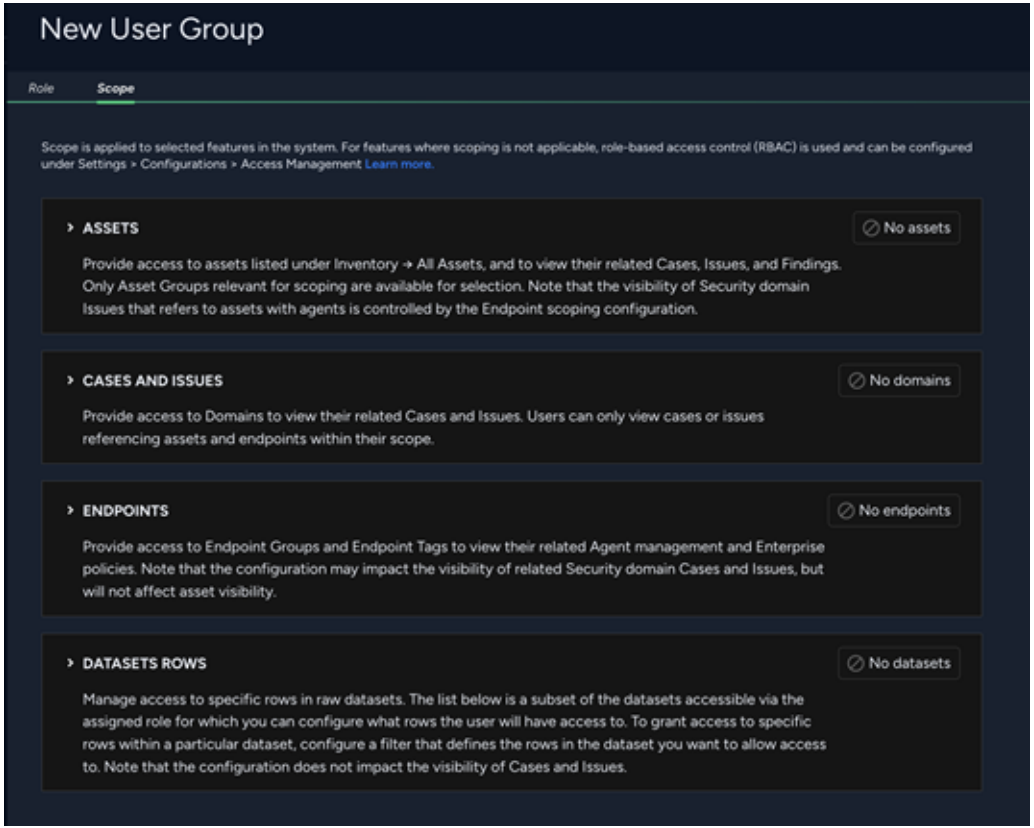
Permission Group	Role
Role	User Group





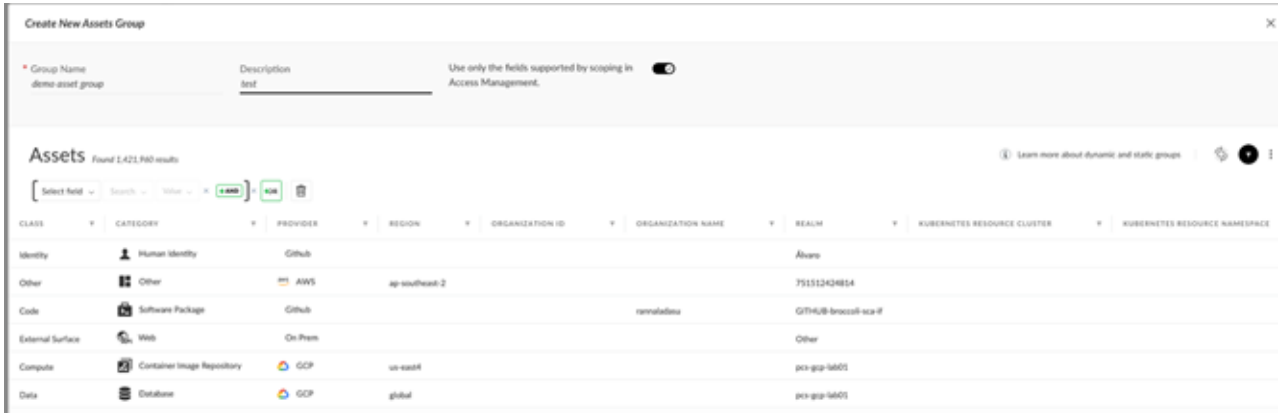
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General Overview - Overview of Components



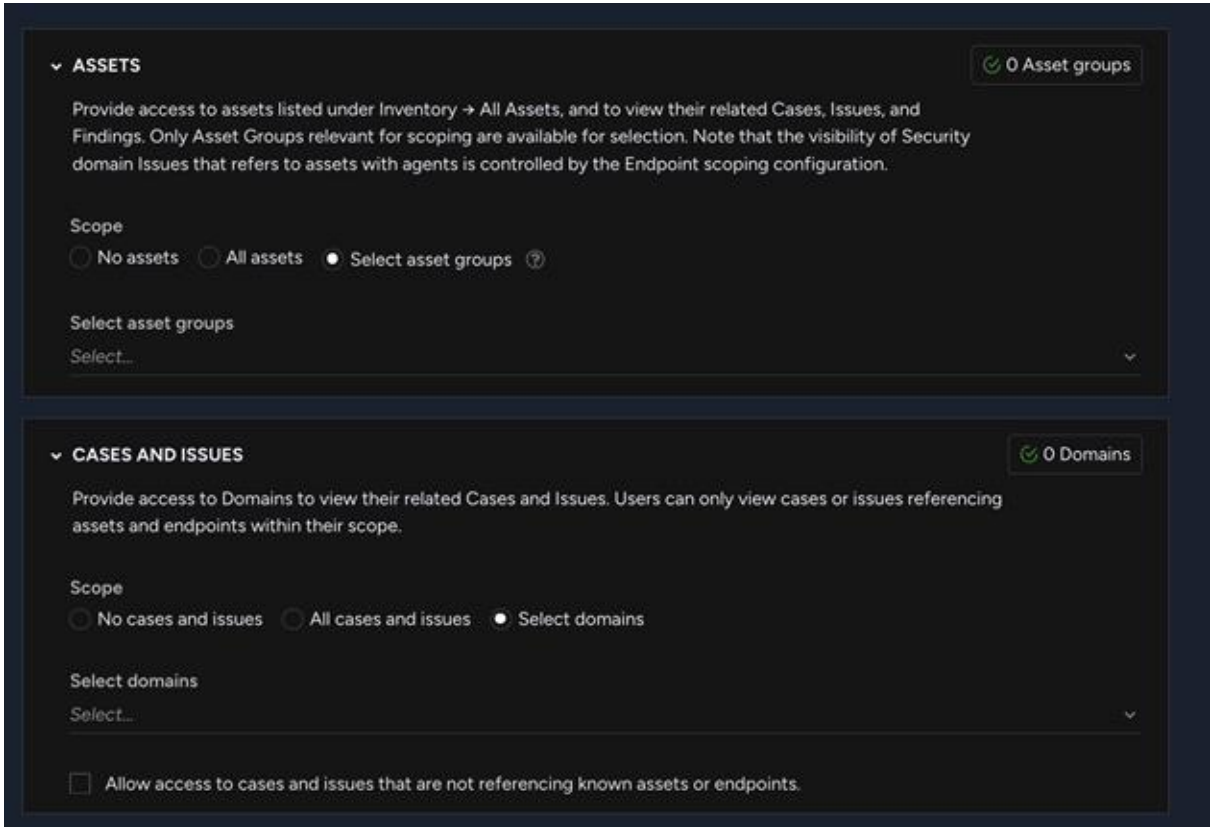
SBAC
Prisma Cloud VS Cortex Equivalents

Cloud Account	Realm
Cloud Type	Provider



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General Overview - Overview of Components



Current Automation Capabilities

RBAC

- Get Roles
- Get user groups
- Get users
- Set a user role

SBAC

- Get asset groups
- Create asset group
- Update asset group
- Delete asset group

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General Overview - Overview of Components

RBAC and SBAC

Teams	Required Permissions
xxxx	xxxx
xxxx	xxxx
xxxx	xxxx
xxxx	xxxx

General Overview

Overview of Teams

Understanding the different cross-functional teams leveraging the platform and their use-cases.

General Overview - Overview of Teams

Teams	Responsibilities
xxxx	xxxx
xxxx	xxxx
xxxx	xxxx
xxxx	xxxx

General Overview - Current Key Team Use-Cases

	Status	Key metrics	Consumption
Use case 1	• TBD	• TBD	• TBD
Use case 2	• TBD	• TBD	• TBD
Use case 3	• TBD	• TBD	• TBD

General Overview - Current Top Challenges Being Faced

Current Top Challenges
xxxx
xxxx
xxxx
xxxx

TSA FIELD ENABLEMENT · WORKSHOP MODULE

DAY 2 - Cortex Cloud Operations

Read the environment. Understand the drivers. Define the operational tasks.

The Platform

Understanding Cortex Cloud

Cloud-Native Application Protection Platform

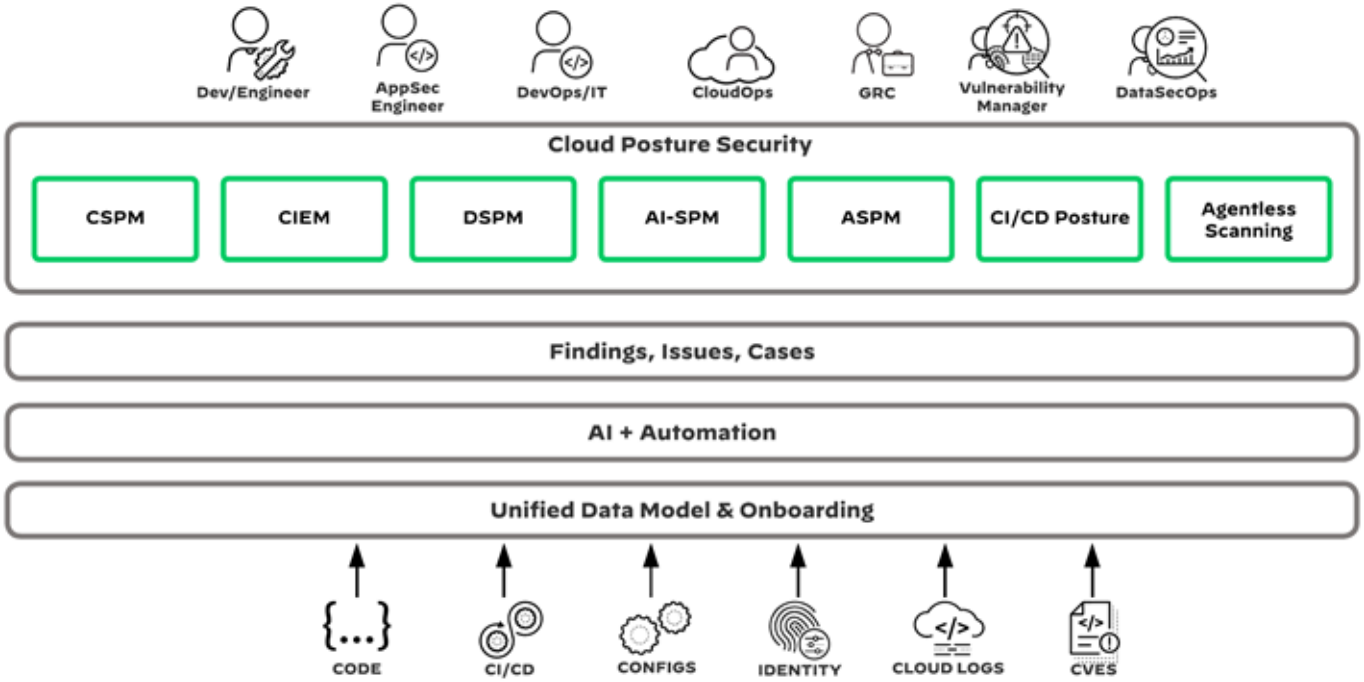
Is an all-in-one security and compliance software solution designed to protect cloud applications and infrastructure. It combines multiple tools: posture management, workload protection, identity governance and application security.

Cortex Cloud

A unified, real-time cloud security platform that combines cloud-native application protection (CNAPP) with cloud detection and response (CDR) built on top of Cortex XSIAM.

The Benefits of Cortex Cloud *— it unifies multi-cloud security from code to runtime, using AI-driven prioritization and automated remediation to stop threats in real time.*

Architecture · Cortex Cloud Components - CSPM



Architecture · Cortex Cloud Components - CSPM - Capabilities (DSPM, ADS, CIEM, ASPM, AI-SPM, KSPM)

1

DSPM

Data Security

2

CIEM

Governance & Privilege Management

3

AI-SPM

AI Security & Posture

4

ASPM

Application Security & Posture

5

ADS

Agentless Scanning

5

KSPM

Kubernetes Security & Posture

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DSPM - Data Security & Posture Management

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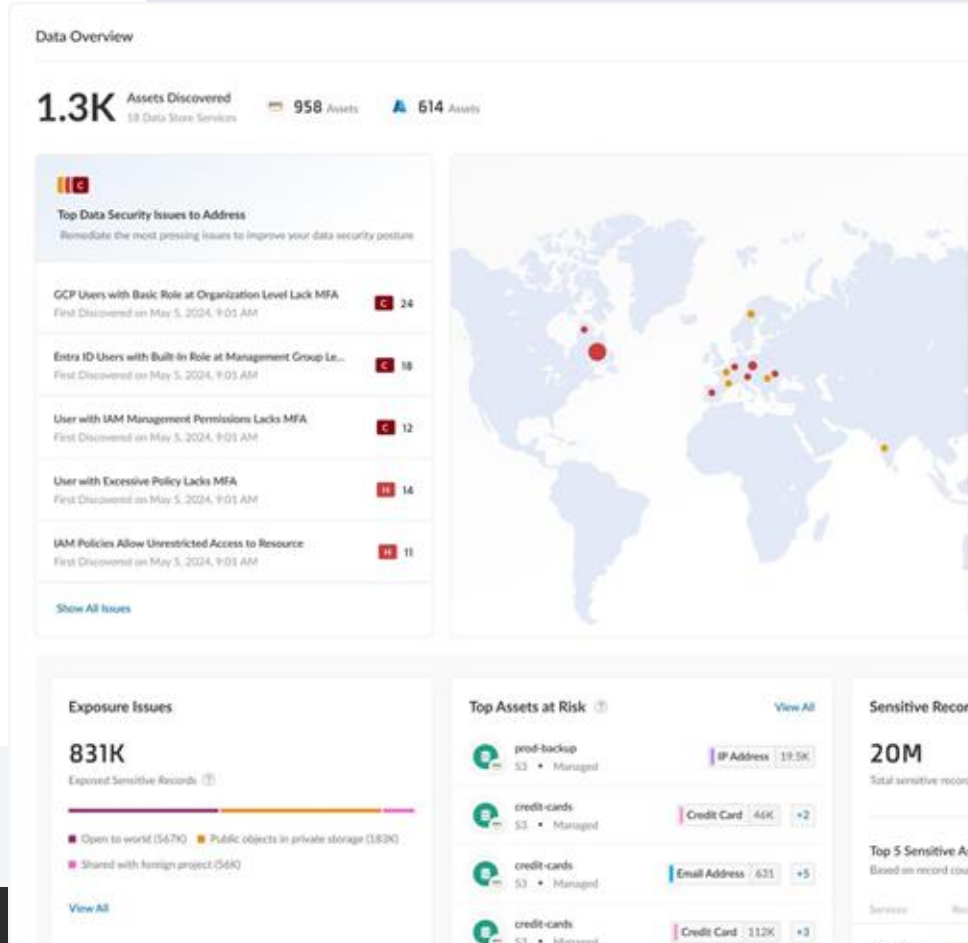
Best-in-class DSPM

Discover, classify, protect, and govern sensitive data across your clouds. Identify risks and stop sensitive data from exfiltration.

- **Conduct automated discovery and classification**
  Leverage hundreds of built-in data classification policies (e.g., PII, PCI, PHI) covering all data assets and mapped to compliance and regulatory frameworks to discover and classify data
- **Carry out risk identification and compliance analysis**
  Identify risks around encryption, logging, retention, and data access governance for data stores containing sensitive information
- **Execute real-time data detection and response (DDR)**
  Monitor, flag, and respond to any action that can lead to data exfiltration or compliance violations

With 100s classifiers
e.g., PCI, PII, PHI, FTC, GDPR, and CCPA

DSPM leadership: GIGAOM | FROST & SULLIVAN



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CIEM - Cloud Infrastructure & Entitlement Management

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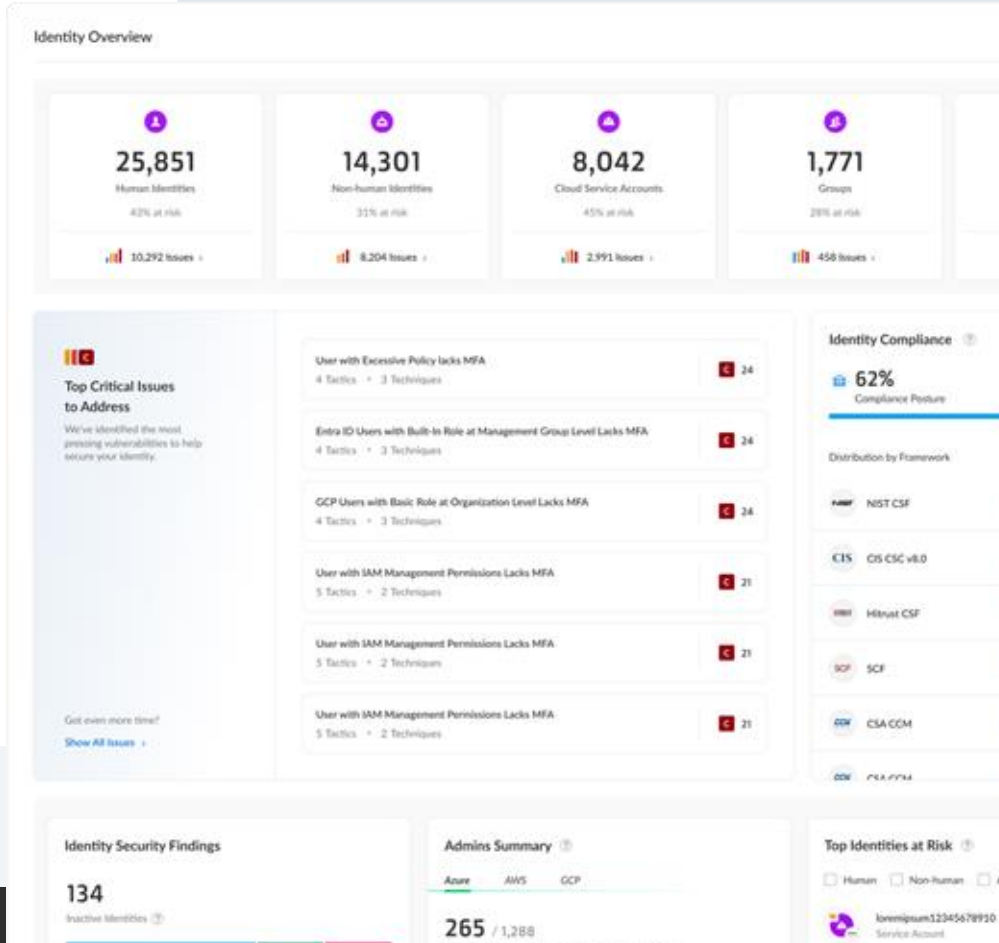
Best-in-class CIEM

Cover human and non-human identities. Gain visibility into who can take actions and with which resources. Follow recommendations to enforce least-privilege access and prevent critical risks leading to attacks.

- **Map net-effective permissions of all identities**
  Understand multicloud and IAM permissions as well as how they contribute to attack paths
- **Rightsize permissions**
  Automate recommendations and actions to enforce least-privilege access
- **Leverage contextual insights to identify attack paths**
  to prioritize critical risks effortlessly

45,000+ high risk actions identification
across AWS, Azure, and GCP

CIEM leadership: kuppingercole ANALYSTS | FROST & SULLIVAN



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AI-SPM - Artificial Intelligence Security Posture Management

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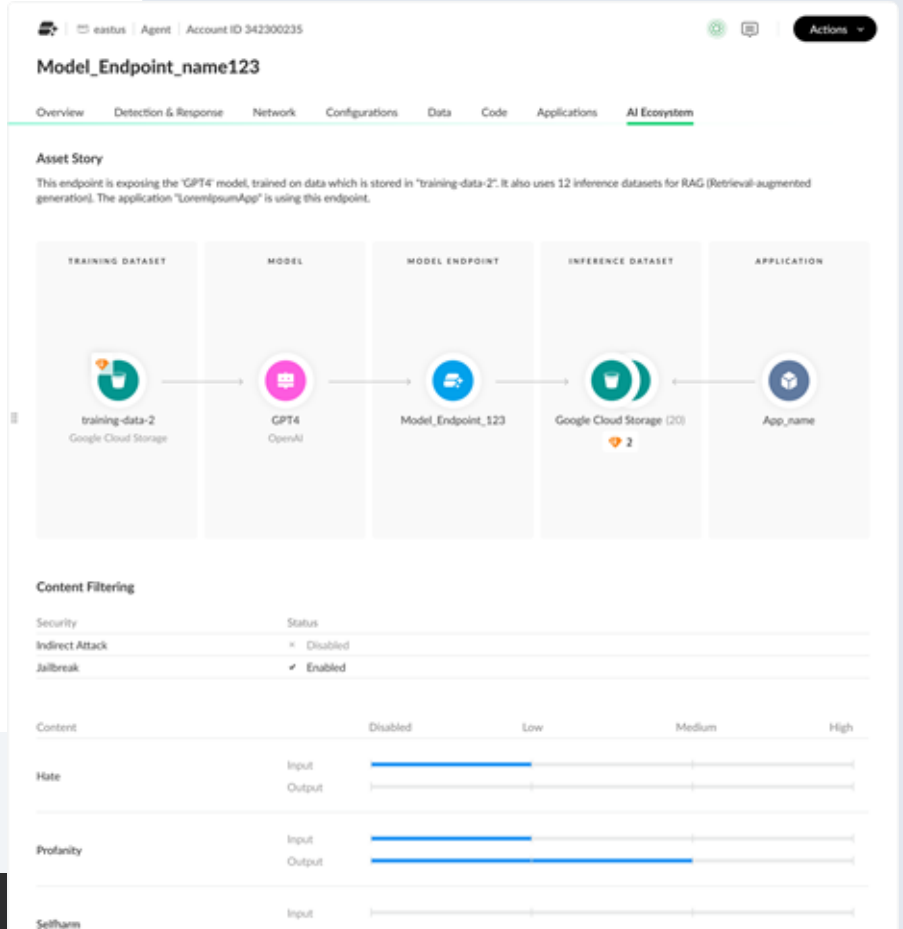
Best-in-class AI-SPM

Manipulating model data can introduce vulnerabilities and biases, expose data, and lead to data privacy violations, compliance, and security risks.

- **Map the AI pipeline**
  Uncover key components of AI systems, including models, agents, endpoints, and the flow of data between them
- **Monitor sensitive data**
  Monitor and govern data exposure, poisoning risk, privacy violations, and security breaches
- **Prioritize AI risks** in your AI pipeline infrastructure and supply chain that could lead to attacks

Measure alignment with industry standards
e.g., NIST 600-1, OWASP top 10 for LLMs

Supported standards: OWASP | NIST



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ASPM - Application Security Posture Management

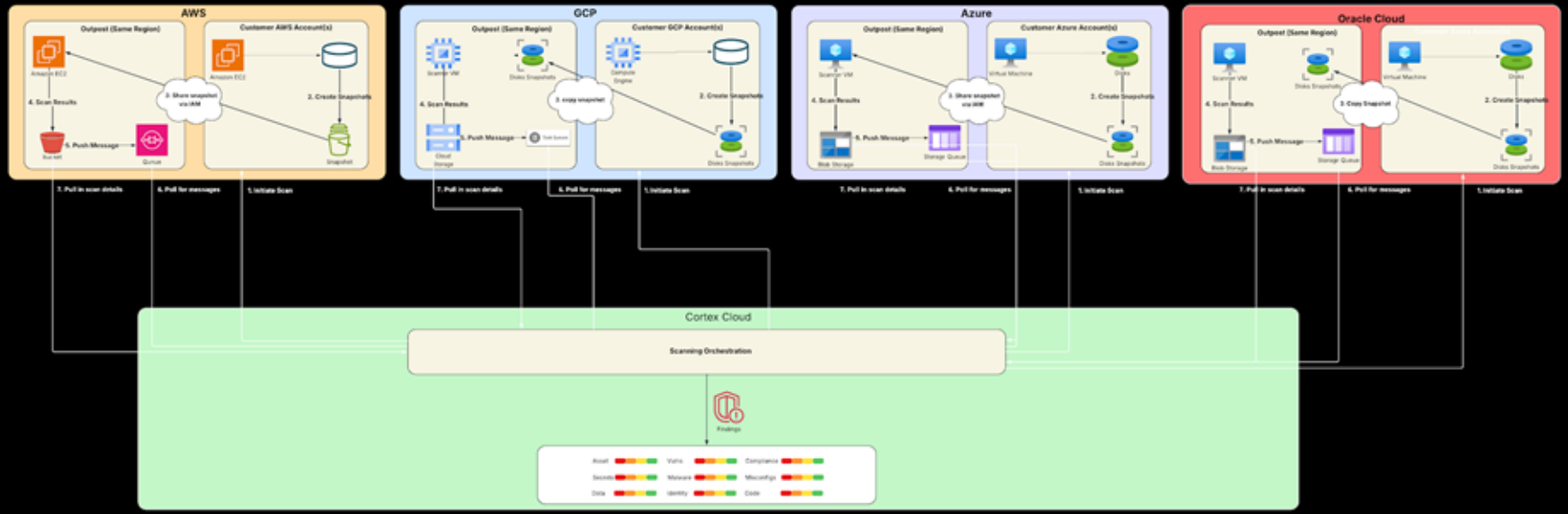
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ADS - Agentless Disk Scanning

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High level overview



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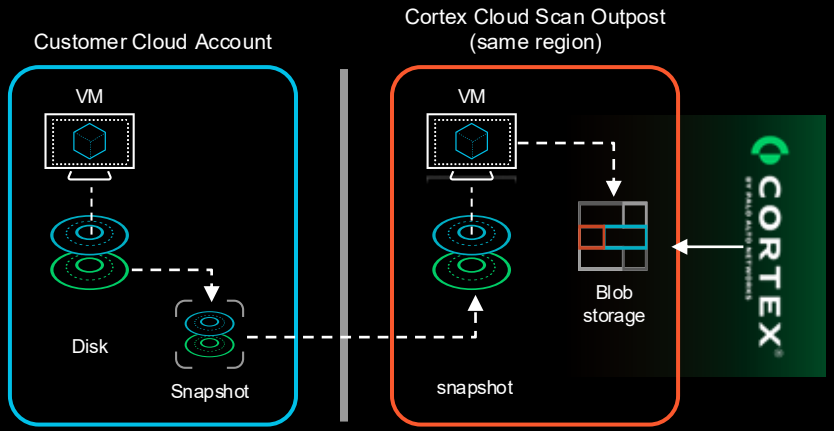
Architecture · Cortex Cloud Components - CSPM - Scanners (Cloud Scanner vs Outpost)

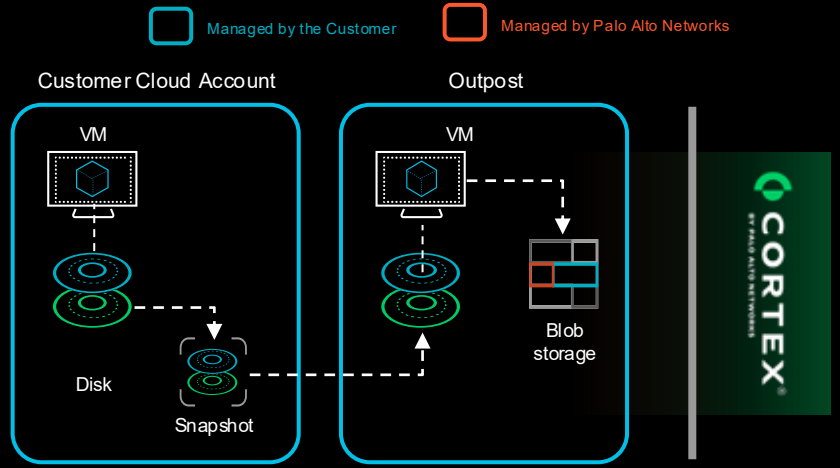
Cloud Scan

- Managed by Cortex/Palo Alto Networks
- No setup and maintenance
- Regular updates and monitoring by Palo Alto Networks

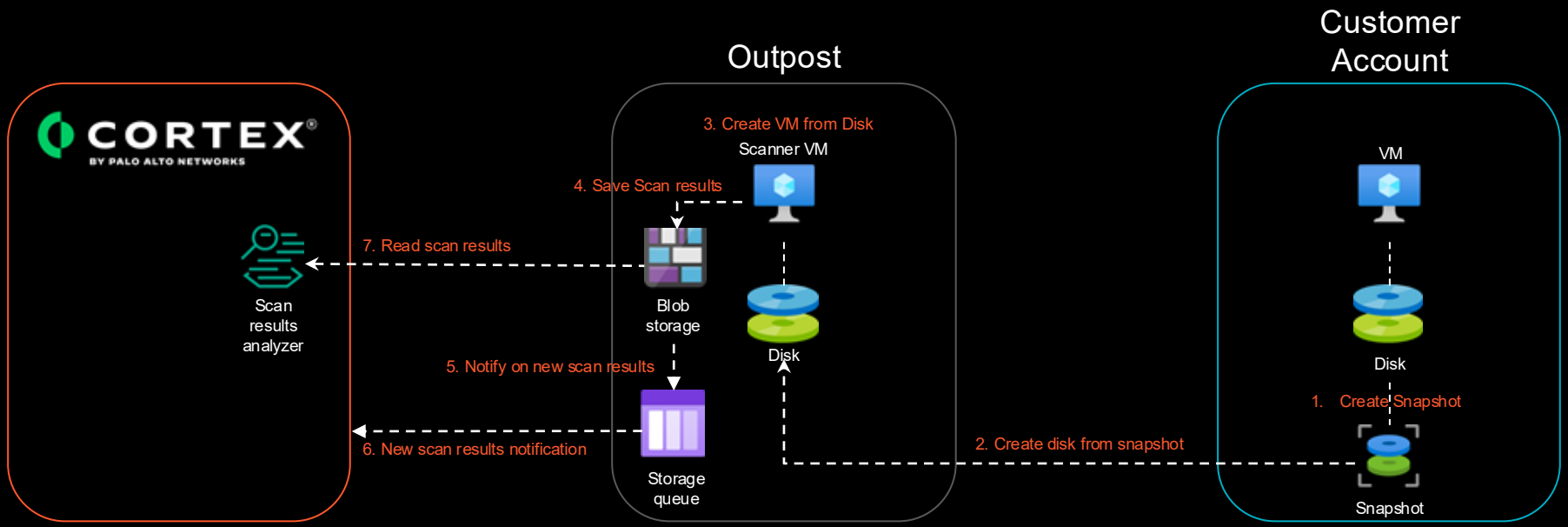
Outpost Scan

- Managed by the customer
- Setup and maintenance requires customer involvement
- The scanning is performed on infrastructure deployed to a CSP account owned by the customer





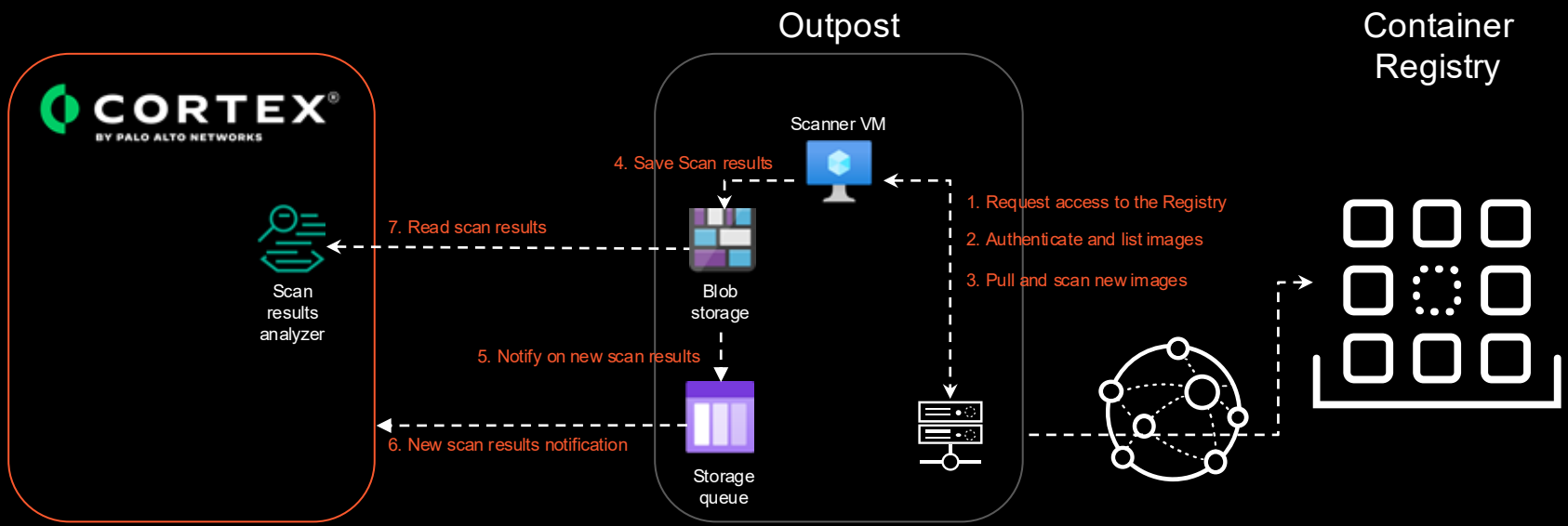
Cortex Cloud Agentless Disk Scanning



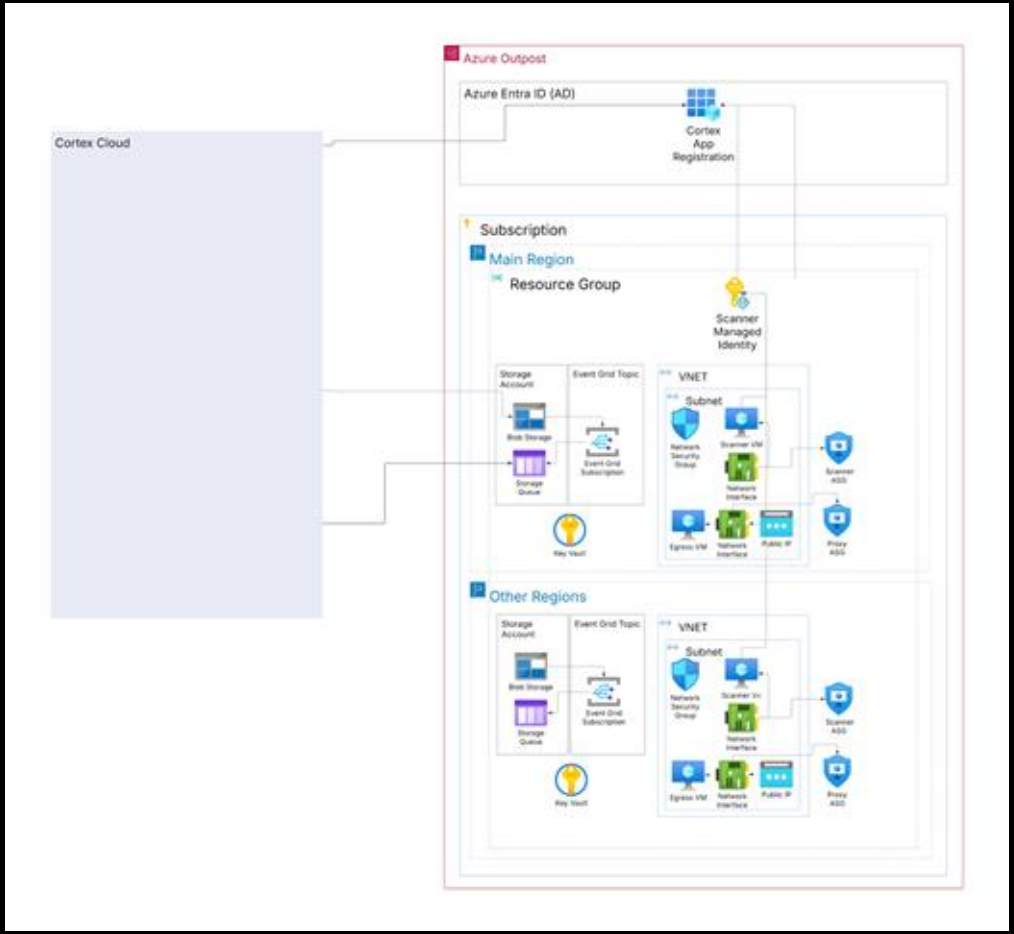
- Snapshots will be deleted when no longer needed

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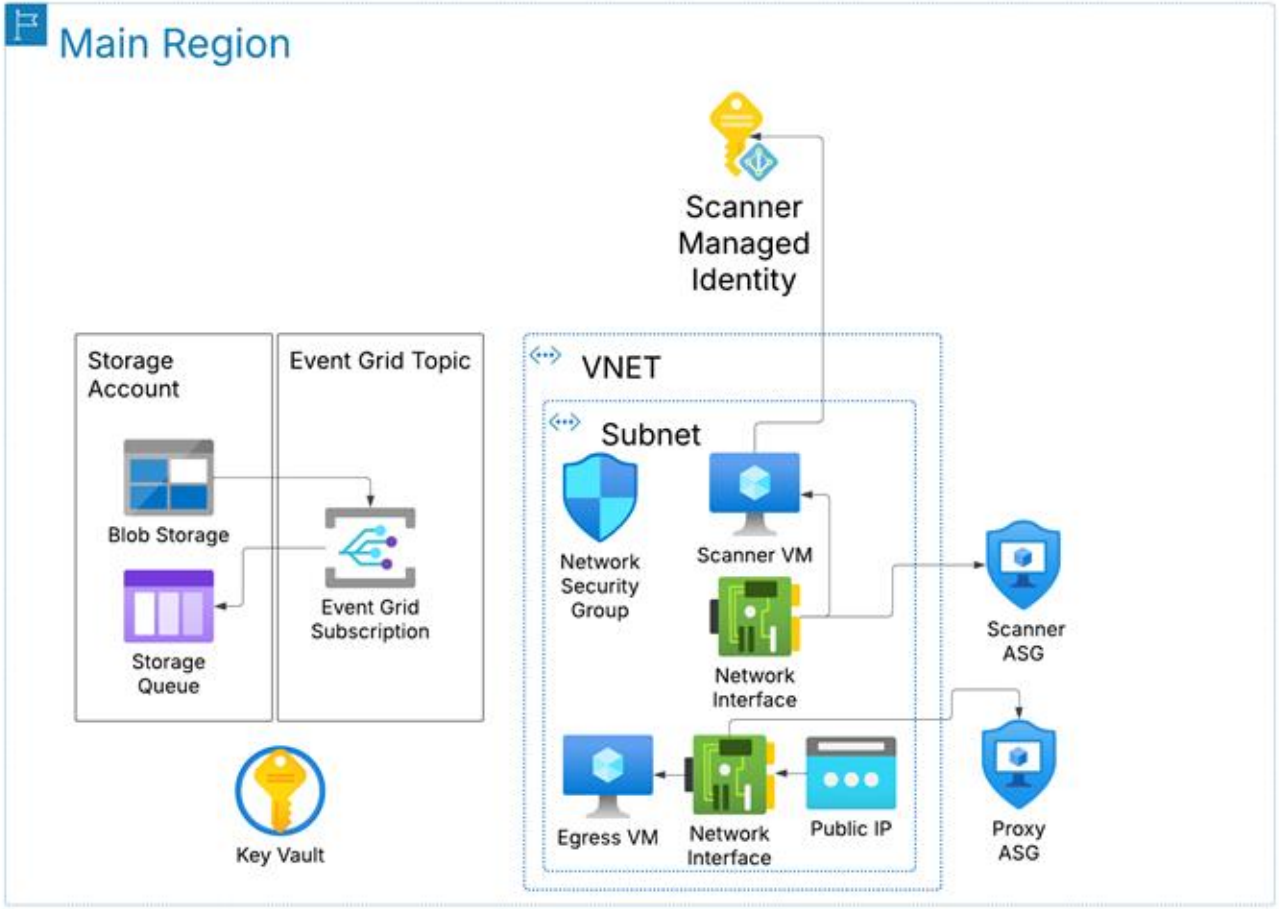
Cortex Cloud Agentless Registry Scanning



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KSPM - Kubernetes Security Posture Management

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Cortex Kubernetes Connector

The **Cortex Kubernetes Connector** is primarily focused on securing Kubernetes environments by providing in-depth posture management and cluster-level visibility.

Key Features and Capabilities:

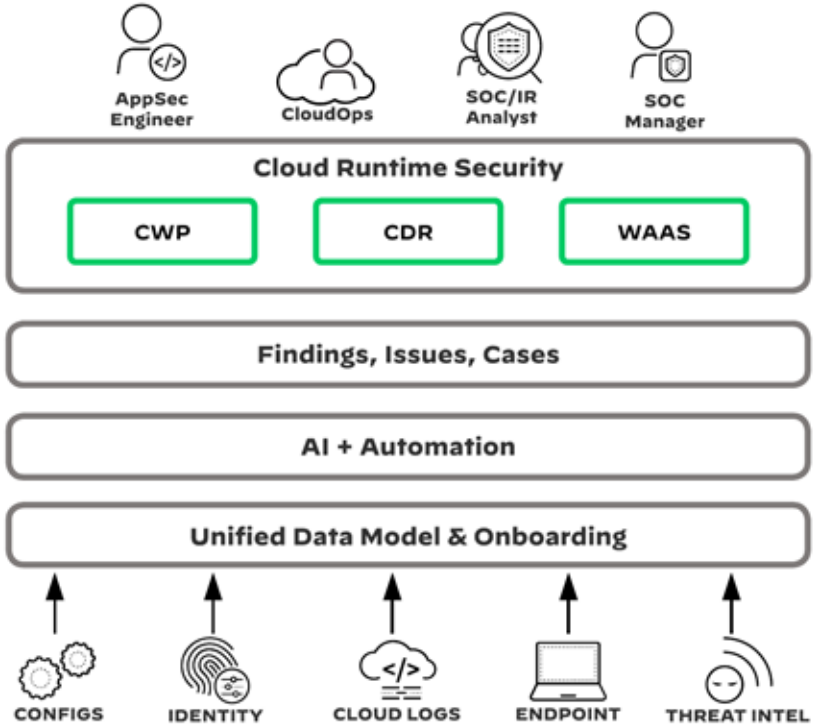
- **Kubernetes Security Posture Management (KSPM)**: The connector provides a comprehensive KSPM offering, bringing visibility into secrets, malware, compliance, misconfigurations, and vulnerabilities within the Kubernetes cluster to the KSPM Dashboard. It can detect secrets within the cluster once deployed.
- **Cluster Resource Discovery**: It automatically discovers in-cluster resources such as namespaces, workloads, pods, and deployments.
- **Policy Management and Enforcement**: It enables centralized cloud workload policy management and enforcement at runtime through its Admission Controller. This allows for scanning compliance checks.
- **Deployment**: The Kubernetes Connector uses a Helm installer and is installed per cluster, supporting AKS, EKS, and GKE clusters.

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Feature/Aspect	Cortex XDR Agent for Cloud	Cortex Kubernetes Connector
Primary Focus	**Real-time runtime protection, detection, and response** for individual workloads (VMs, containers, serverless).	**Kubernetes Security Posture Management (KSPM)**, cluster resource discovery, and policy enforcement.
Visibility Scope	Deep visibility into operating system-level activities, system calls, network I/O, file system tampering on individual nodes/workloads.	Visibility into cluster resources (namespaces, workloads, pods), secrets, malware, compliance, misconfigurations, and vulnerabilities at the cluster level.
Capabilities	AI-driven prevention and detection, NGAV, EDR, host firewall, device control, disk encryption. Can detect and stop specific attacks like reverse shells.	Admission Controller for policy enforcement, centralized workload policy management. Supports compliance checks.
Deployment	Deployable via daemonsets/helm	Helm installer, typically installed per cluster.
Data Type	Real-time telemetry and event data from workloads.	Configuration, posture, and compliance data from Kubernetes resources.
Coexistence	Running multiple EDR agents on the same workload is generally not recommended due to potential conflicts and resource issues.	Works alongside the XDR agent for comprehensive security in Kubernetes.

Architecture · Cortex Cloud Components - CSPM - Automation & Response Techniques

Architecture · Cortex Cloud Components - CWP



Architecture · Cortex Cloud Components - CWP - Capabilities (CWP, CDR & WAAS)

1

CWP

Securing Compute Layer

2

CDR

Real-Time Defense

3

WAAS

Web & API Protection

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CWP - Cloud Workload Protection

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Cortex Agent (Cortex XDR Agent for Cloud)

The **Cortex XDR Agent for Cloud** is a unified agent designed for both endpoint and cloud security, recognized for its best-in-class protection capabilities.

Key Features and Capabilities:

- **Real-time Protection and Detection**: The agent provides real-time prevention and detection of known and unknown threats, including zero-days, malware, malicious processes, vulnerability exploits, and API abuse. It uses an AI-driven approach, combining multiple data sources to detect and prevent runtime attacks with over 7,000 detectors, including 2,400+ ML models.
- **Comprehensive Visibility**: It offers deep visibility into operating system-level application changes such such as system calls, network I/O, and file system tampering. It collects extensive telemetry data for SOC teams to perform detection and response activities and stitch together data for anomalous activities and threats.
- **Workload Coverage**: The agent supports various cloud workloads, including virtual machines (VMs), containers, and serverless functions.
- **Unified Agent**: It is the same industry-leading Cortex XDR agent used for traditional endpoints and cloud workloads, ensuring consistent security. It delivers Next-Generation Antivirus (NGAV), Endpoint Detection and Response (EDR), host firewall, device control, and disk encryption.

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Cortex Agent (Cortex XDR Agent for Cloud)

Key Features and Capabilities (continued):

- **Deployment**: For container orchestrator environments, it is deployable via Helm charts. For serverless functions, it can be deployed as a layer or embedded.
- **Management**: The agent is managed from the Cortex console, where users can view, investigate, and respond to runtime security alerts.

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Cortex Cloud Secures all the Compute Options Enterprises Use



Virtual Machines



Containers



Containers-as-a-Service



On-Demand Containers



Serverless

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XDR agent requirements

REQUIREMENT	MINIMUM SPECIFICATION
Processor	Processor 2.3 GHz dual-core processor
RAM	4GB; 8GB recommended
Hard disk space	10 GB
Architecture	x86_64 (x86 64bit) For aarch64 (ARM 64 bit) see Cortex XDR Agent for Linux Requirements for details.
Operating system versions	See the Cortex XDR Compatibility Matrix.

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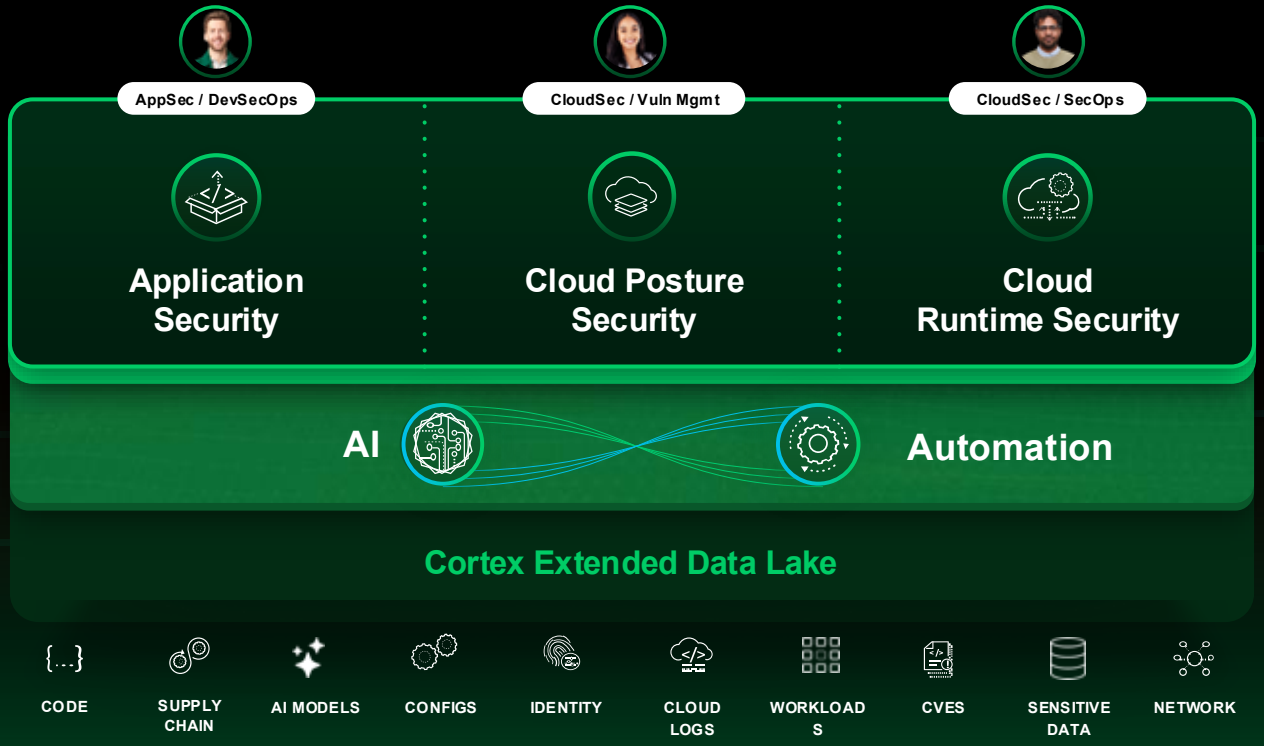
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CDR - Cloud Detection & Response

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Introducing Cortex Cloud

The industry's leading CNAPP merged with best-in-class CDR for real-time cloud security



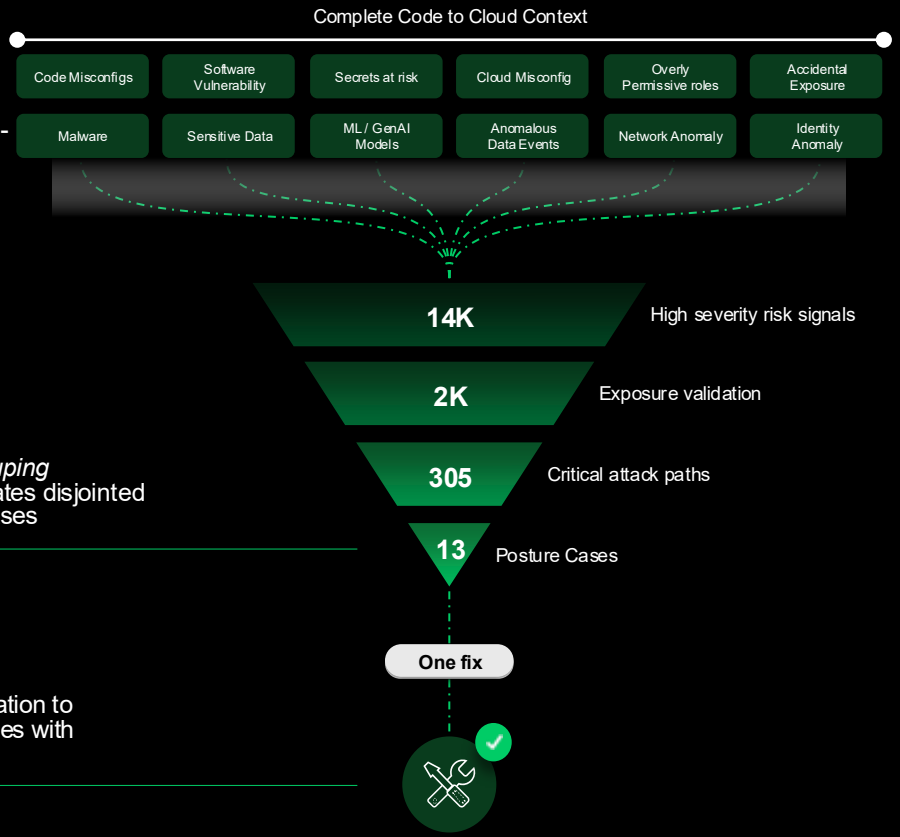
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Prioritize & Remediate Posture Issues at Scale

Centralize all risk findings across multi-cloud

AI-driven *SmartGrouping* automatically correlates disjointed alerts into holistic cases

Best-in-class automation to resolve multiple issues with a single fix



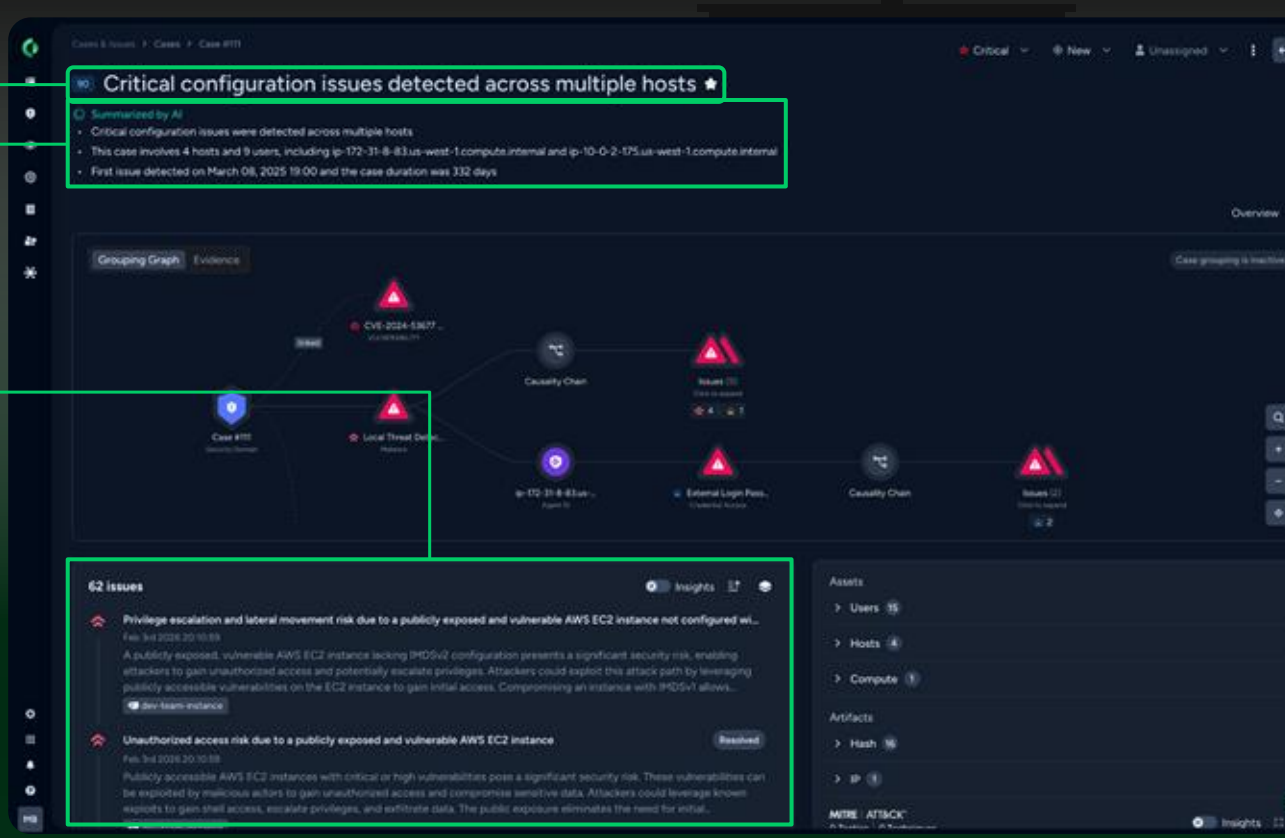
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Prioritize and automate response with AI-driven CDR

SmartScoring helps prioritize **one critical case**

SmartGrouping elevates **61+ alerts** from **8 sources**

AI Agents automatically summarized case details for **actionable** response



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WAAS - Web Application & API Security

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Architecture · Cortex Cloud Components - CWP - WAAS

Cortex Web and API Security (WAAS) capabilities offer comprehensive protection of APIs across integrated API gateways and web-based applications and APIs running on Linux-based workloads.

Use Cases

- **API Visibility**: APIs are discovered through traffic mirroring and API Gateway logs. Cortex scans and collects metadata including domains, paths, HTTP methods, authentication types, protocol schemas (HTTP/HTTPS), and request/response content types. The source of discovery comes from analyzing these key elements.
- **Posture Management and Risk Insights**: WAAS assesses discovered APIs for internet exposure, sensitive data transmissions, weak authentication, lack of encryption, and specification drift. This evaluation provides critical posture management and risk insights to ensure the security of the APIs.
- **Threat Detection**: WAAS identifies API-specific threats, including SQL injection (SQLi), Cross-site scripting (XSS), CVE exploit attempts, authentication bypass, sensitive data leakage, bot and scanner activity, and traffic anomalies. This detection capability enhances threat identification and response measures for safeguarding APIs. Refer to Monitor and investigate API threats for more information on the threats.

API Security

Discover, classify, and secure APIs on cloud-native applications. Govern and manage API security risk.

- **API visibility**
  Automatically discover cloud-native APIs from development to production for a complete, up-to-date inventory of your API landscape.
- **API posture and compliance**
  Assess all APIs to find and fix critical risks like sensitive data exposure, misconfigurations, compliance controls and shadow APIs.
- **API protection**
  Stop real-time attacks in the OWASP API Security Top 10. Accelerate investigations and response with deep, contextual data to dramatically reduce your Mean Time to Respond (MTTR)



API Integrations

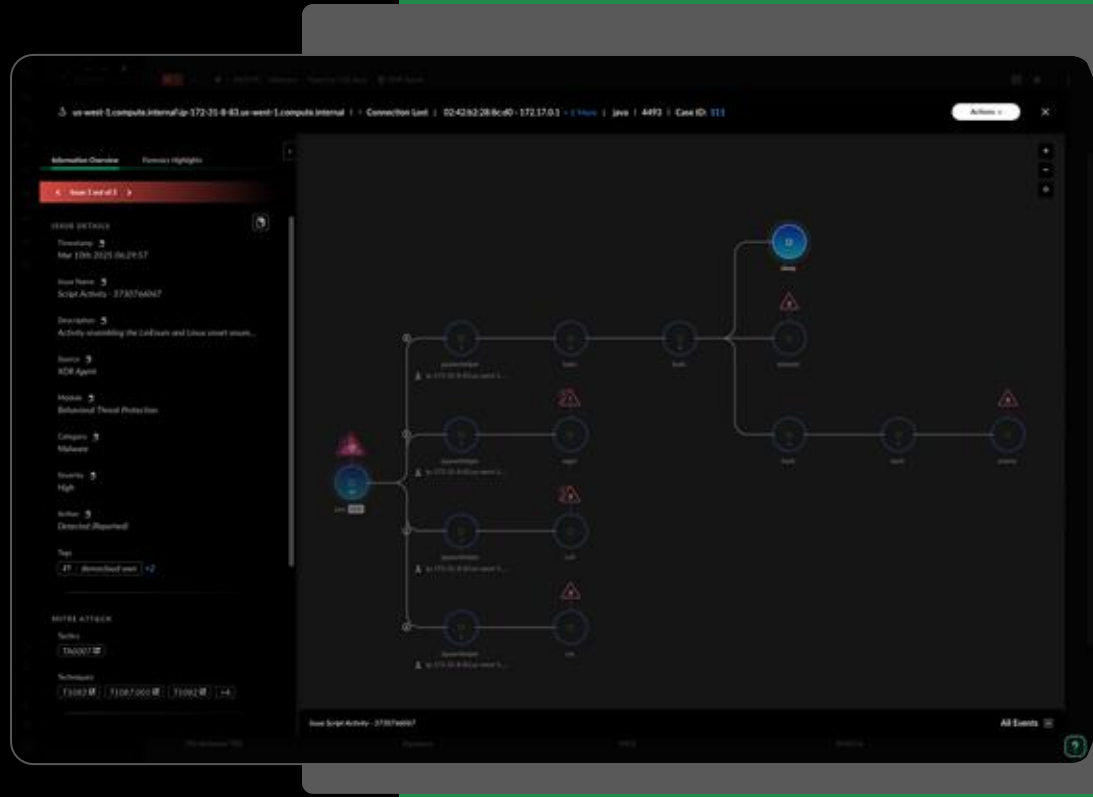
AWS API Gateway, Azure API Management, Kong, F5, Google Apigee

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Runtime Protection

Stop cloud attacks before they become breaches

- **Exploit Prevention**
  Prevent unpatched vulnerabilities from becoming breaches with real-time by detecting suspicious behavior in memory, processes, or system calls
- **Behavioral Threat Protection**
  Detect advanced attacks using machine learning and behavioral analytics to stop attacks such as malicious process chains, credential theft, lateral movement, or privilege escalation
- **Malware Protection**
  Block known and unknown malware with ML-powered local analysis and dynamic sandboxing from WildFire - the largest malware engine



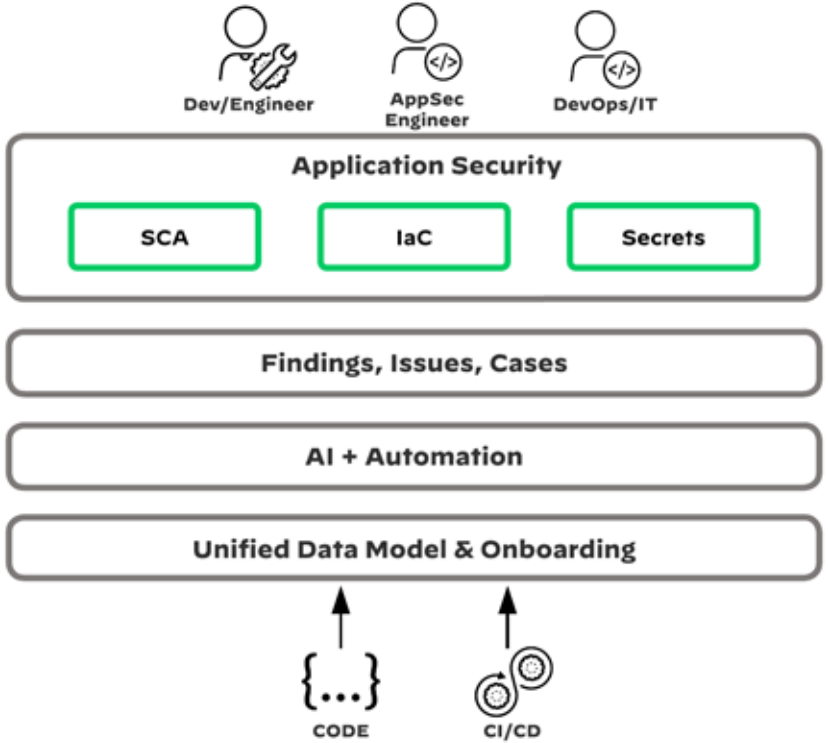
Easy to deploy in minutes

Vast support for all cloud workloads including Hosts, Containers, Kubernetes and Serverless fucntions

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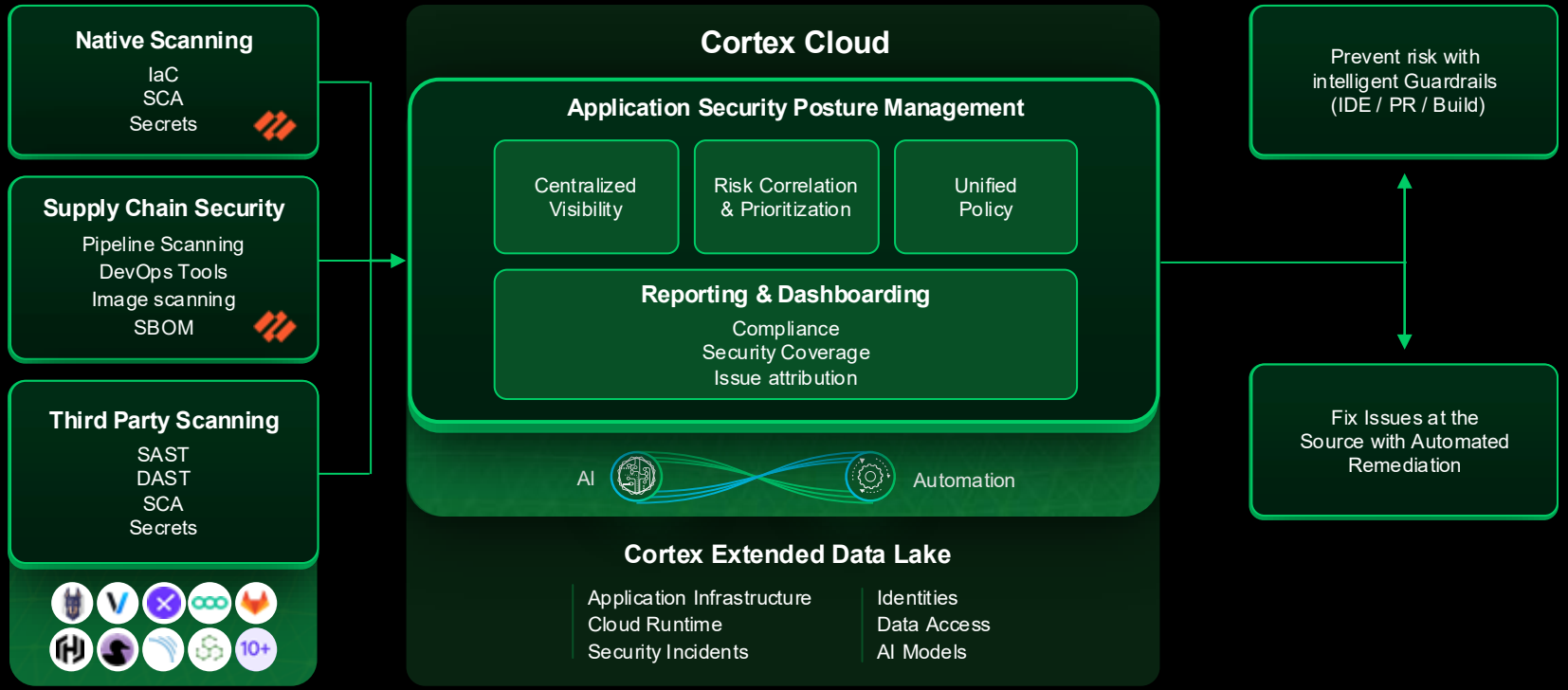
Architecture · Cortex Cloud Components - CWP - Automation & Response Techniques

Architecture · Cortex Cloud Components - AppSec



Introducing Cortex Cloud Application Security

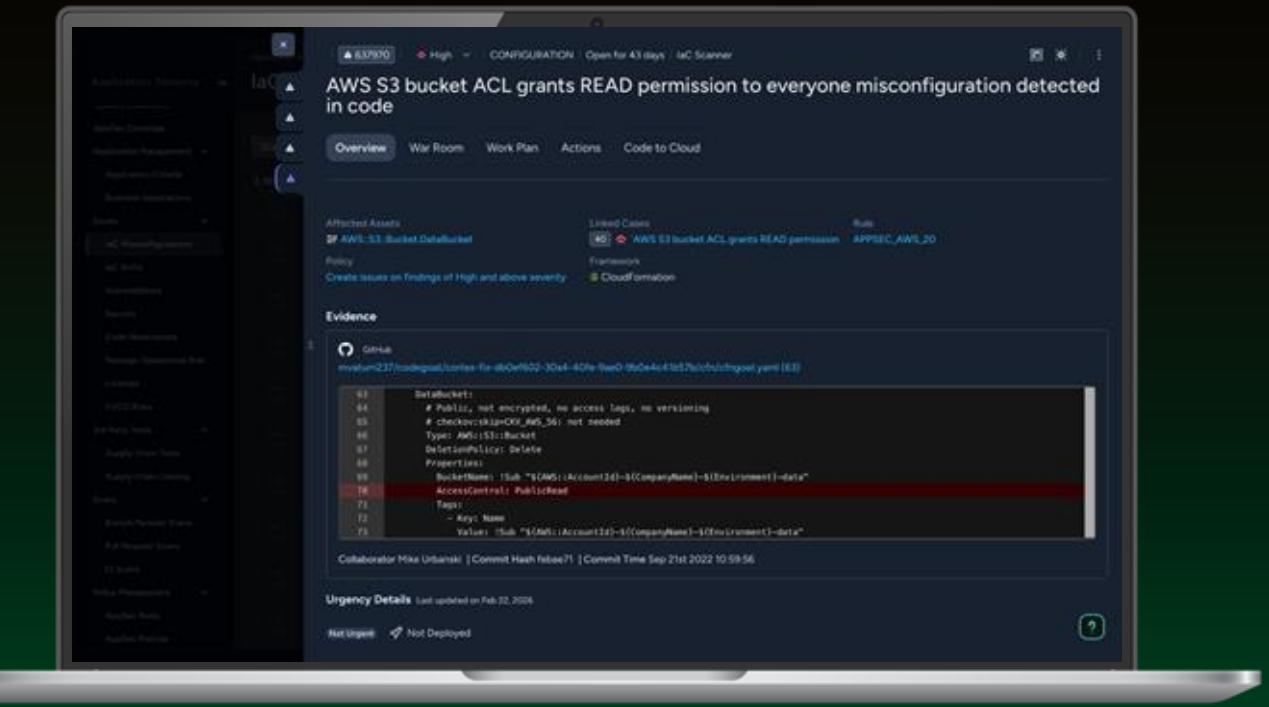
Accelerate secure development with prevention-first ASPM



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Secure Your Code



IaC Security

Secure cloud infrastructure directly in code before deployment

Software Composition Analysis

Automatically identify vulnerabilities and operational risks in open-source packages and their entire dependency trees

Secrets Security

Identify and eliminate secrets before they reach production

3rd Party Scanning Tools

Natively integrate findings from Veracode, Semgrep, Snyk and SonarQube or via SARIF for central management and consistent security policy

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Track AppSec Coverage



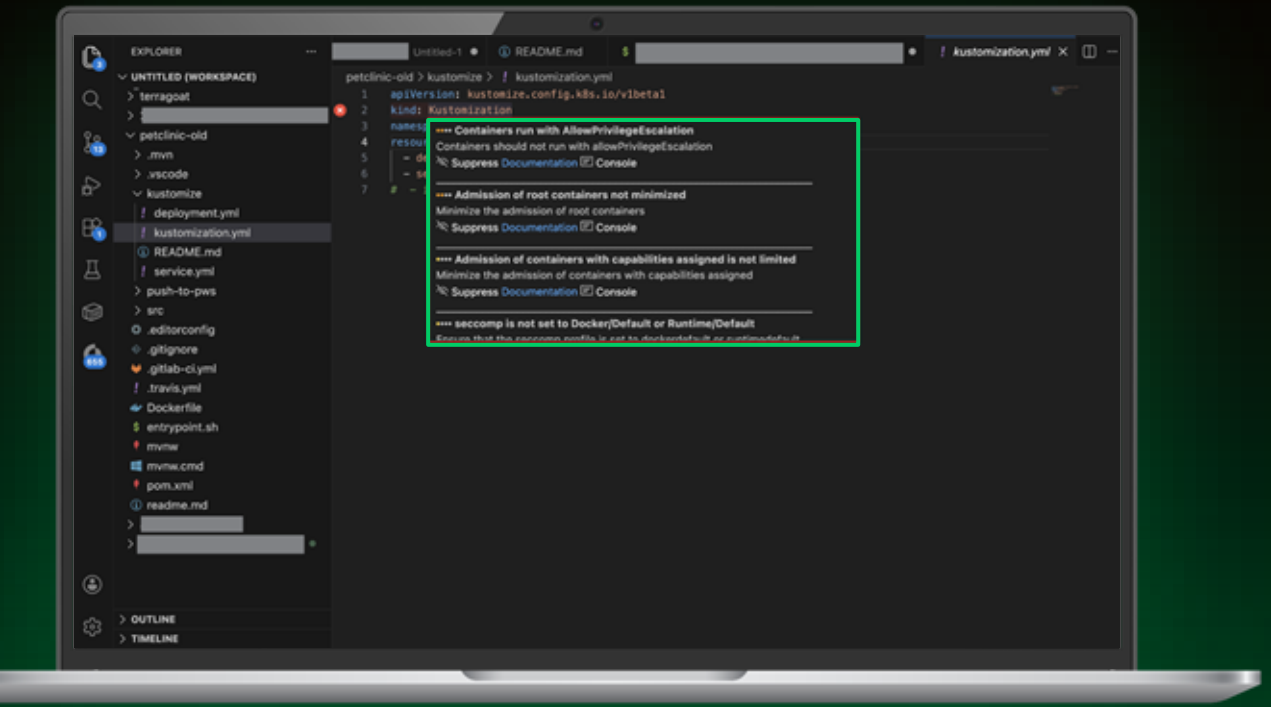
Track AppSec scanning (SAST, SCA, IaC & Secrets) coverage across repos and pipelines

Identify and eliminate security blind spots

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Integrate Security into Development Workflows



Meet developers where they work

Provide clear, actionable insights within development tools so developers can easily and quickly fix issues

Solve issues early in development lifecycle
Fixing issues in development is quicker, easier and less expensive

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Architecture · Cortex Cloud Components - AppSec - Capabilities (Secrets, SCA, AI-Security, CI/CD & IaC)

1

Secrets

Secrets Scanning

2

SCA

Software Composition Analysis

3

IaC

Infrastructure as Code Scanning

4

CI/CD

Pipeline and Supply Chain Security

1

AI-Security

AI Guardrails

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Secrets

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SCA - Software Composition Analysis

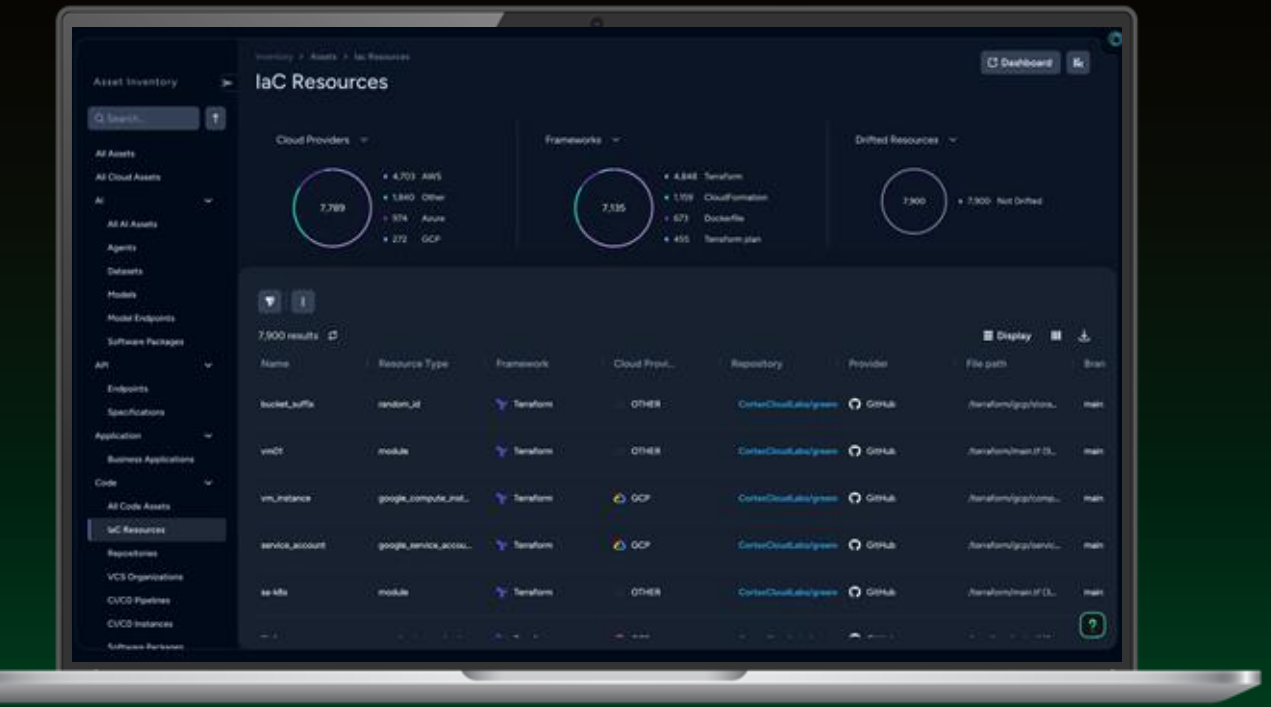
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IaC - Infrastructure as Code

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Monitor Configurations for Security Drift



Continuously monitors deployed cloud resources and compares them against secure IaC baselines

Only creates drift issues when changes **introduce security misconfigurations**

Supports scoped policies (cloud account, region, resource type, repository)

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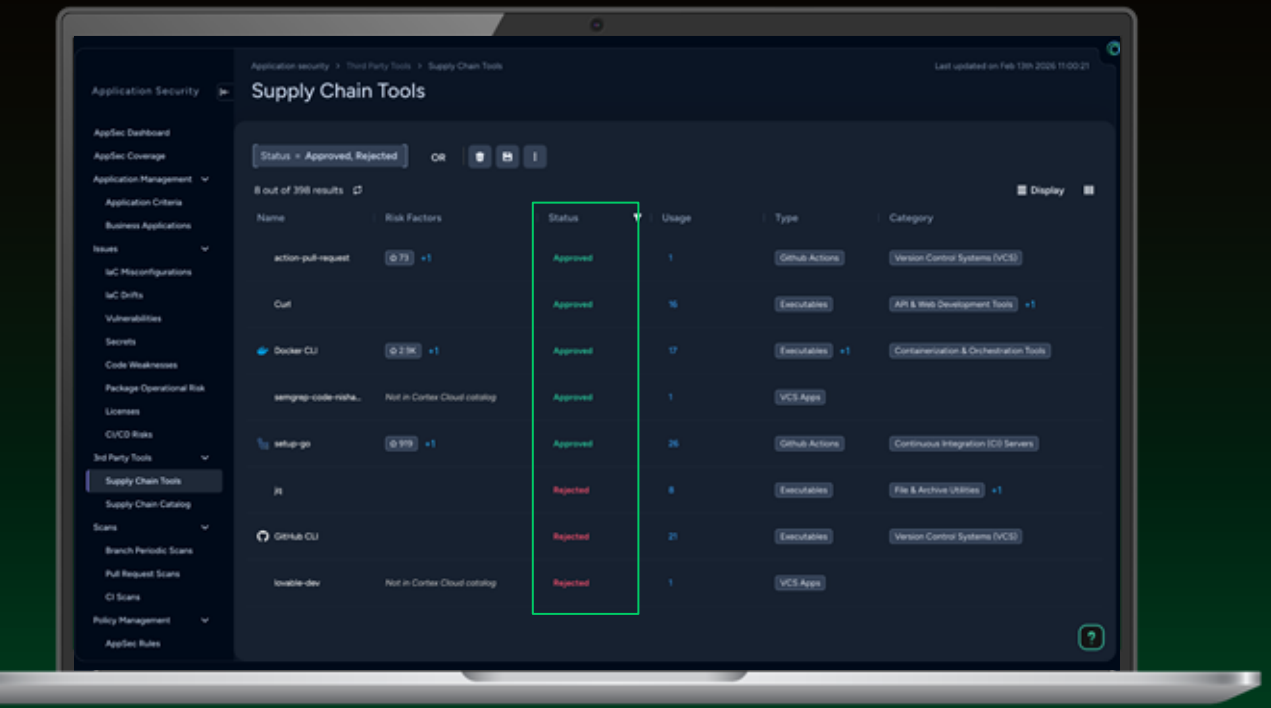
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CI/CD - Supply Chain & Pipeline Security

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Software Supply Chain Security



Deploy secure code with confidence

Unlock visibility across the entire software supply chain—from **developer tools and code identities** to registries, orchestrations, and SBOMs.

Automate compliance reporting for industry standards like CIS Github/Gitlab and OWASP Top 10 for CI/CD

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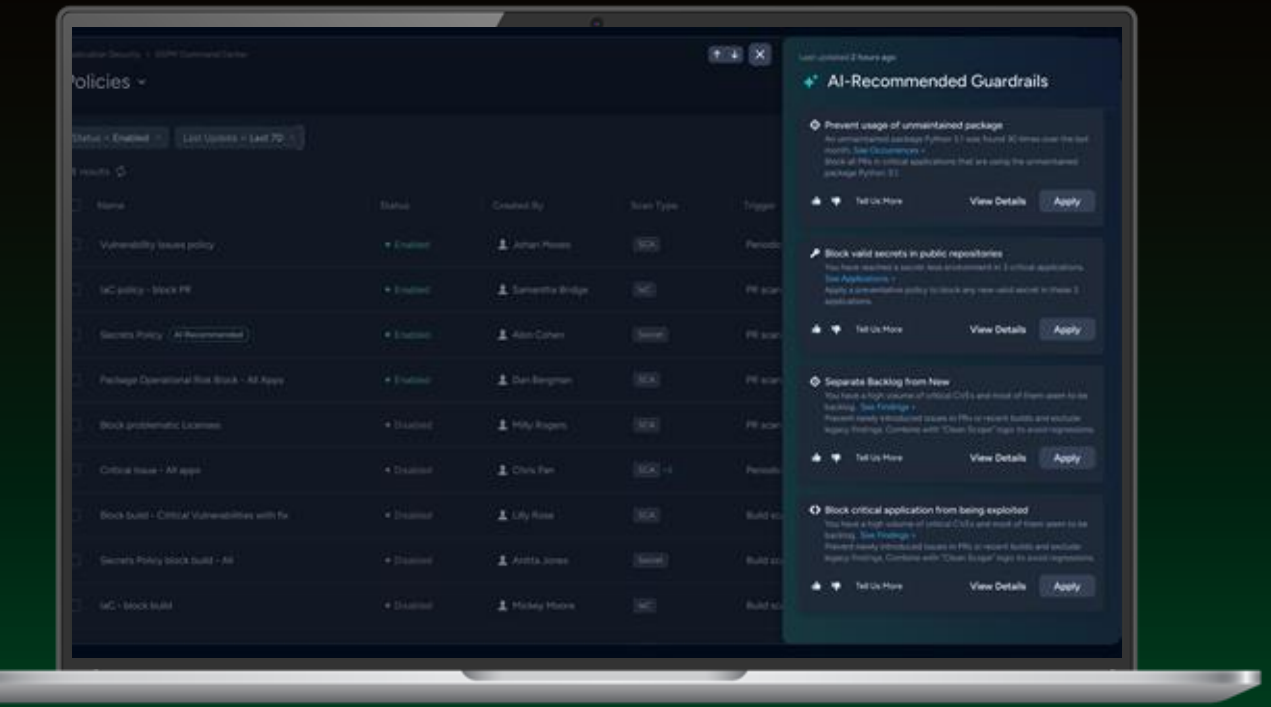
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AI-Security

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Prevent Risks with AI-Powered Guardrails



Use AI to analyze development and production environments and generate prevention policies tailored to your organization

Differentiate between new and existing issues to stop risk from being introduced while working through backlogs

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Architecture · Cortex Cloud Components - AppSec - Automation & Response Techniques

Asset Hierarchy

Understanding the Hierarchy of Assets

Understanding the hierarchy of assets will help make more sense of roles and responsibilities and the actions required when doing policy creation, investigations, remediation and quarantine scenarios.

1

Class

The Domain (high-level)

i.e. Compute, Network, Data

2

Category

The Function (normalized)

i.e. Virtual Machine, Container

3

Type

The Specifics (provider)

i.e AWS EC2 Instance

Asset Hierarchy - Primary Asset Classes

- **Compute:** VMs, Containers, Serverless, K8s.
- **Data:** DBs, Buckets, sensitive data.
- **Identity:** Users & Service identities.
- **Network:** LBs, Interfaces, Subnets.
- **AI:** Models, Infrastructure, & Endpoints.
- **Code:** Repos, IaC, & CI/CD pipelines.
- **External Surface:** Domains & Certs.
- **Security Services:** Managed tools.

Accelerating remediation: Automated fix suggestions and manual remediation guidance enable developers to resolve code weaknesses directly in the source repository without context-switching to external https://cortex-docs.paloaltonetworks.com/cortex-cloud-posture-management/cloud-inventory-and-environment/asset-classes

Understanding Platform Taxonomy

Knowing the type of taxonomy used for data being ingested in the platform will help during investigations as well as understanding case management use-cases.

1



Findings

Basic Unit (raw data)

Informational descriptions of the environment state. (e.g., Exposed bucket).

2



Issues

Actionable Tasks

Something that requires user/automation handling. Linked to findings.

3

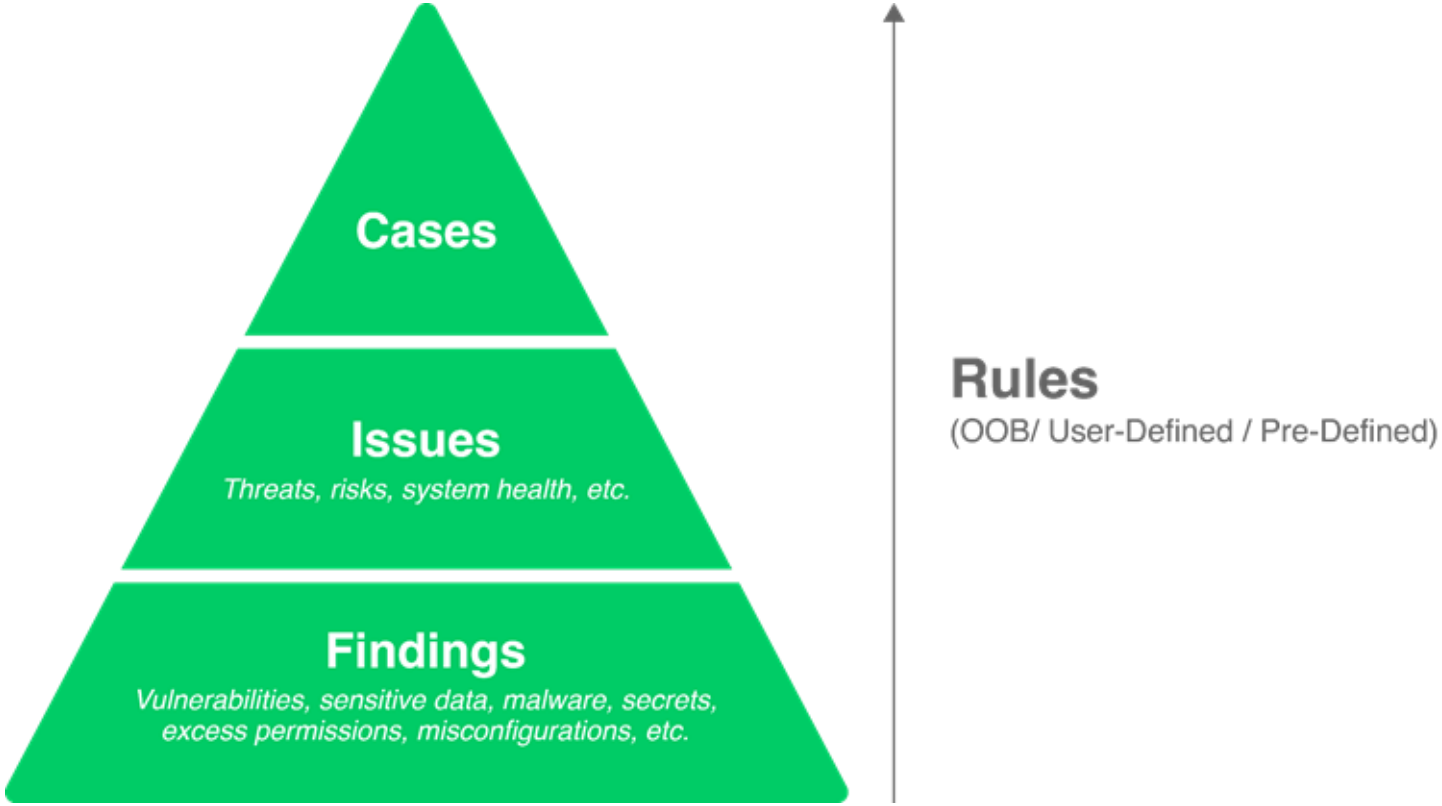


Cases

Issue Grouping

High-level investigations with assignees and playbooks.

Platform Taxonomy - Ingestion Flow



Cortex Cloud Operationalization

How To Operationalize Cortex Cloud

Determining the “best practice” for Cortex Cloud operationalization based on Persona

THE SAID

Cloud Security Engineer

YOU DESIGN “I use Cortex Cloud to continuously monitor our multi-cloud infrastructure, identify and remediate posture misconfigurations and vulnerabilities, and enforce runtime threat defense across our workloads, APIs, and identity environments.”

THEY SAID

Cloud DevOps Engineer

YOU DESIGN “I use Cortex Cloud to integrate automated Infrastructure-as-Code and container security scanning directly into our CI/CD pipelines, allowing me to catch and fix misconfigurations and vulnerabilities early in the development cycle before code ever reaches production.”

Operationalization of Cortex Cloud - Pillars

Key Pillars Of Cloud Operationalization

01

Data Ingestion

Validation and accuracy of data within the platform to support use-cases.

Operational Value

High-fidelity data sources are critical in Cortex Cloud and Cortex XSIAM because AI models, analytics engines, and automated response playbooks require rich, structured, and un-truncated telemetry—rather than raw, low-quality log text or isolated alerts—to accurately identify and remediate threats.

02

Detections

Policy and Rule creation to support misconfiguration, vulnerability and threat detections.

Operational Value

Developing tailored detection rules and policies for misconfigurations, vulnerabilities, and active threats delivers operational value by converting raw security noise into actionable, prioritized intelligence, enabling security teams to systematically eliminate risk vectors and automate incident response.

03

Remediation

Deterministic Remediation tactics - quarantine, code fixes, vulnerability patching, corrective permissions, etc …

Operational Value

Deterministic remediation—executing pre-scripted, predictable, and rule-based corrective actions in response to specific security findings—transforms SecOps from passive monitoring into active risk elimination. By replacing manual, ad-hoc triage with deterministic tactics, organizations ensure that threats are remediated consistently, correctly, and at machine speed.

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Operations Through DevSecOps

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Operations - DevSecOps

High-Value Operational Tasks for DevOps Engineers

Automating IaC & Container Security Gates

Enabling Inline Pull-Request (PR) Feedback

Supply Chain Security & Secrets Elimination

Managing Policy as Code (PaC) & Exception Rules

Infrastructure as Code (IaC) Onboarding

Operations - DevSecOps

Operational Capability	In-Product (Cortex Cloud Console & Native Engine)	3rd-Party DevOps & Integration Approach
IaC & Code Security Scanning	Centralizes Application Security Posture Management (ASPM), aggregates repository risks, maps code dependencies, and auto-generates pull-request fixes.	Executes checkov or cortex-cli scans within **GitHub Actions**, **GitLab CI**, or **Jenkins** pipelines to enforce step pass/fail thresholds.
Infrastructure & Fleet Management	Displays onboarded cloud accounts, active runtime defenders, and infrastructure health across multi-cloud environments.	Uses **Terraform** or **OpenTofu** modules to onboard AWS/Azure/GCP accounts, deploy workload agents, and manage policy settings as code.
Developer Workflow & Real-Time Feedback	Correlates runtime threats back to code source lines and developer identities (CODEOWNERS).	Installs plugins into **VS Code** / **JetBrains IDEs** for local linting and configures Git **pre-commit hooks** to catch secrets before push.

Operations - DevSecOps

Operational Capability	In-Product (Cortex Cloud Console & Native Engine)	3rd-Party DevOps & Integration Approach
Issue Triage & Ticket Routing	Groups misconfigurations by risk severity and maps pipeline vulnerabilities to cloud runtime exposure.	Triggers **Jira**, **ServiceNow**, or **Slack** webhooks to automatically assign tickets to responsible engineering teams based on repository ownership metadata.

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DevOps Policy Management “Best Practice”

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Operations - DevSecOps - Policy Management

Structure Policy as Code & Multi-Tier Governance

- **Leverage Out-of-the-Box (OOTB) Frameworks**

  Establish baseline coverage using Cortex Cloud's pre-built OOTB rules across Infrastructure-as-Code (IaC), secrets, SAST, SCA, and CI/CD security. These rules automatically align with compliance standards such as CIS, NIST, and PCI-DSS.

- **Manage Policies via Infrastructure-as-Code**

  Scale policy management programmatically using the Cortex Cloud Terraform provider or Public API. Managing custom rules via HCL or JSON payloads ensures auditability, version control, and multi-tenant synchronization.

- **Validate Rules Before Deployment**

  Run syntax and logic checks using validation endpoints or *terraform plan* before publishing new custom rules to prevent logic errors or alert storms across repository fleets.

Operations - DevSecOps - Policy Management

Drive Context-Aware Prioritization (Code-To-Cloud)

- **Prioritize by Runtime Reachability**

  Correlate static code findings (SAST/SCA) with live runtime telemetry. Focus immediate developer effort on vulnerabilities in packages or code paths actively executing in production containers or hosts that are exposed to the internet.

- **Automate Ownership & Service Mapping**

  Group repositories into Business Applications and map security issues directly to *CODEOWNERS* metadata. This ensures findings automatically route to the correct engineering pod without manual SecOps intervention.

Operations - DevSecOps - Policy Management

Implement Staged Guardrails & Shift-Left Workflows

- **Adopt a Phased Enforcement Strategy**

  Phase in policy enforcement using a three-stage rollout: **Inform** (IDE plugins and developer notifications), **Warn** (automated PR comments without blocking builds), and **Enforce** (blocking CI/CD pipeline runs via *cortex-cli* for Critical/High severity policy failures).

- **Enable Smart-Fix Pull Requests**

  Configure automated fix suggestions so developers can accept single-click PR commits for IaC misconfigurations and exposed secrets directly inside GitHub, GitLab, or Bitbucket.

Operations - DevSecOps - Policy Management

Enforce Software Supply Chain Security (SSCS)

- **Automate Continuous SBOM Tracking**

  Mandate continuous manifest and image scanning to maintain up-to-date Software Bills of Materials (SBOMs), identifying open-source package risks and license compliance issues across all dependencies.

- **Harden CI/CD & VCS Configurations**

  Apply policies that audit build system workflow files (e.g., GitHub Actions, GitLab CI) for exposed API tokens, over-privileged deployment keys, and unpinned third-party build actions.

Operations - DevSecOps - Policy Management

Enforce Governance & Time-Bound Exception Management

- **Enforce Scoped Risk Acceptance**

  Require all developer suppressions or policy bypasses to include documented business justifications, approvers, and mandatory expiration dates (e.g., 30–90 days) to prevent permanent security debt.

- **Set and Monitor SLA Targets**

  Define strict Service Level Agreements for issue resolution based on vulnerability severity and business impact (e.g., Critical: 48 hours, High: 14 days) and track adherence via the ASPM Command Center.

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DevOps Issue Management “Best Practice”

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Operations - DevSecOps - Issue Management

Contextual Risk Triaging & Prioritization

- **Correlate Code with Runtime Telemetry**

  Prioritize Application Security (AppSec) issues based on real-world reachability. Vulnerabilities in open-source packages (SCA) or code (SAST) should be elevated to top priority if the affected component is running in production, exposed to the internet, or loaded in memory.

- **Group Findings by Root Cause**

  Avoid overwhelming engineering teams with thousands of individual alert instances. Group issues by root-cause repository, container base image, or shared module so developers can fix multiple issues with a single update or pull request.

- **Incorporate Threat Intelligence**

  Elevate issue priority dynamically when a detected vulnerability (CVE) has active, published exploits (e.g., CISA KEV list) or high EPSS (Exploit Prediction Scoring System) scores.

Operations - DevSecOps - Issue Management

Automated Ownership & Workstream Routing

- **Map Repositories to CODEOWNERS**

  Utilize *CODEOWNERS* metadata, Git commit histories, and Application Boundaries in Cortex Cloud to automatically map every discovered issue to the specific owner, squad, or developer accountable for that codebase.

- **Integrate Bi-Directionally with Ticketing Systems**

  Sync Cortex Cloud issues directly with existing developer tools (e.g., Jira, GitHub Issues, ServiceNow). Ensure bi-directional synchronization so closing a ticket or merging a fix automatically updates the issue state in Cortex Cloud.

- **Establish Notification Thresholds**

  Configure Slack, Teams, or email alerts to trigger only for high-severity or actionable issues to avoid alert fatigue, while routing routine lower-severity issues into sprint backlogs.

Operations - DevSecOps - Issue Management

SLA Enforcement & Lifecycle Management

- **Define Risk-Based Remediation SLAs**

  Establish strict, enforceable timeframes for issue resolution based on asset criticality and risk severity (e.g., Critical/Internet-Exposed: 48 hours, High: 14 days, Medium: 30–60 days).

- **Automate Status Transitions & Re-Testing**

  Configure Cortex Cloud to automatically verify fixes upon pipeline re-runs or container rebuilds, automatically transitioning resolved issues to "Closed" without requiring manual SecOps sign-off.

- **Track Remediation Velocity & SLA Compliance**

  Monitor Mean Time to Remediate (MTTR) across different engineering departments using ASPM dashboards to identify bottlenecks and recognize top-performing security leads.

Operations - DevSecOps - Issue Management

Developer-First Remediation & Self-Service

- **Provide One-Click "Smart-Fix" Pull Requests**

  Enable automated fix generation so developers can review and merge pre-formatted code commits, dependency upgrades, or IaC parameter updates directly in their pull requests.

- **Deliver Actionable Context & Fix Guidance**

  Ensure issue tickets contain explicit remediation instructions, precise code lines, file paths, and recommended library version upgrades rather than generic vulnerability descriptions.

- **Remediate at the Golden Image / Template Level**

  Address recurring vulnerabilities by updating base container images in container registries (ECR, ACR, Docker Hub) or central IaC modules, remediating issues across dozens of downstream applications at once.

Operations - DevSecOps - Issue Management

Risk Acceptance & Suppression Governance

- **Enforce Time-Bound Risk Acceptance**

  Require formal business justifications, technical approvers, and mandatory expiration dates (e.g., 30, 60, or 90 days) for all accepted risks or issue suppressions.

- **Audit Active Suppressions Continuously**

  Conduct monthly audits of suppressed or muted issues to ensure expired workarounds or temporary exceptions are reviewed and revoked if no longer valid.

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DevOps Supported Integrations

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Architecture · Cortex Cloud Components - AppSec - Supported Integrations

Supported Version Control Systems (VCS)	Supported CI/CD Systems & Pipeline Scanners
• **GitHub:** GitHub SaaS (Cloud) and GitHub Enterprise (On-Prem). • **GitLab:** GitLab SaaS and GitLab Self-Managed (On-Prem). • **Bitbucket:** Bitbucket Cloud and Bitbucket Data Center. • **Azure DevOps:** Azure Repos. • **AWS:** AWS CodeCommit.	• **Native CI/CD Pipeline Scanning:** GitHub Actions, GitLab CI/CD, Jenkins, CircleCI, and Azure Pipelines. • **CLI & Scripted Pipeline Integration:** cortex-cli for embedding pass/fail security checks into any runner (e.g., Bitbucket Pipelines, CircleCI, harness, or custom runners). • **Infrastructure-as-Code Workflow Tooling:** Terraform Cloud / HCP Terraform Run Tasks.

Architecture · Cortex Cloud Components - AppSec - Supported Integrations

Supported Integrated Development Environments (IDEs)	Supported Container & Package Registries
• **Visual Studio Code (VS Code):** Native Extension. • **JetBrains IDE Suite:** IntelliJ IDEA, PyCharm, WebStorm, GoLand, Rider, and CLion plugins. • **Visual Studio:** Extension for local development linting. • **Eclipse:** Native IDE plugin.	• **Amazon Web Services:** Amazon Elastic Container Registry (ECR). • **Microsoft Azure:** Azure Container Registry (ACR). • **Google Cloud:** Google Artifact Registry (GAR) / Google Container Registry (GCR). • **Third-Party & On-Prem Registries:** Docker Hub, Quay.io, and JFrog Artifactory.

Architecture · Cortex Cloud Components - AppSec - Supported Integrations

Supported Operational, Ticketing & Collaboration Platforms	Supported Infrastructure-as-Code (IaC) Frameworks (Policy-as-Code)
• **Ticketing & ITSM:** JIRA Cloud, JIRA Data Center, and ServiceNow. • **Notification & Webhooks:** Slack, Microsoft Teams, and PagerDuty. • **Programmatic Integration:** Public REST APIs, Custom Webhooks, and Transporter over Broker VM for secure on-premise connectivity.	• HashiCorp Terraform & OpenTofu • AWS CloudFormation & AWS SAM • Kubernetes Manifests, Helm Charts & Kustomize • Azure Resource Manager (ARM) & Bicep • Dockerfiles & Container Specifications • Serverless Framework Templates

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Issue & Case Management Lifecycle

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Incident Management

Case Management
Organize & Track Incidents Efficiently



Investigation Flow
Streamlined & Guided Investigations



Queue Management
Prioritize & Assign Tasks Automatically



Cases/Issues (Customization)
Flexible & Adaptable Workflows



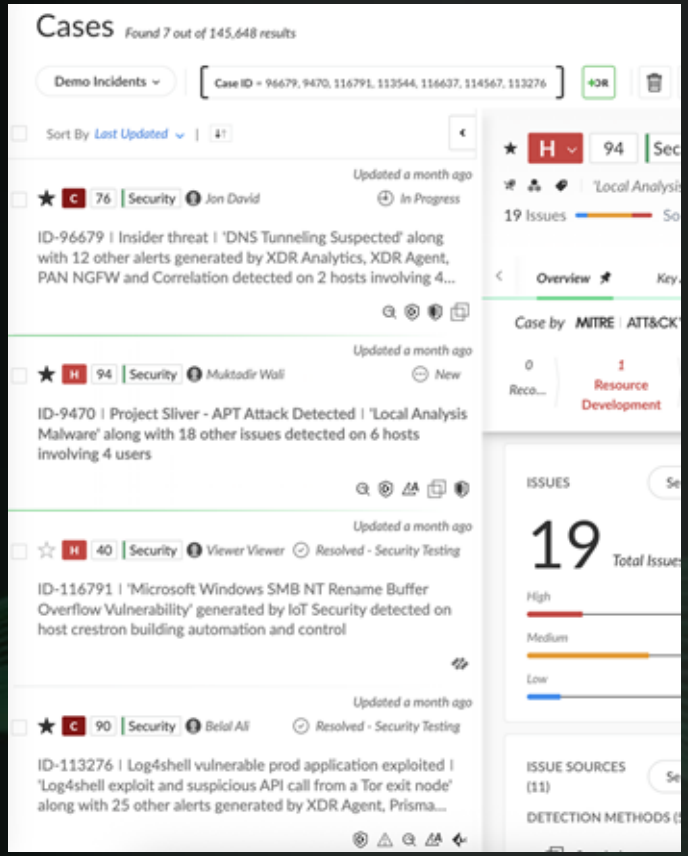
Incident SmartScores & Severity
ML-Driven Risk Scoring & Prioritization

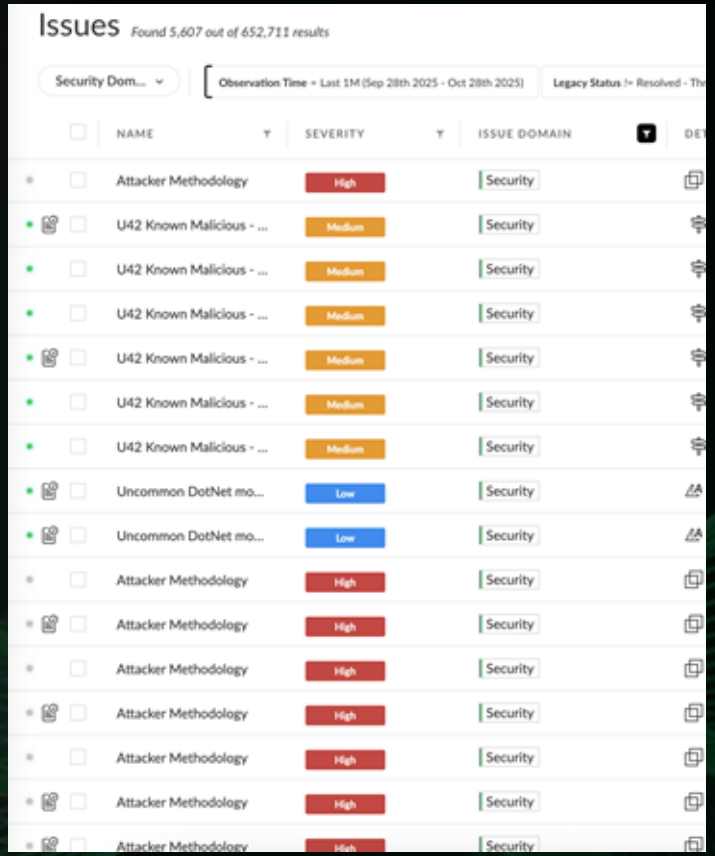


Exceptions
Manage & Document Outliers



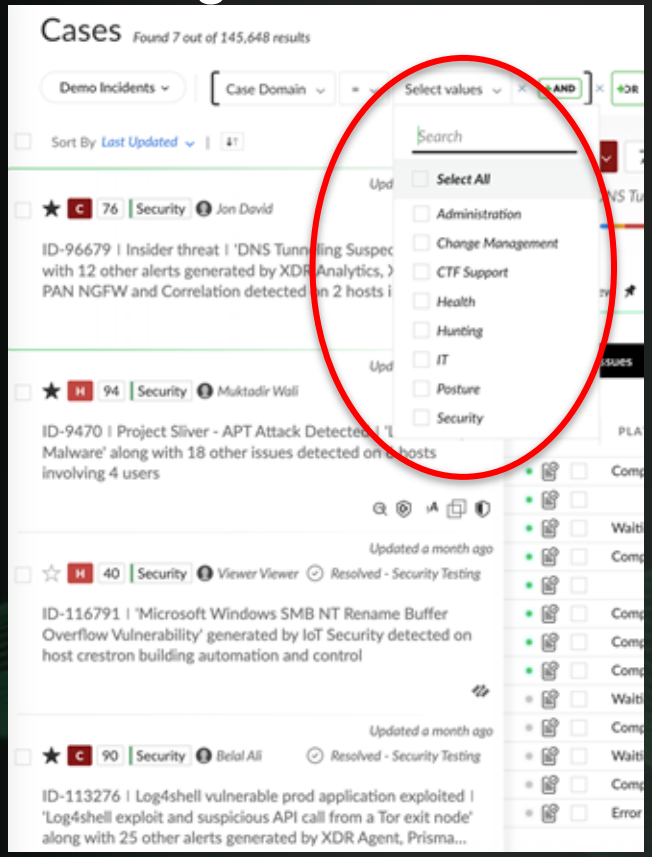
View Cases and/or Issues

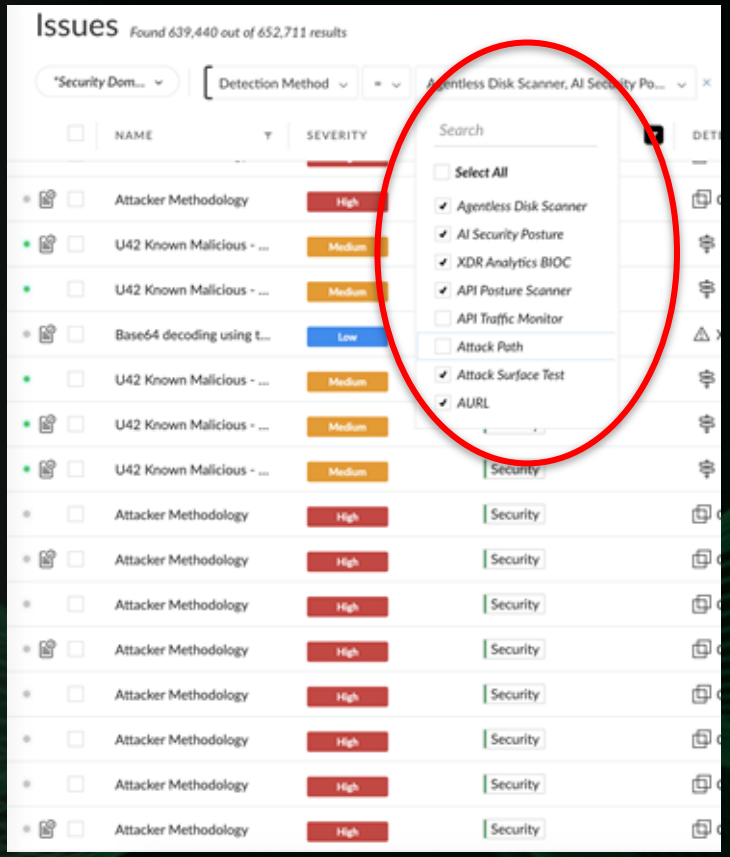




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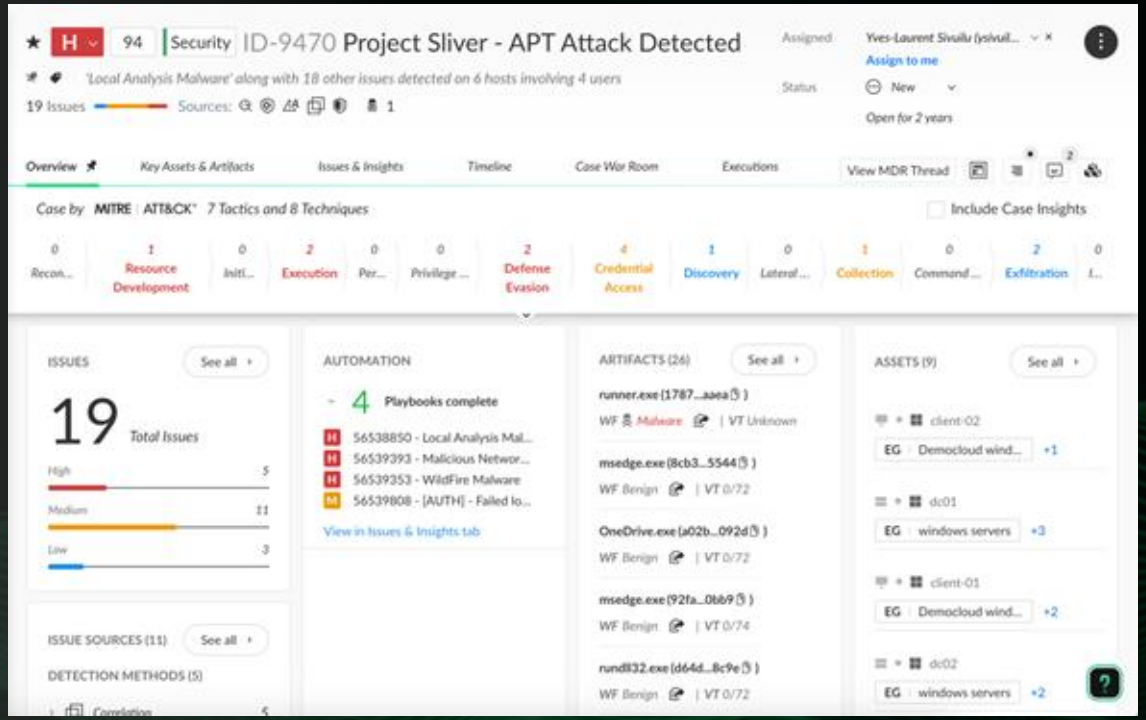
Filtering within Cases and/or Issues



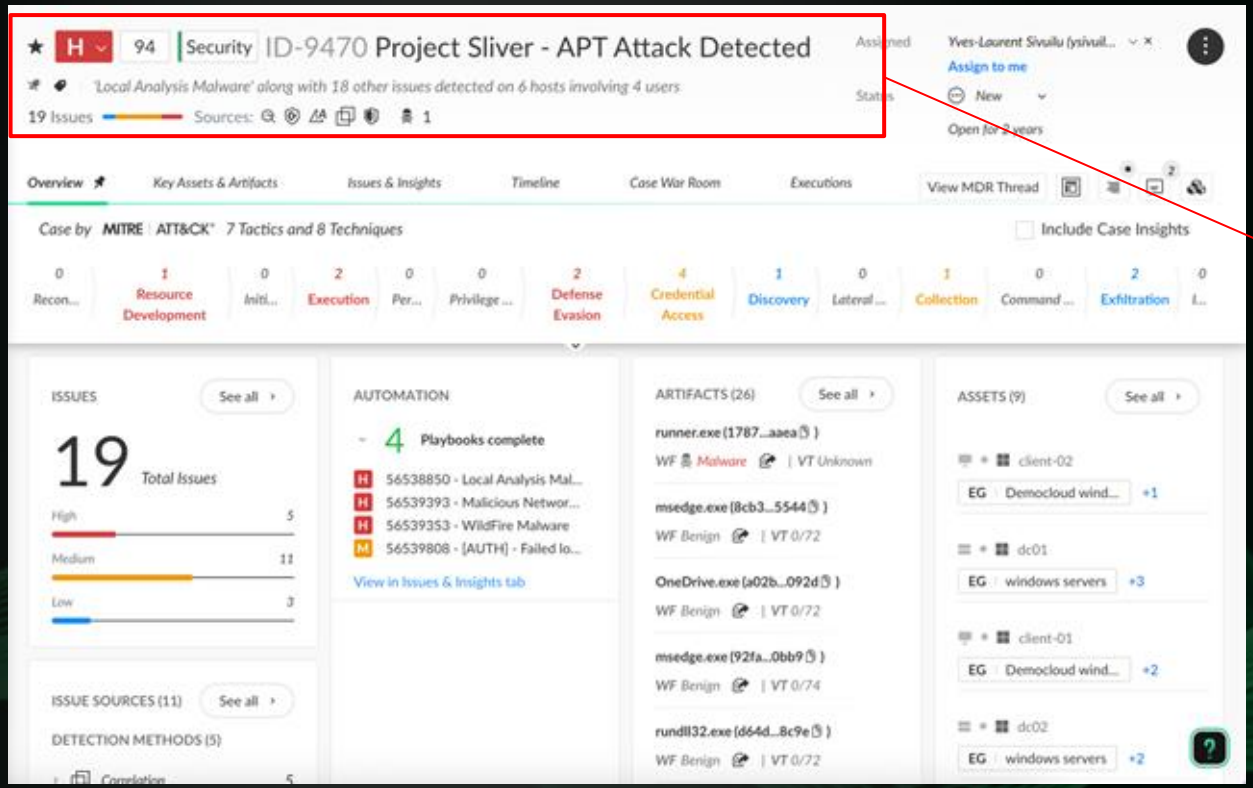


Singular View of Case That Includes…

- Causality
- Stitched Data
- Grouped Issues
- Artifacts
- Indicators
- Automation
- War Room(s)
- Assignments
- Issue Sources
- Data Sources
- Timeline
- And More…



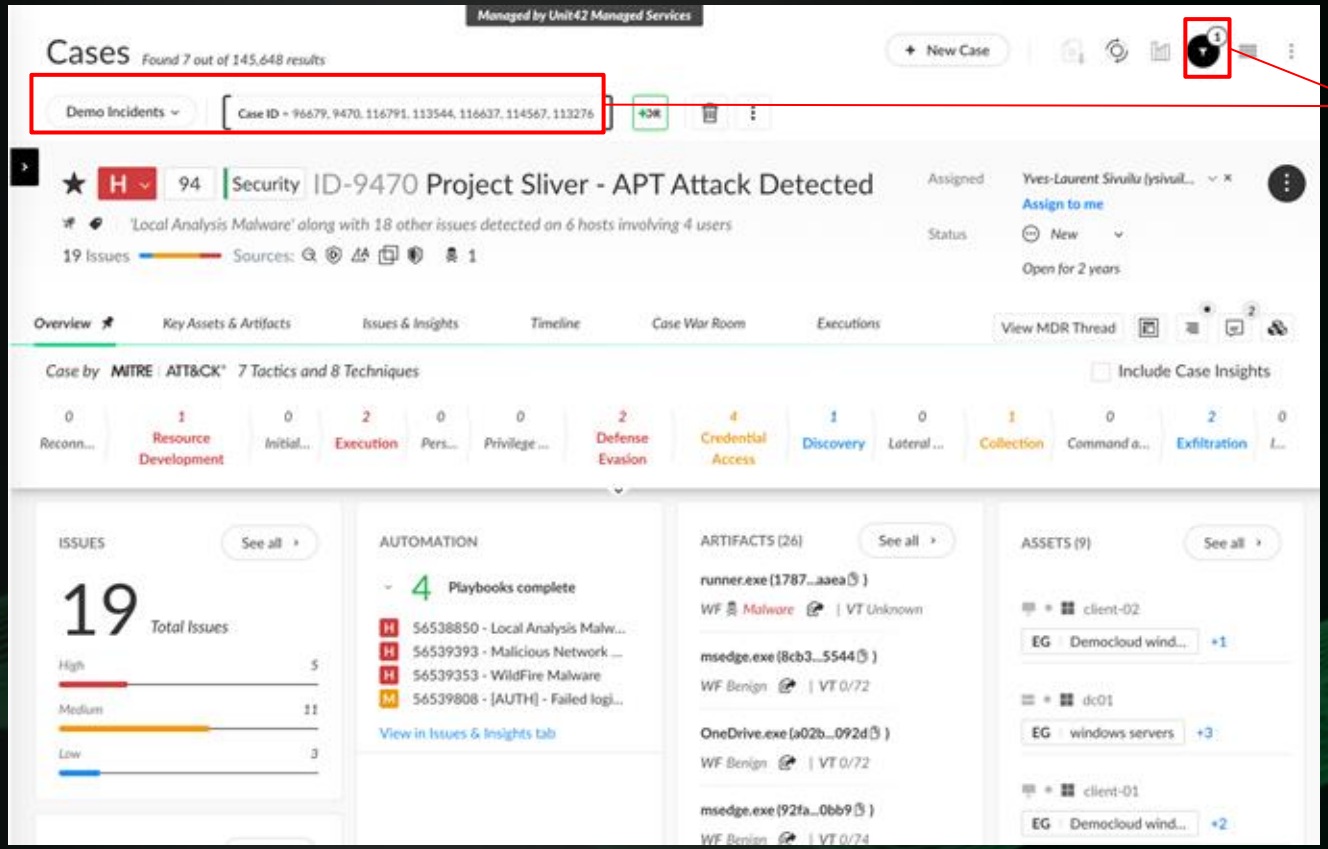
Case Breakdown



Title Block

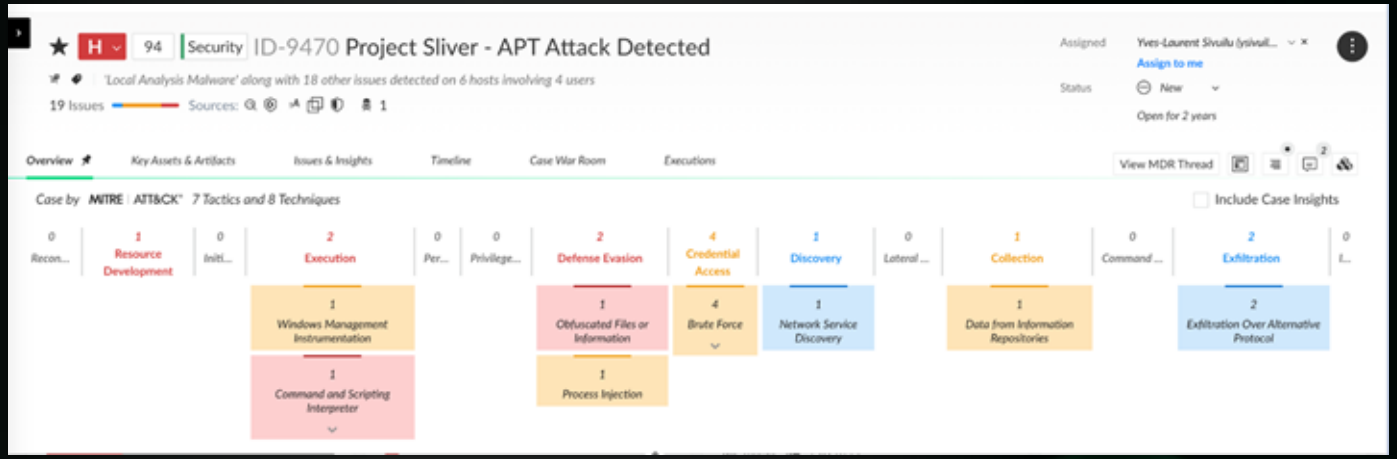
- Starred
- Smart Score
- Case ID
- Title
- Description
- Grouping Lock
- Tags
- Summary

Case Breakdown



Filtering

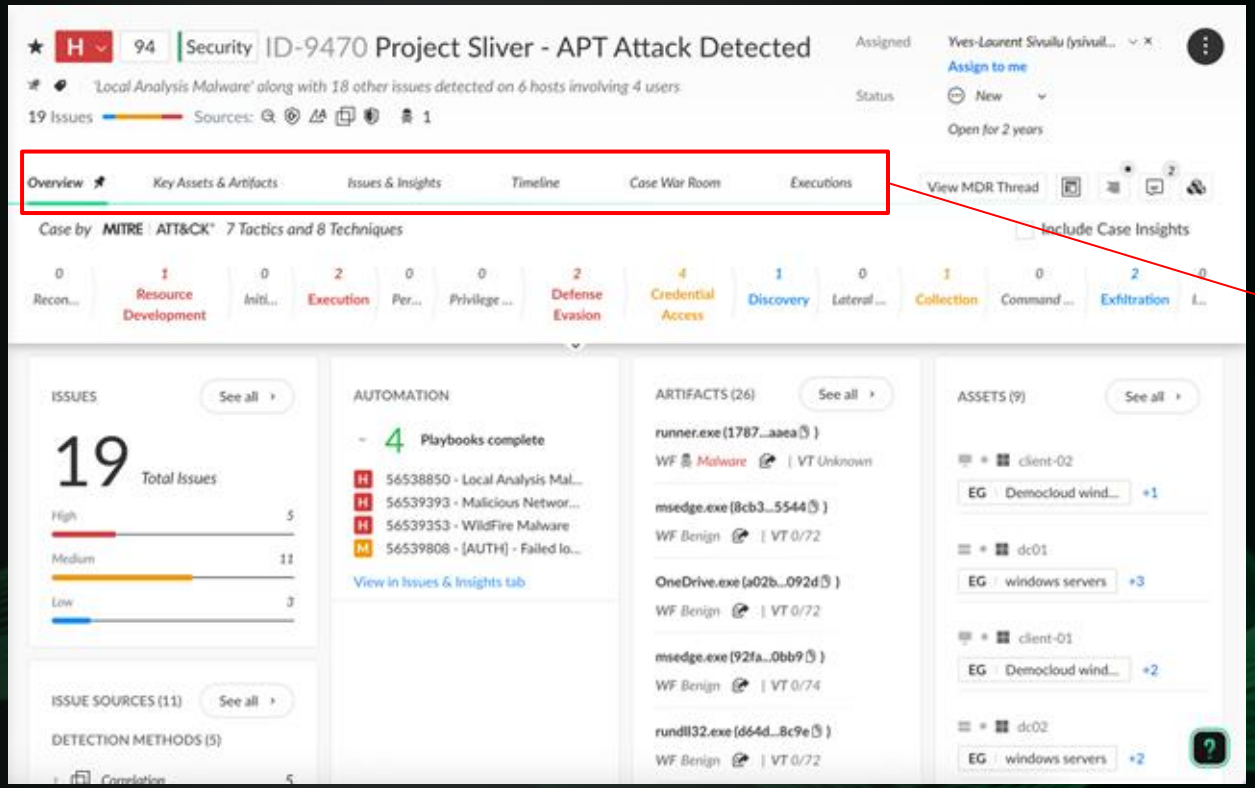
Case Breakdown



MITRE

- Tactics
- Techniques
- Drilldown
- Roll-Up Reporting

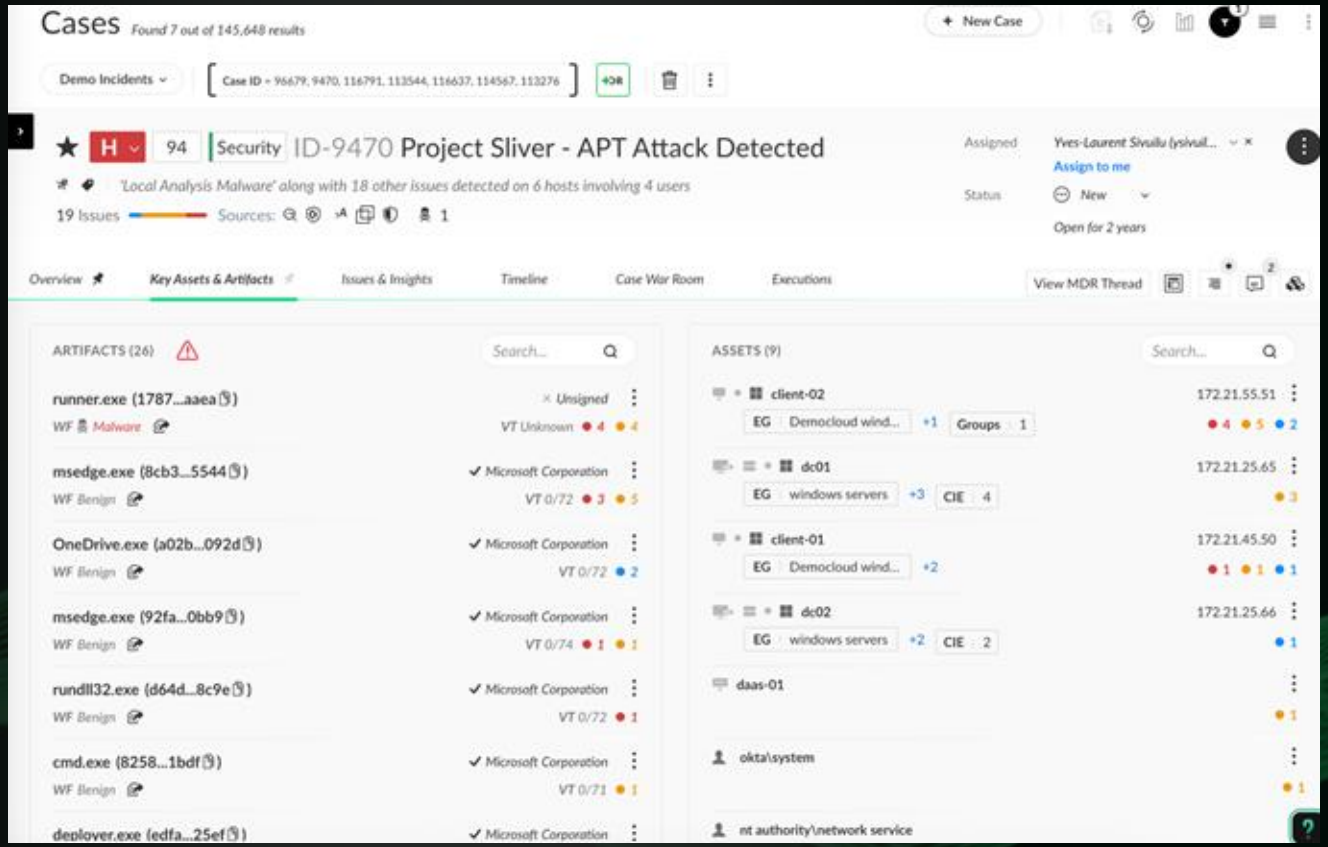
Case Breakdown



Case Navigation

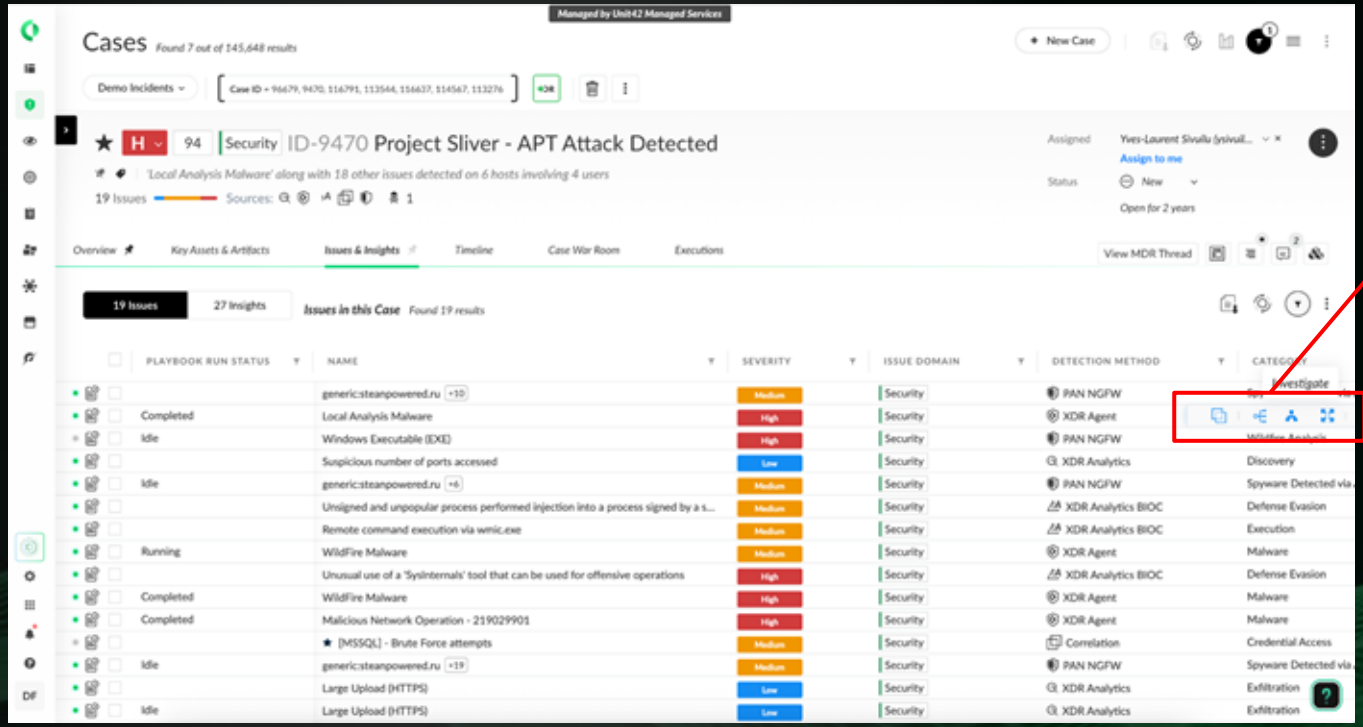
- Overview
- Assets and Artifacts
- Issues and Insights
- Timeline
- Case War Room
- Executions (Causality)

Case Breakdown



Key Assets & Artifacts

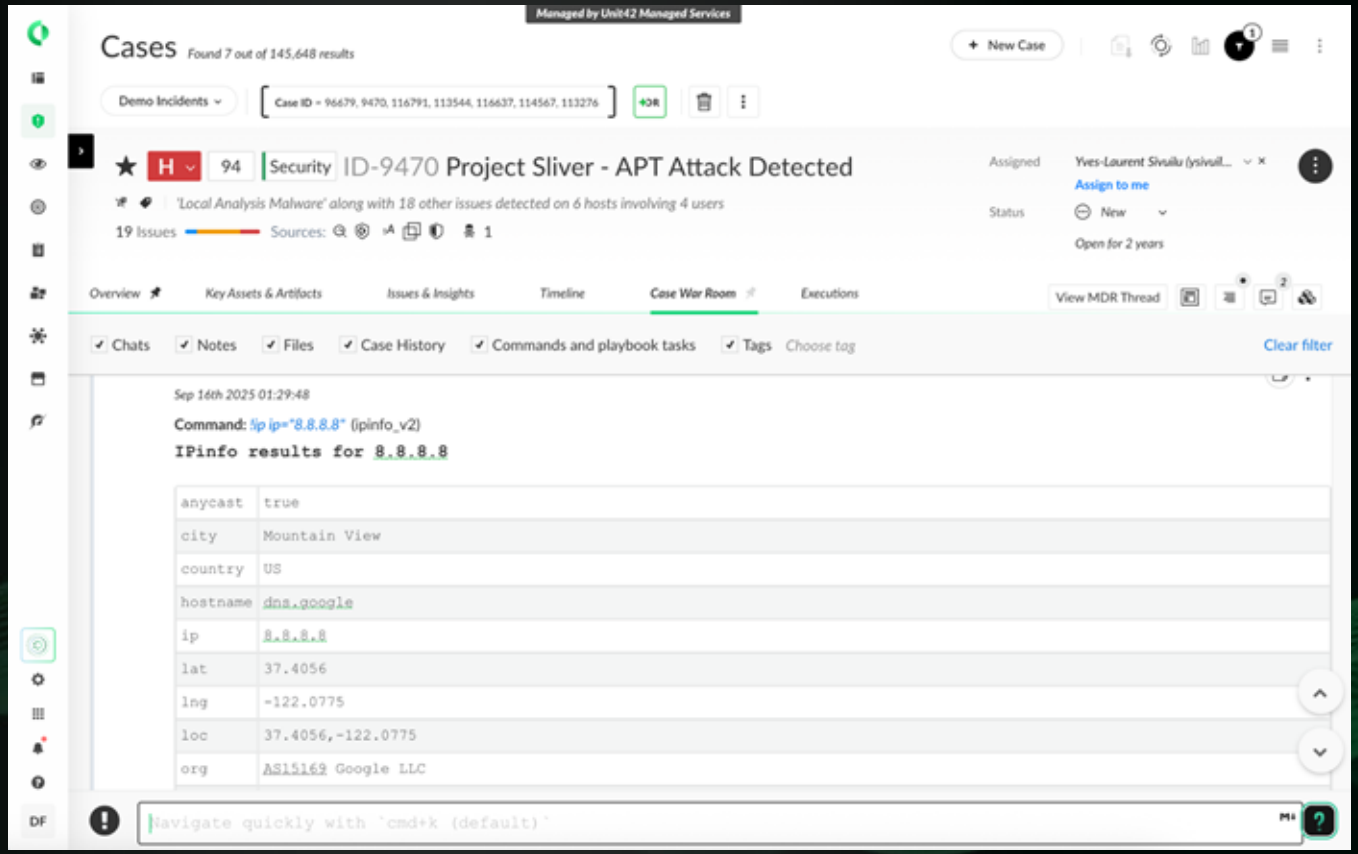
Case Breakdown



Issues and Insights

- Fly out menu
- Insights

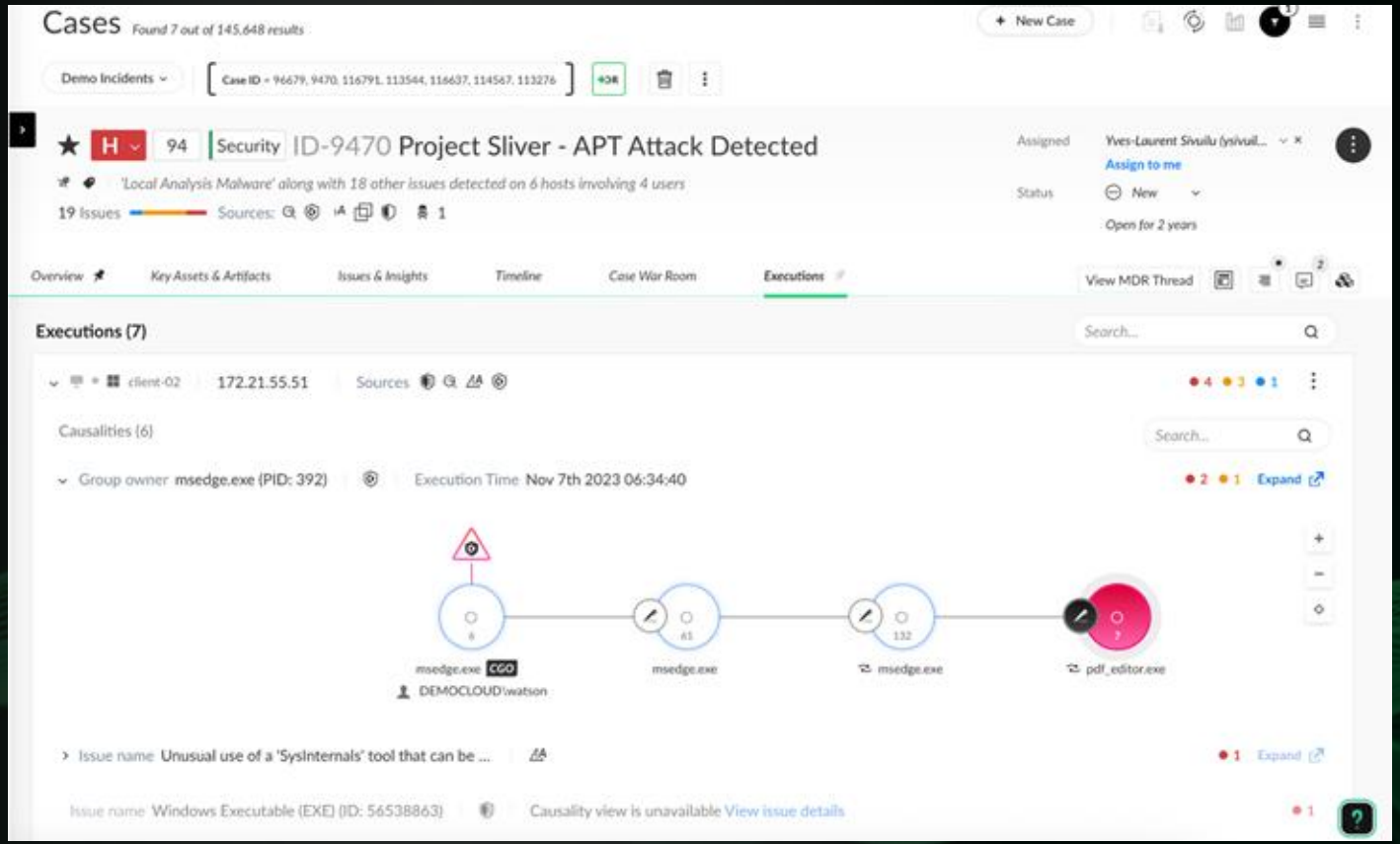
Case Breakdown



Incident War Room

- Collaborate
- Execute
- Automation Results
- Files
- Tags

Case Breakdown

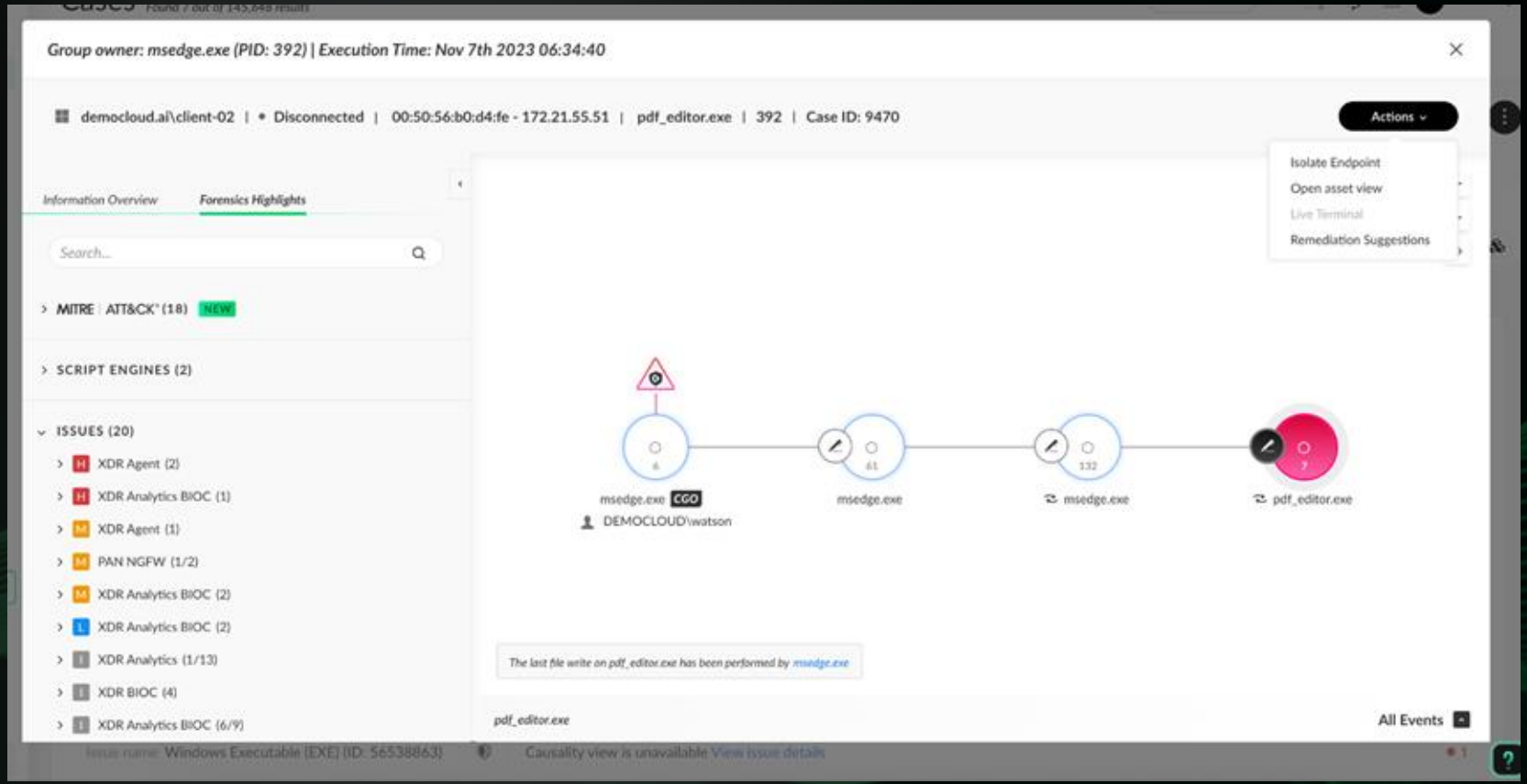


Executions (Causality)

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Case Breakdown

Executions - Expanded

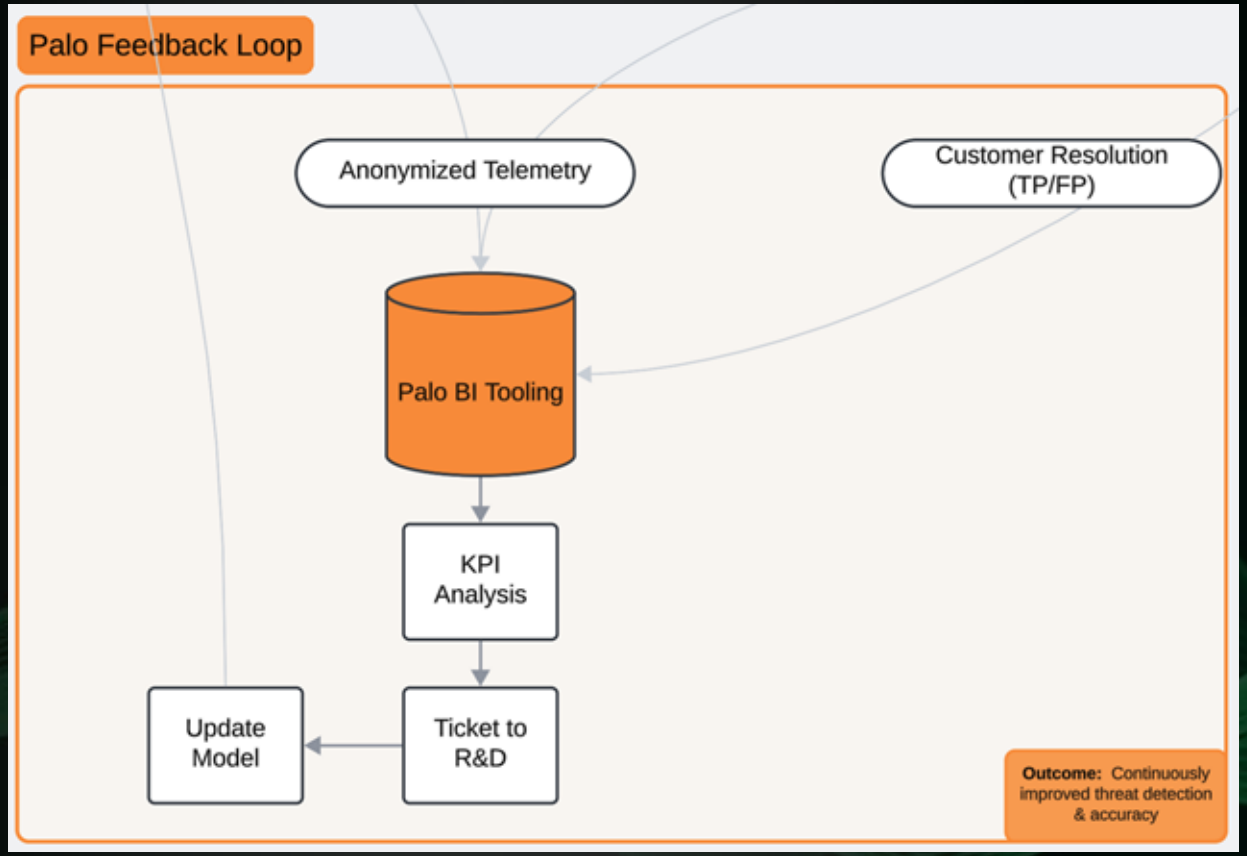


Case Management Best Practices

- Case hygiene
- Automation first (Use case planning)
- Optimal queries
- MITRE Classification

Exceptions

For Analytics Exceptions - Remember…



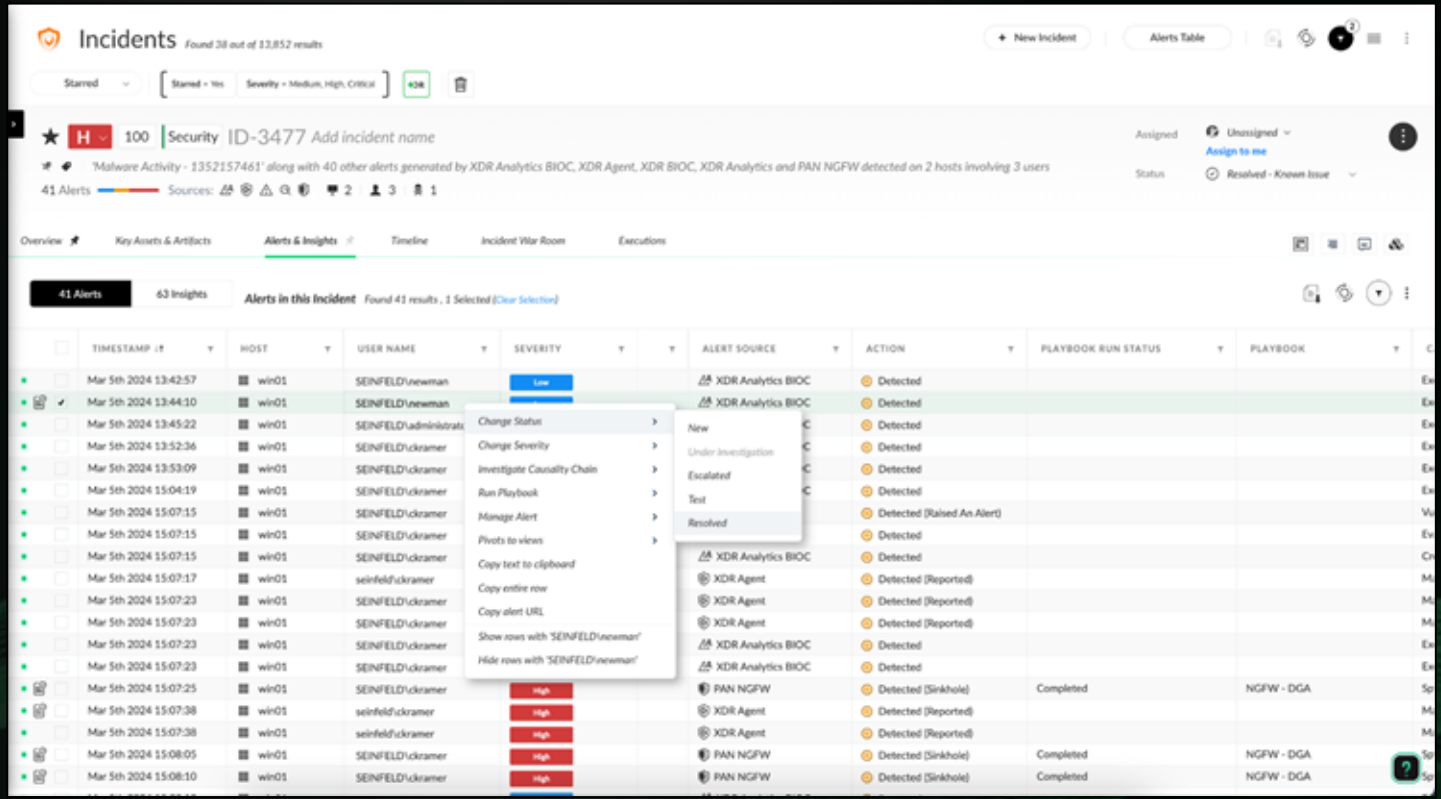
Your control over analytics based alerts is:

Incident Resolution (TP/FP)

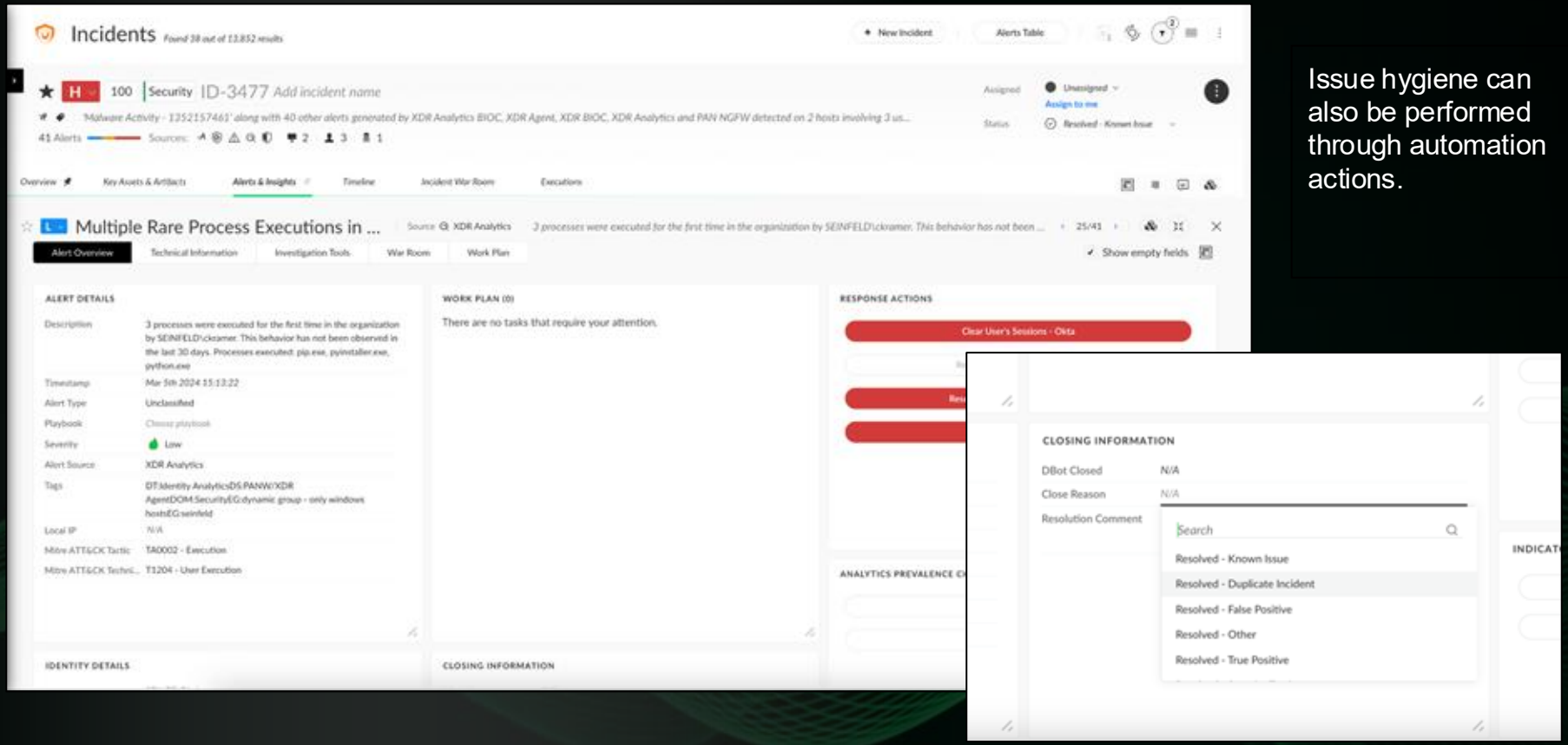
Exclusions (we will cover exclusions)

For ABIOC's you can also disable/enable rules

Case Hygiene - Issues

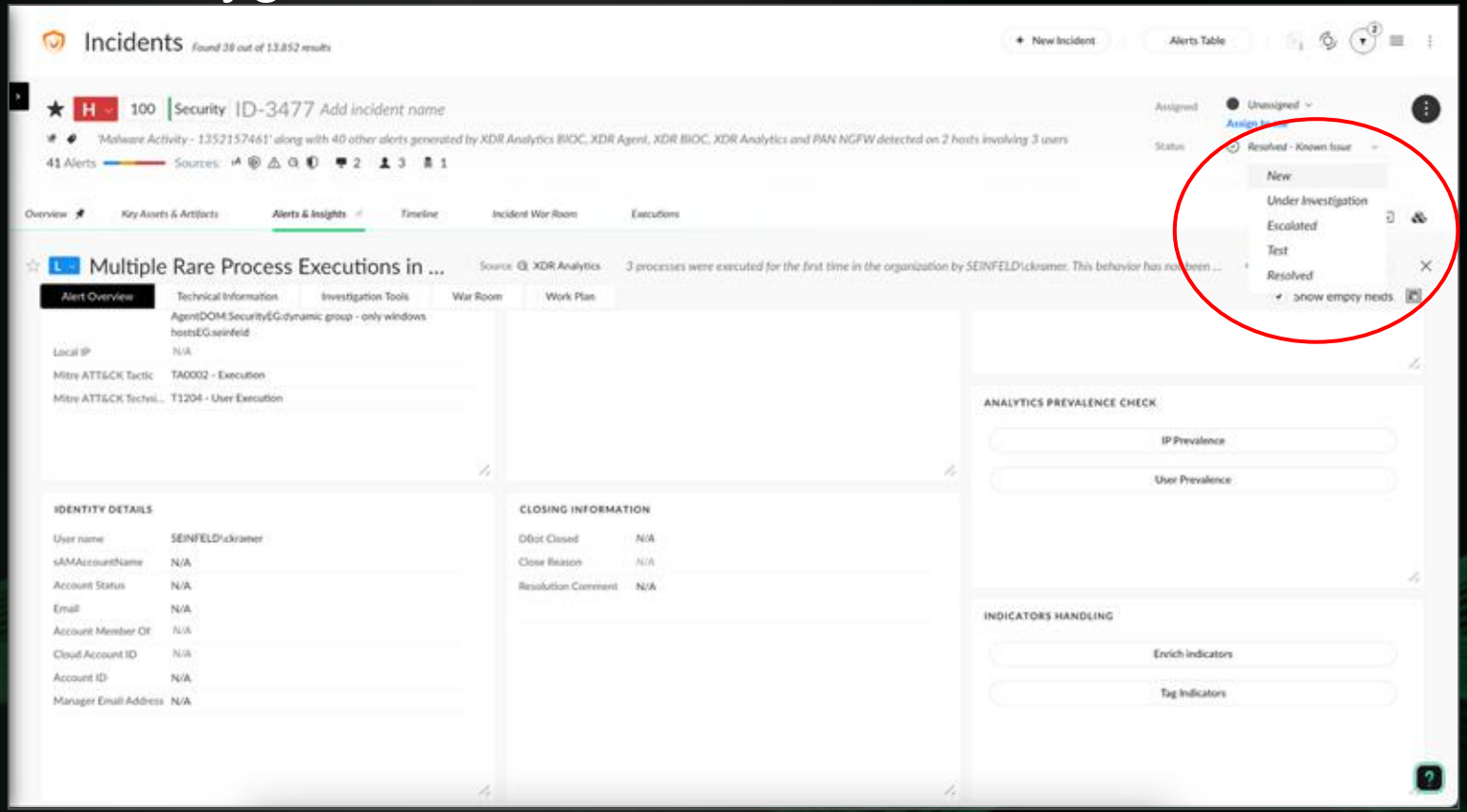


Case Hygiene - Issues



Issue hygiene can also be performed through automation actions.

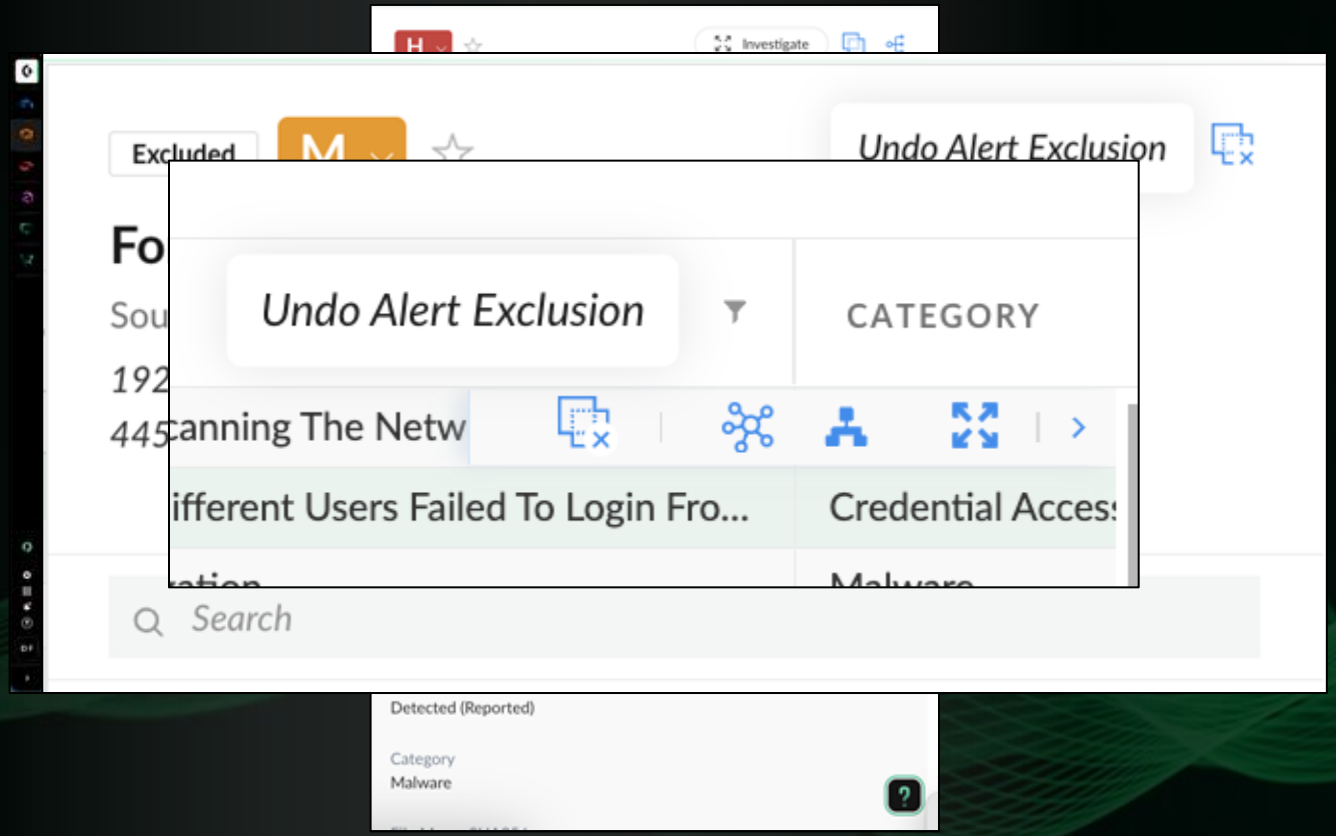
Case Hygiene - Case Level



The case resolution can be pushed to issues.

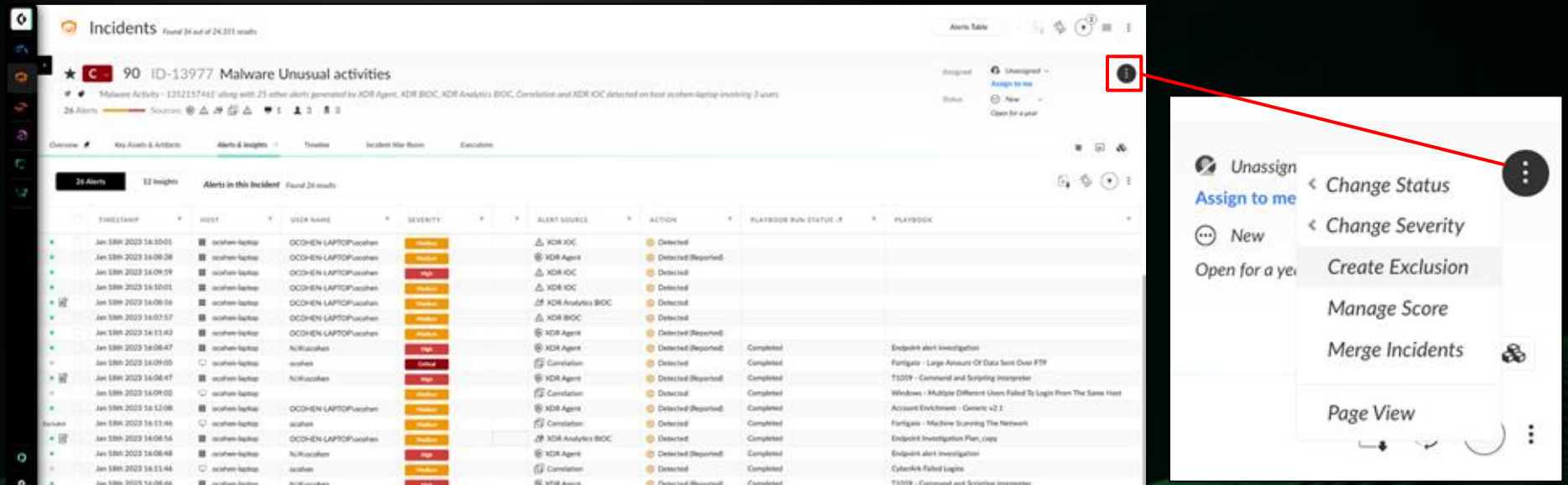
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Issue Exclusions (Exclude Issues)

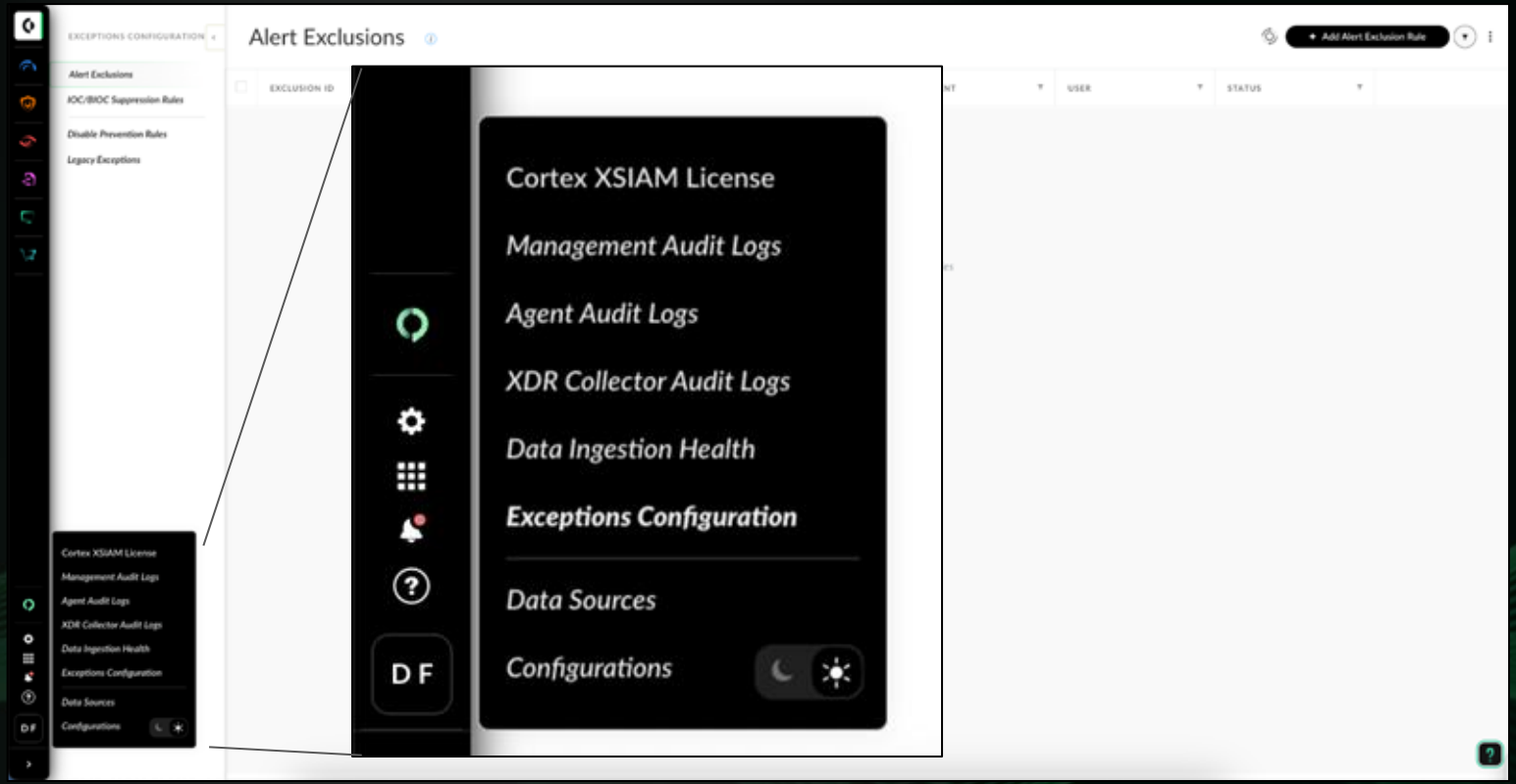


Exclude Issues

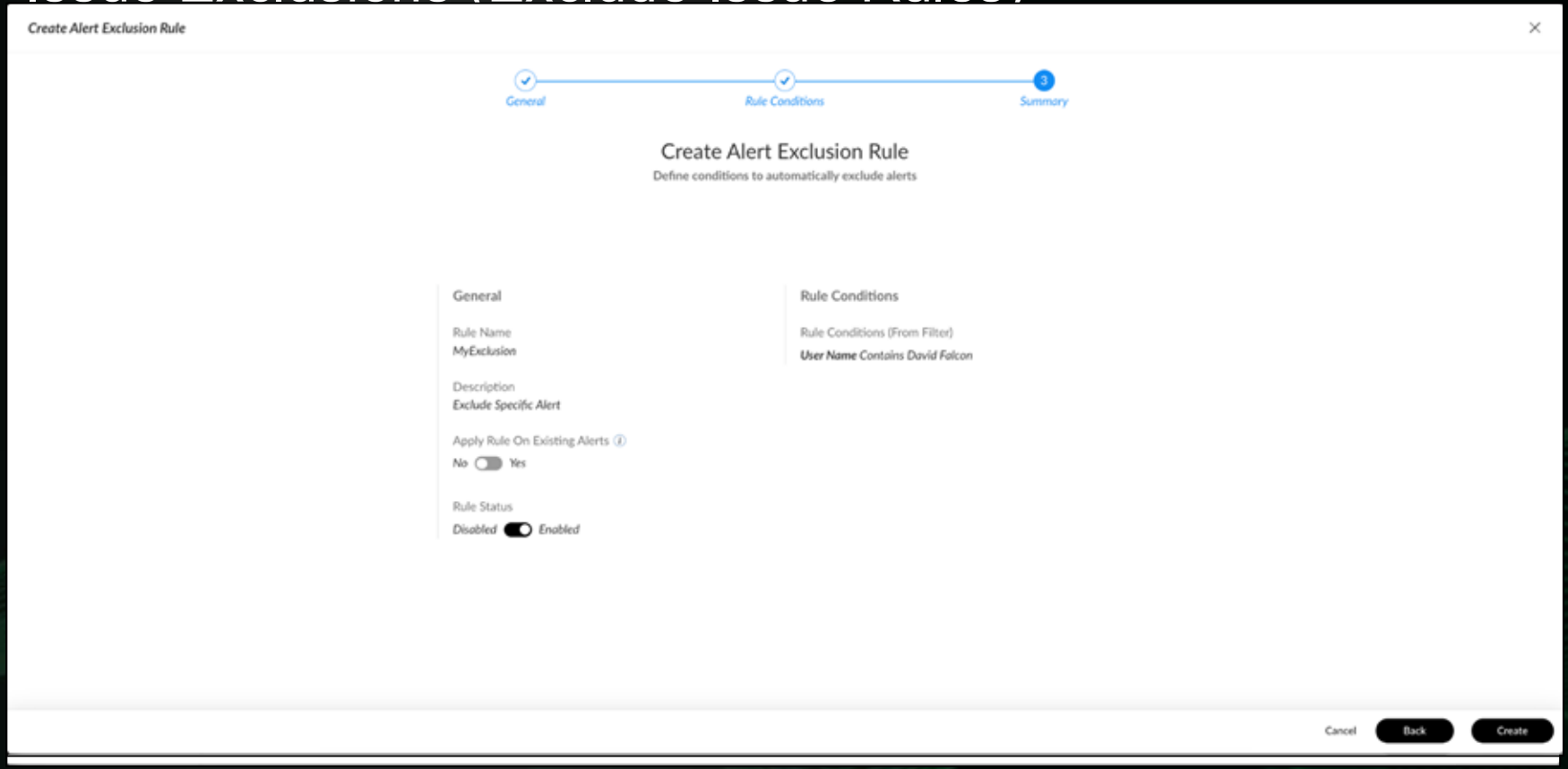
Issue Exclusions (Exclude Issue Rules)



Issue Exclusions (Exclude Issue Rules)



Issue Exclusions (Exclude Issue Rules)

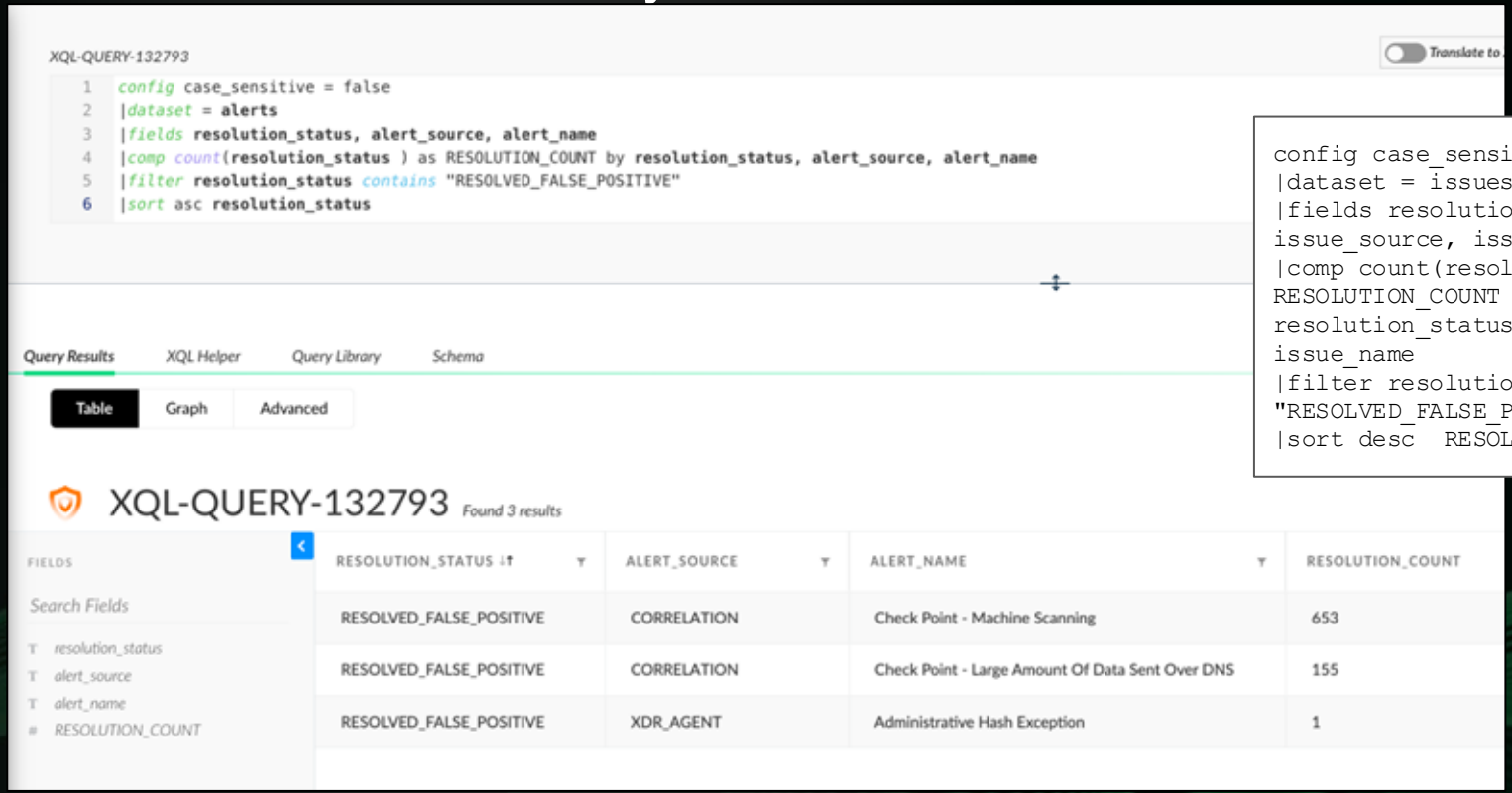


Issue Resolution Analysis




```
config case_sensitive = false
|dataset = issues
|fields resolution_status
|comp count(resolution_status ) as
RESOLUTION_COUNT by resolution_status
|filter resolution_status contains "RESOLVED"
|sort asc resolution_status
| view graph type = pie subtype = full
show_callouts = `true` show_callouts_names =
`true` xaxis = resolution_status yaxis =
RESOLUTION_COUNT
```

Issue Resolution Analysis




```
config case_sensitive = false
|dataset = issues
|fields resolution_status,
issue_source, issue_name
|comp count(resolution_status ) as
RESOLUTION_COUNT by
resolution_status, issue_source,
issue_name
|filter resolution_status contains
"RESOLVED_FALSE_POSITIVE"
|sort desc  RESOLUTION_COUNT
```

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Integrations - ITSM (SNOW & JIRA)

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Integration - ITSM (SNOW & JIRA)

	Implemented	Requirement	Use-Case
SNOW	• TBD	• TBD	• TBD
JIRA	• TBD	• TBD	• TBD

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Reporting

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Reporting

Teams	Required Reports
xxxx	xxxx
xxxx	xxxx
xxxx	xxxx
xxxx	xxxx

TSA FIELD ENABLEMENT · WORKSHOP MODULE

DAY 3 - XSIAM Operations (Based on Entitlements)

Read the environment. Understand the drivers. Define the operational tasks.

General Overview - Cortex XSIAM

Overview of Modules (In Entitlements)

We will walk through your entitlements, and the platform to get a better understanding of how it’s being operationalized currently.

General Overview - Overview of Modules (Based On Entitlements)

Entitlement Items	Used	Not-Used	Why	Team Ownership
xxxx	xxxx	xxxx	xxxx	xxxx
xxxx	xxxx	xxxx	xxxx	xxxx
xxxx	xxxx	xxxx	xxxx	xxxx
xxxx	xxxx	xxxx	xxxx	xxxx

XSIAM Operations

XSIAM Operations

Operations within Cortex XSIAM with Cortex Cloud consolidate multi-cloud posture management, runtime workload defense, and cloud detection and response (CDR) into a unified, AI-driven SecOps platform to continuously monitor assets, automate cross-cloud incident triage, and accelerate remediation at machine speed.

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Datasets

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Data Modelling

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Modeling vs. Parsing

Modeling Rules

- Post ingestion
- Per dataset
- Mapped on read (search)
- Limited to XDM schema
- Automatically mapped to the alert table
- Auto enrichments on selected fields
- Simple search support
- Aliases support

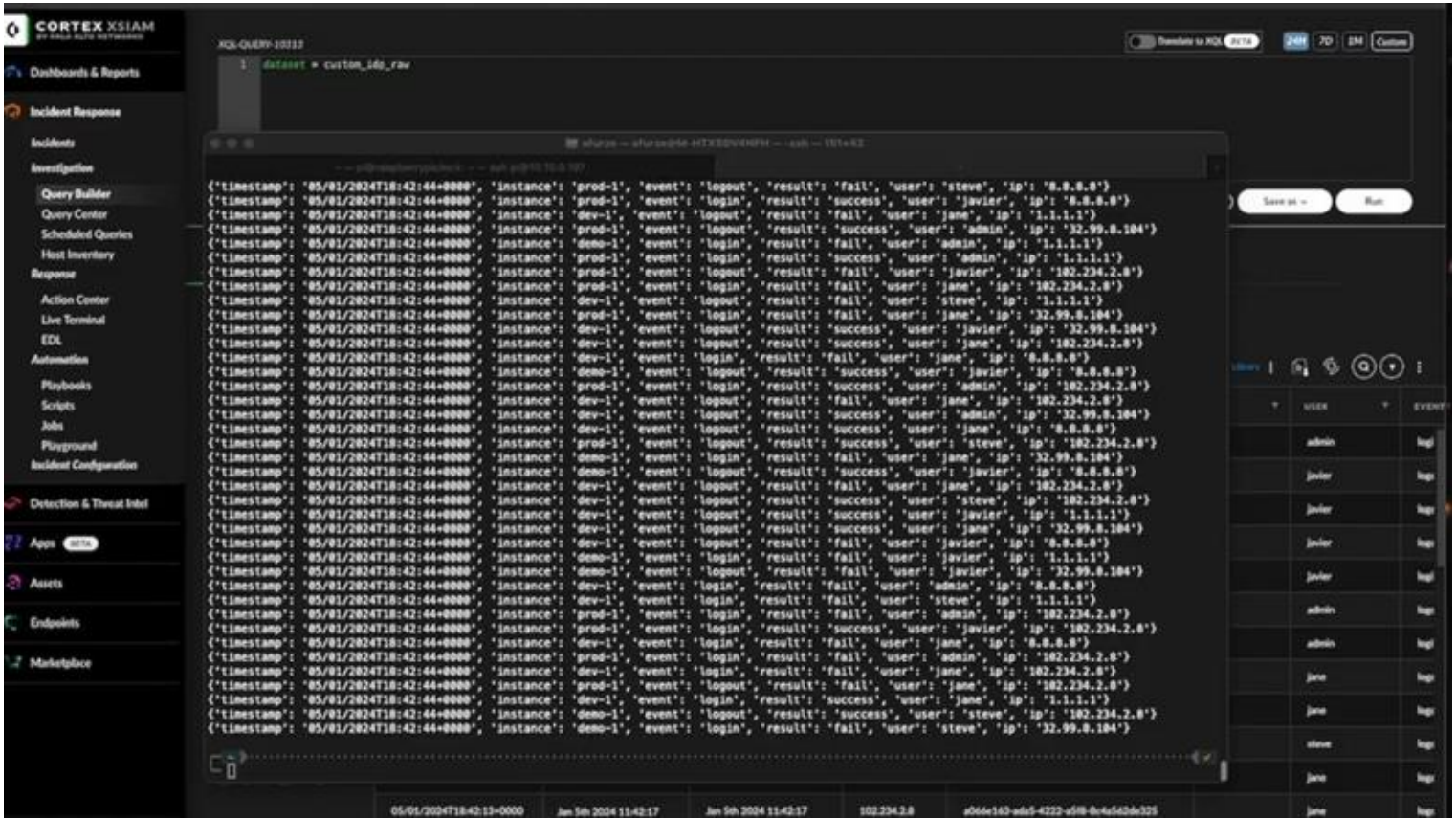


Parsing Rules

- Occurs during ingestion
- Assigned to Vendor & Product pair.
- **Parsing rules can't be undone**
- **Altering the data**
- No schema limitations
- Current OOTB usage is for timestamps modifications only

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Demo: Parsing Rule



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XDM Mapping - Good Practices

1. XQL Reference Guide - Know the Language.
2. Get to know the XDM data model.
3. Search XSIAM marketplace Github for examples.
4. Check the XSIAM Marketplace for OOTB packs.
5. Review documentation provided by the vendor.
6. Identify filters to segment sections in your modeling rule.
7. Note the type of the field - “string”, “int”, etc.
8. Aggregate “alter” usages - A single “alter” can perform actions on multiple fields.
9. Parsing VS Modeling - modeling will work on search time while parsing will modify the stored data (use with caution)
10. 5 Tuples for Network Analytics - source & target IP’s and Ports + Protocol.
11. Prefer UTC timezone when configuring logging.
12. Test your Regex patterns across multiple logs via Regex 101.

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Data Fusion Engine

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XSIAM Detection Engines

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Issues and Severities

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Cases and Severities

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Detection Engines

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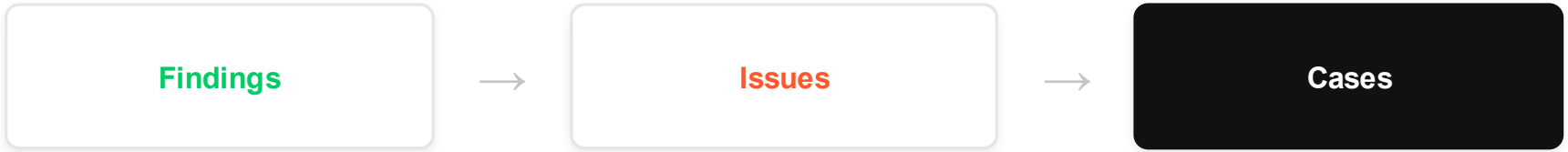
Issues and Cases

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General Overview - Overview of XSIAM Capabilities - Taxonomy

Cases and Issues

Cortex XSIAMs taxonomy categorizes raw posture and asset state into non-actionable **Findings**, escalates risk into actionable **Issues** and automatically correlates issues into unified **Cases** to streamline incident investigation and response.

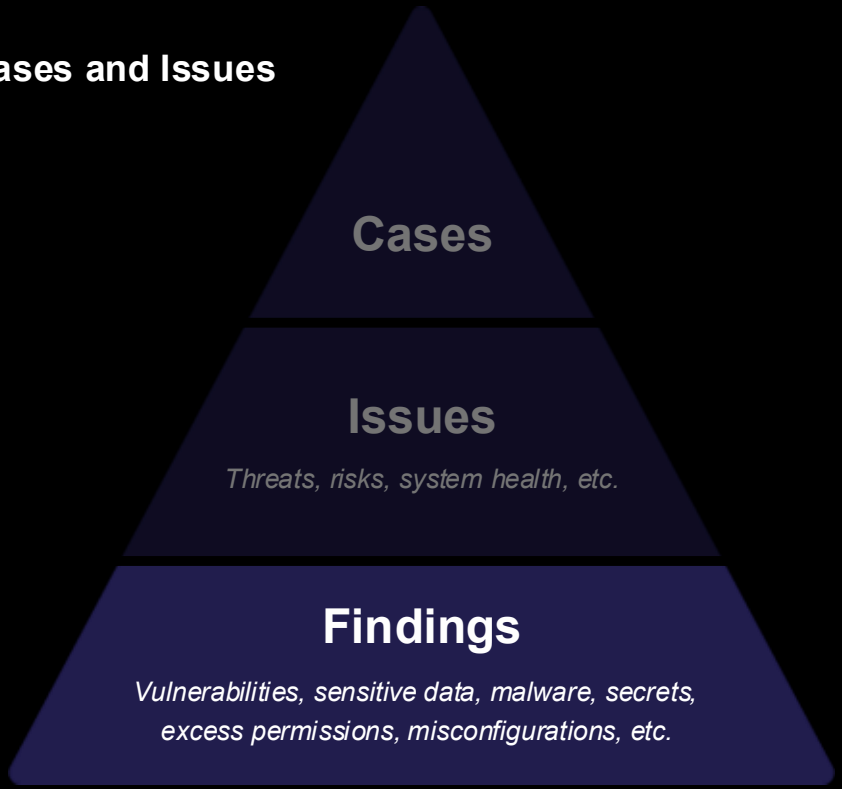


EXAMPLE

Finding: A routine cloud security scan detects an AWS S3 bucket configured with public read permissions (raw posture discovery).
Issue: Cortex XSIAM flags that the bucket contains confidential customer data, breaching your compliance policy and escalating the posture finding into an actionable **Issue** (*"Publicly Exposed Sensitive Data Store"*).
Case: An hour later, an untrusted IP address downloads a large volume of files from that bucket; XSIAM automatically groups the original exposure **Issue** and the subsequent data exfiltration **Issue** into a single **Case** (*"Unsanctioned Cloud Data Exfiltration"*) for the SOC team to investigate and auto-remediate.

General Overview - Overview of XSIAM Capabilities - Taxonomy

Cases and Issues



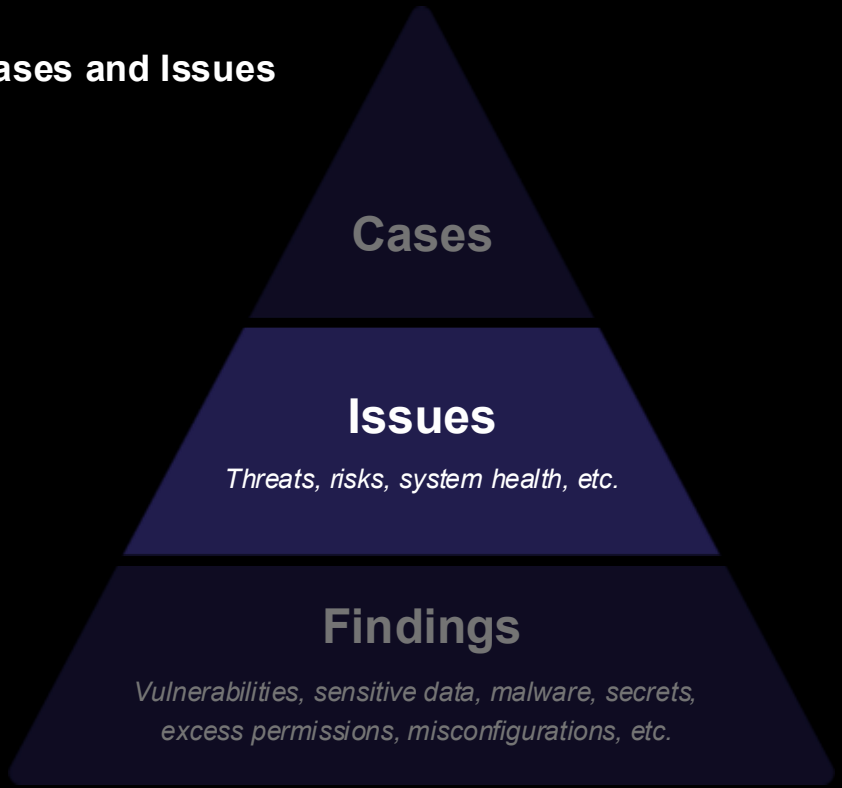
Findings

Information deduced by the platform from raw cloud data.

- Building block for other concepts (issues, dashboard widgets etc.).
- Not necessarily security gaps or actionable.

General Overview - Overview of XSIAM Capabilities - Taxonomy

Cases and Issues



Findings

Information deduced by the platform from raw cloud data.

- Building block for other concepts (issues, dashboard widgets etc.).
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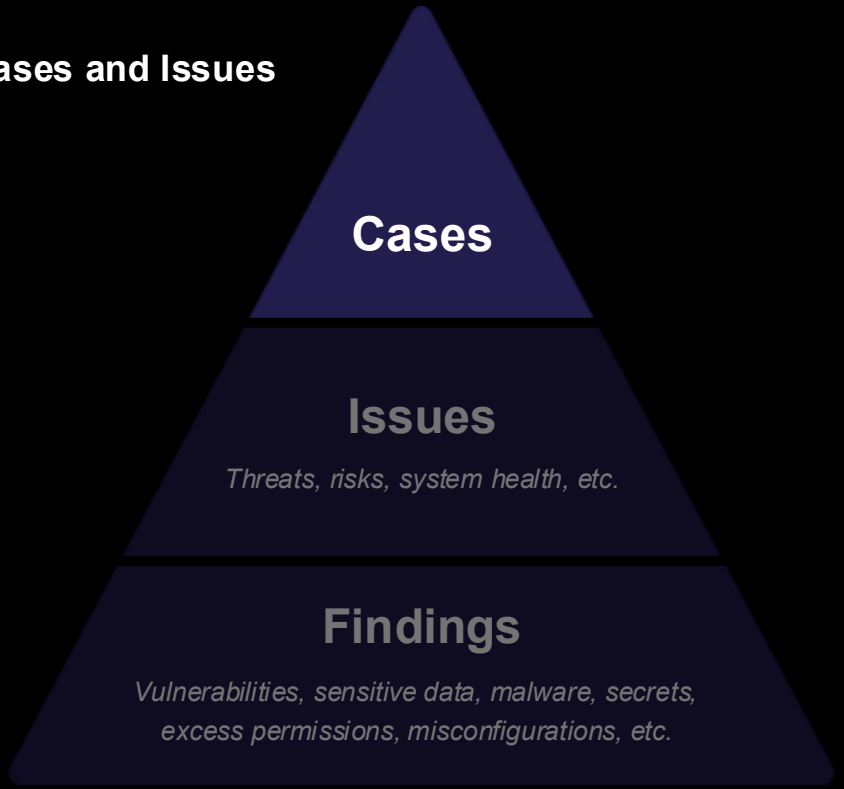
Issues

Basic unit of an actionable task - a security gap in your environment.

- Has dedicated evidence and remediation guides.
- Can be assigned, have a status, trigger playbooks etc.
- Built based on findings, attributes, permissions query.
- Created from detection rules, which can be customized.

General Overview - Overview of XSIAM Capabilities - Taxonomy

Cases and Issues



Findings

Information deduced by the platform from raw cloud data.

- Building block for other concepts (issues, dashboard widgets etc.).
- Not necessarily security gaps or actionable.

Issues

Basic unit of an actionable task - a security gap in your environment.

- Has dedicated evidence and remediation guides.
- Can be assigned, have a status, trigger playbooks etc.
- Built based on findings, attributes, permissions query.
- Created from detection rules, which can be customized.

Cases

Groups of issues

- Can be assigned, have a status, trigger playbooks etc..

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Issue Domains

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RBAC and SBAC

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Dashboards and Widgets

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Custom Layouts and Buttons

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Issue Grouping & 50K Issues a Day Per Module

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TSA FIELD ENABLEMENT · WORKSHOP MODULE

DAY 4 - Optimization and Architecture Considerations

Read the environment. Understand the drivers. Define the operational tasks.

Optimization and Next Steps With Recommendations

Optimization

Optimization within Cortex XSIAM with Cortex Cloud is looking at continuously tuning multi-cloud data ingestions, AI detection models, and automated playbooks to eliminate alert noise, streamline SOC workflows, and maximize operational efficiency across cloud environments.

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Leveraging Agentix and AI

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Cortex AgentiX



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What Good Looks Like

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Optimization and Next Steps With Recommendations - Parking Lot

Optimization and Next Steps With Recommendations

Next Steps and Recommendations

Taking into consideration our knowledge of your environment we will help define a blueprint for next steps and key recommendations that need to be addressed.

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Parking Lot Items

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Roadmap

Optimization and Next Steps With Recommendations - Roadmap

THE PRINCIPLE

Enablement is a prescription.
Diagnose first.

Read the environment · understand the drivers · prescribe the sessions that close the gap.